Uyuni 2026.08

管理指南

UYUNI

章 1. 前言

Administration Guide + Uyuni 2026.08

本指南将介绍维护、监控以及自定义 Uyuni 服务器等管理任务。

发布日期： 2026-08-05

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章 2. 操作

您可以通过多种不同的方式管理对客户端执行的操作：

- 可以安排自动重复性操作，以按照指定的日程安排将 Highstate 或任意一组自定义状态应用于客户端。
- 可以将重复性操作应用于单个客户端、系统组中的所有客户端或整个组织。
- 可以通过创建操作链来设置要按特定顺序执行的操作。
  - 可以提前创建和编辑操作链，并将其安排为在适当的时间运行。
- 还可以在一个或多个客户端上执行远程命令。
  - 使用远程命令可以向单个客户端或者与搜索词匹配的所有客户端发出命令。

2.1. 重复性操作

可以对单个客户端、系统组、组织中的所有客户端应用重复性操作。

Uyuni 目前支持将以下操作类型作为重复性操作：

- **Highstate:** Execute the highstate.
- **Custom state:** Execute a set of custom states. A custom state can be either an internal state provided by Uyuni, or a configuration channel created by a user.

有关配置通道的详细信息，请参见 **Client Configuration Guide › Configuration Management**。

过程：创建新的重复性操作

1. To apply a recurring action to an individual client, navigate to **Systems**, click the client to configure schedules for, and navigate to the **Recurring Actions** tab.
2. To apply a recurring action to a system group, navigate to **Systems › System Groups**, select the group to configure schedules for, and navigate to **Recurring Actions** tab.
3. 单击 **[创建]**。
4. Select an action type from the **Action Type** dropdown.
5. 键入新日程安排的名称。
6. 选择重复性操作的频率：
   - **Hourly:** Type the minute of each hour. For example, **15** runs the action at fifteen minutes past every hour.
   - **Daily:** Select the time of each day. For example, **01:00** runs the action at 0100 every day, in the timezone of the Uyuni Server.
   - **Weekly:** Select the day of the week and the time of the day, to execute the action every week at the specified time.
   - **Monthly:** Select the day of the month and the time of the day, to execute the action every

month at the specified time.

  ◦ **Custom Quartz format:** For more detailed options, enter a custom quartz string. 例如，要在每个月的每个星期六 02:15 运行重复性操作，请输入：

```
0 15 2 ? * 7
```

7. OPTIONAL: Toggle the **Test mode** switch on to run the schedule in test mode.
8. For actions of type **Custom state**, select the states from the list of available states and click [**Save Changes**]. This will only save the current state selection and not the schedule.
9. 在下一个窗格中，拖放所选状态使其按执行顺序排列，然后单击 [**确认**]。
10. 单击 [**创建日程安排**] 保存设置，并查看现有日程安排的完整列表。

组织管理员可为组织中的所有客户端设置和编辑重复性操作。导航到**首页 › 我的组织 › 重复性操作**，即可看到适用于整个组织的所有重复性操作。

Uyuni 管理员可为所有组织中的所有客户端设置和编辑重复性操作。导航到**管理 › 组织**，选择要管理的组织，然后导航到**状态 › 重复性操作**选项卡。

2.2. 操作链

如果您需要对客户端执行多个有序操作，可以创建一个操作链以确保遵循该顺序。

默认情况下，大多数客户端会在发出命令后立即执行操作。在某些情况下，操作需要很长时间，这可能意味着之后发出的操作会失败。例如，如果您指示客户端重引导，然后发出第二个命令，则第二个操作可能会失败，因为重引导仍在进行。为确保操作按正确的顺序进行，请使用操作链。

> 对于事务更新系统，操作链会在单个快照内执行，直到进行重引导操作。这可能会产生某些限制。
>
> 有关详细信息，请参见 **Client Configuration Guide › Clients Slemicro** 和 **Client Configuration Guide › Clients Opensuseleapmicro**。

可以在所有客户端上使用操作链。操作链可以包含任意数量、采用任意顺序的以下操作：

- **系统细节 › 远程命令**
- **系统细节 › 安排系统重引导**
- **系统细节 › 状态 › Highstate**
- **系统细节 › 软件 › 软件包 › 列出/去除**
- **系统细节 › 软件 › 软件包 › 安装**
- **系统细节 › 软件 › 软件包 › 升级**

* **系统细节 › 软件 › 补丁**
* **系统细节 › 软件 › 软件通道**
* **系统细节 › 配置**
* **映像 › 构建**

> 操作链因用户而异。要在 Web UI 中查看操作链，您必须以创建相应操作链的用户身份登录。

过程：创建新操作链

1. In the Uyuni Web UI, navigate to the first action you want to perform in the action chain. For example, navigate to **System Details** for a client, and click **[Schedule System Reboot]**.
2. Check the **Add to** field and select the action chain you want to add to:
   - If this is your first action chain, select **new action chain**.
   - 如果该操作链已存在，请从列表中选择它。
   - 如果您已有操作链，但想要创建一个新操作链，请键入新操作链的名称以创建该操作链。
3. 确认操作。 该操作不会立即执行，而是创建新的操作链，并在屏幕顶部显示一个确认此行为的蓝色条。
4. Continue adding actions to your action chain by checking the **Add to** field and selecting the name of the action chain to add them to.
5. 添加完操作后，导航到**日程安排 › 操作链**并从列表中选择操作链。
6. 通过将操作拖放到正确的位置来重新排列操作顺序。 单击蓝色加号查看要对其执行操作的客户端。单击 **[保存]** 以保存更改。
7. 安排操作链的运行时间，然后单击 **[保存并安排]**。 如果您在不单击 **[保存]** 或 **[保存并安排]** 的情况下离开页面，将丢弃所有未保存的更改。

> 如果操作链中的某个操作失败，操作链将会停止，且不再执行其他操作。

可以通过导航到**日程安排 › 待执行的操作**来查看操作链中已安排的操作。

2.3. 远程命令

请按照此过程通过 Salt 运行远程命令。

在开始之前，请确保您的客户端已订阅适用于其中所安装操作系统的工具子通道。有关订阅软件通道的详细信息，请参见 **Client Configuration Guide › Channels**。

> * 对于事务更新系统，需考虑到远程命令会在单个快照内运行。这可能会产生某些限制。有关详细信息，请参见 **Client Configuration Guide › Clients Slemicro** 和 **Client Configuration Guide › Clients Opensuseleapmicro**。
> * Remote commands are run from the **/tmp/** directory on the client. To ensure that remote commands work accurately, do not mount **/tmp** with the **noexec**

option. For more information, see **Administration Guide › Troubleshooting › Troubleshooting Mounting /tmp with noexec**.

- All commands run from the **Remote Commands** page are executed as root on clients. Wildcards can be used to run commands across any number of systems. Always take extra care to check your commands before issuing them.

Procedure: Running Remote Commands on Clients

1. 导航到 **Salt › 远程命令**。
2. In the first field, before the **@** symbol, type the command you want to issue.
3. In the second field, after the **@** symbol, type the client you want to issue the command on. You can type the **minion-id** of an individual client, or you can use wildcards to target a range of clients.
4. 单击 **[查找目标]** 查看指定为目标的客户端，并确认使用了正确的细节。
5. 单击 **[运行命令]** 向目标客户端发出命令。

章 3. Ansible 集成

Ansible 是一款计算机客户端系统管理工具。有关详细信息，请访问 https://www.ansible.com。

Uyuni 支持管理 Ansible 控制节点。有关详细信息，请参见 **Administration Guide › Ansible Setup Control Node › At.ansible.overview**。

The supported version for the Ansible Control Node is Ansible 11.3. It should be obtained from the operating system vendor’ s official repositories. For example, on SUSE Linux Enterprise 15 SP6 and SP7, Ansible is available through the **Systems Management Module.** For Control Nodes running operating systems other than SUSE Linux Enterprise, use Ansible shipped together with that distribution.

Ansible 软件同样适用于 Uyuni 代理及 Uyuni for Retail 分支服务器。

3.1. 功能概述

Uyuni 可让系统管理员操作其 Ansible 控制节点。支持的功能包括：

- 清单文件检查
- 发现剧本
- 执行剧本

更多信息：

- 清单是指受管 Ansible 节点的排序列表。有关如何整理清单的详细信息，请参见 https://docs.ansible.com/ansible/latest/inventory_guide/intro_inventory.html。
- 剧本用于描述应如何管理清单。有关剧本的详细信息，请参见 https://docs.ansible.com/ansible/latest/playbook_guide/playbooks_intro.html。

3.2. 要求和基本配置

To use Ansible features, you need to register the already existing Ansible Control Node to the Uyuni Server. In the Web UI, on the **System Details › Properties** page of the registered system, you must enable the **Ansible Control Node** system type of the **Add-on System Types** list.

Enabling the **Ansible Control Node** system type ensures that the **ansible** package is installed on the system by adding it in the highstate and activates the Ansible features under the **System Details › Ansible** tab.

As the next step, configure the paths to your Ansible playbook directories and inventory files on the **System Details › Ansible › Control Node** page. As an inventory path, you can use the standard Ansible inventory path **/etc/ansible/hosts**. As a playbook directory, you can use any directory on the control node, where your playbook files are stored. A playbook directory either contains **.yml** files or subdirectories with **.yml** files.

有关如何安装和设置 Ansible 控制节点的信息，请参见 **Administration Guide › Ansible Setup Control Node**。

3.3. 清单检查

定义清单路径后，您可以使用 Uyuni 检查其中的内容。

过程：从 Web UI 检查清单

1. 在 Uyuni Web UI 中，导航到**系统细节 › Ansible › 清单**
2. 单击某个清单路径，在控制节点上实时执行清单检查。

3.4. 剧本发现

定义剧本目录后，可以在**系统细节 › Ansible › 剧本**页面上发现剧本。

与清单检查一样，剧本发现操作也是在控制节点上实时运行。

3.5. 剧本执行

You can schedule a playbook execution from the **System Details › Ansible › Playbooks** page. After selecting the playbook you wish to execute, you can select the inventory file for the execution from the **Inventory Path** drop-down menu of the **Schedule Playbook Execution** dialog. If you do not select any item, the default inventory configured in your Control Node will be used. The drop-down menu is populated with the inventories you defined in your Inventory paths and with inventories that have been locally discovered in your playbook directories. These are displayed as **Custom Inventory** items in the playbook details. You can also enter an arbitrary inventory path.

You can also pass variables to the playbook execution to override previously defined variables. To pass variables, locate the **Playbook Content** section and click **[Edit variables]**. In the **Edit Variables** dialog, you can add or edit variables as a list, dictionary, string, or boolean. Alternatively, click **[Edit YAML]** to edit the variables directly as YAML. Click **[Save]** to apply the variables to the playbook content.

然后，您可以选择剧本执行时间或选择操作链。最终，Uyuni 会在控制节点上将剧本作为操作来执行。可以在操作细节页面上查看操作结果。

3.6. 设置 Ansible 控制节点

要设置 Ansible 控制节点，请在 Uyuni Web UI 中执行以下步骤。

> To configure a client as the Ansible Control Node, the Ansible package must be installed on that system. Usually, the Ansible package should be obtained from the operating system vendor’ s official repositories. For example, on SUSE Linux Enterprise 15 SP6 and SP7, Ansible is available through the **Systems Management Module.**

Procedure: Setting up Ansible Control Node on a SUSE Linux Enterprise 15 SP6 or SP7 system

1. In the Uyuni Web UI, navigate to **Admin › Setup Wizard › Products**, verify that **SUSE Linux Enterprise Server 15 SP6 x86_64** (or later) with the **Systems Management Module** and the required **Python 3 Module** are selected and synchronized.
2. 部署 SUSE Linux Enterprise 15 SP6（或更高版本）客户端。
3. In the Uyuni Web UI, navigate to the **Systems › Overview** page of the client. Select **Software › Software Channels** and subscribe the client to the **SUSE Linux Enterprise Server 15 SP6 x86_64** (or later SP), **Systems Management Module** and **Python 3 Module** channels.
4. Select **Details › Properties** of your client. From the **Add-On System Types** list enable **Ansible Control Node** and click [**Update Properties**].
5. 导航到客户端概览页面，选择**状态 › Highstate**，然后单击 [**应用 Highstate**]。
6. 选择**事件**选项卡并校验 Highstate 的状态。

> If you want to install a newer Ansible on a SUSE Linux Enterprise 15 SP4 or SP5 client, you must enable the **Python 3 Module**.

> 较新的 Ansible 版本已不再支持管理使用过时 Python 版本的节点。如果受管节点仍然默认使用较旧 Python 版本，在运行剧本时可能会遇到连接错误或执行失败。为解决此问题，用户应在条件允许的情况下升级受管节点上的 Python，并在 Ansible 清单或配置中设置正确的 Python 解释器。

3.6.1. 创建 Ansible 清单文件

Ansible 集成工具会将剧本部署为清单文件。请为表 1 中列出的每个操作系统创建一个清单文件。

过程：创建 Ansible 清单文件

1. Create and add your hosts to an inventory file to be managed by Ansible. The default path for an Ansible inventory is **/etc/ansible/hosts**.

   列表 1. 清单示例

```
client240.mgr.example.org
client241.mgr.example.org
client242.mgr.example.org
client243.mgr.example.org
ansible_ssh_private_key_file=/etc/ansible/some_ssh_key
```

```
[mygroup1]
client241.mgr.example.org
client242.mgr.example.org

[mygroup2]
client243.mgr.example.org

[all:vars]
ansible_ssh_private_key_file=/etc/ansible/my_ansible_private_key
```

2. In the Uyuni Web UI, from the **Ansible** tab navigate to **Ansible › Control Node** to add inventory files to the control node.
3. Under the **Playbook Directories** section add **/usr/share/scap-security-guide/ansible** to the **Add a Playbook Directories** field and click **[Save]**.
4. Under **Inventory Files** add your inventory file locations to the **Add an Inventory file** field and click **[Save]**.

列表 2. 示例

```
/etc/ansible/sles15
/etc/ansible/sles12
/etc/ansible/centos7
```

有关更多剧本示例，请参见 https://github.com/ansible/ansible-examples。

3.6.2. 与 Ansible 节点建立通信

Procedure: Establishing communication with Ansible Nodes

1. 创建您要在清单中使用的 SSH 密钥。

```
ssh-keygen -f /etc/ansible/my_ansible_private_key
```

2. 将生成的 SSH 密钥复制到 Ansible 受管客户端。示例：

```
ssh-copy-id -i /etc/ansible/my_ansible_public_key
root@client240.mgr.example.org
```

3. Declare the private key in **/etc/ansible/ansible.cfg** as follows:

```
private_key_file = /etc/ansible/my_ansible_private_key
```

Replace **my_ansible_private_key** with the name of the file containing the private key.

4. 通过从控制节点执行以下命令来测试 Ansible 是否正常运行：

```
ansible all -m ping
ansible mygroup1 -m ping
ansible client240.mgr.example.org -m ping
```

现在您可以运行更新。有关详细信息，请参见 **Administration Guide › Ansible Compliance As Code**。

3.7. 合规性即代码

本文档提供了有关使用 Ansible 剧本运行合规性即代码修复的深入信息。

有关使用 bash 脚本运行合规性即代码修复的详细信息，请参见**Administration Guide › 修复**。

要执行修复，您需要在 Ansible 控制节点上安装 SCAP 安全指南软件包。

过程：安装 SCAP 安全指南软件包

1. 在**系统 › 概览**中选择客户端。然后单击**软件 › 软件包 › 安装**。
2. Search for **scap-security-guide** and install the package suitable for your system. See the following table for package distribution requirements:

表格 1. SCAP 安全指南软件包要求

Package name	Supported Systems
scap-security-guide	openSUSE, SLES12, SLES15
scap-security-guide-redhat	CentOS 7, CentOS 8, Fedora, Oracle Linux 7, Oracle Linux 8, RHEL7, RHEL8, RHEL9, Red Hat OpenStack Platform 10, Red Hat OpenStack Platform 13, Red Hat Virtualization 4, Scientific Linux
scap-security-guide-debian	Debian 12
scap-security-guide-ubuntu	Ubuntu 22.04

For Debian 13, Ubuntu 24.04, and Ubuntu 26.04, SUSE does not provide custom security guide packages. You must use upstream packages (**ssg-debian** and **ssg-debderived** respectively) from their respective community repositories. Please note that these upstream guides are not officially supported by SUSE.

3.7.1. 使用 Ansible 剧本进行修复

需要一个 Ansible 控制节点。有关详细信息，请参见 **Administration Guide › Ansible Setup Control Node**。

以下过程将介绍如何使用 Ansible 剧本运行修复任务。

过程：使用 Ansible 剧本运行修复

1. 在控制节点系统菜单中选择 **Ansible › 剧本**，然后单击某个剧本，例如：

   ```
   sle15-playbook-stig.yml
   ```

2. To run the playbook, select the **Inventory Path** for the clients, for example:

   ```
   /etc/ansible/sles15
   ```

   单击 **[日程安排]**。

3. Check the status of the scheduled event under the **Events** tab.

In case playbooks are in a different directory, you can follow the link to **Setup Ansible Control Node** to find out how to add it.

章 4. 身份验证方法

Uyuni 支持多种不同的身份验证方法。本章介绍可插入身份验证模块 (PAM) 和单点登录 (SSO)。

4.1. 使用单点登录 (SSO) 进行身份验证

Uyuni 通过实现安全声明标记语言 (SAML) 2 协议来支持单点登录 (SSO)。

单点登录是一种身份验证过程，它允许用户使用一组身份凭证访问多个应用程序。SAML 是一套基于 XML 的标准，用于交换身份验证和授权数据。SAML 身份服务提供者 (IdP) 向服务提供者 (SP) 提供身份验证和授权服务，例如 Uyuni。Uyuni 公开三个端点，必须启用它们才能进行单点登录。

Uyuni 中的 SSO 支持：

- 使用 SSO 登录。
- 使用服务提供者发起的单点注销 (SLO) 和身份服务提供者单点注销服务 (SLS) 注销。
- 声明和 nameId 加密。
- 声明签名。
- 使用 AuthNRequest、LogoutRequest 和 LogoutResponses 进行消息签名。
- 启用声明使用者服务端点。
- 启用单点注销服务端点。
- 发布 SP 元数据（可签名）。

Uyuni 中的 SSO 不支持：

- 身份服务提供者 (IdP) 的产品选择和实现。
- 对其他产品的 SAML 支持（请查看相关的产品文档）。

有关 SSO 的示例实现，请参见 **Administration Guide › Auth Methods Sso Example**。

> If you change from the default authentication method to single sign-on, the new SSO credentials apply only to the Web UI. Client tools such as **mgr-sync** or **spacecmd** continue to work with the default authentication method only.

4.1.1. 先决条件

在开始之前，需要事先使用这些参数配置一个外部身份服务提供者。请查看 IdP 文档获取说明。

> The mapping between the IdP user and the Uyuni user is specified in a SAML:Attribute. The SAML:Attribute must be configured in the IdP and must be passed to Uyuni in the SAML authentication. The attribute must be named **uid** and must contain the Uyuni user mapped to it after login. The Uyuni user must be created before you activate single sign-on.

需要以下端点：

- 声明使用者服务 (ACS)：接受 SAML 消息以建立与服务提供者的会话的端点。 Uyuni 中 ACS 的端点是：https://server.example.com/rhn/manager/sso/acs
- 单点注销服务 (SLS)：从 IdP 发起注销请求的端点。 Uyuni 中 SLS 的端点是：https://server.example.com/rhn/manager/sso/sls
- 元数据：用于检索 SAML 的 Uyuni 元数据的端点。 Uyuni 中元数据的端点是：https://server.example.com/rhn/manager/sso/metadata

After the authentication with the IdP using the user **orgadmin** is successful, you are logged in to Uyuni as the **orgadmin** user, provided that the **orgadmin** user exists in Uyuni.

4.1.2. 启用 SSO

SSO 与其他类型的身份验证是互斥的：请要么启用，要么禁用 SSO。默认已禁用 SSO。

Use **mgrctl term** before running steps inside the server container.

过程：启用 SSO

1. 如果您的用户在 Uyuni 中尚不存在，请先创建他们。
2. Edit **/etc/rhn/rhn.conf** and add this line at the end of the file:

```
java.sso = true
```

3. Find the parameters you want to customize in **/usr/share/rhn/config-defaults/rhn_java_sso.conf**. Insert the parameters you want to customize into **/etc/rhn/rhn.conf** and prefix them with **java.sso**. For example, in **/usr/share/rhn/config-defaults/rhn_java_sso.conf** find:

```
onelogin.saml2.sp.assertion_consumer_service.url = https://YOUR-PRODUCT-HOSTNAME-OR-
IP/rhn/manager/sso/acs
```

To customize it, create the corresponding option in **/etc/rhn/rhn.conf** by prefixing the option name with **java.sso.**:

```
java.sso.onelogin.saml2.sp.assertion_consumer_service.url = https://YOUR-PRODUCT-
HOSTNAME-OR-IP/rhn/manager/sso/acs
```

To find all the occurrences you need to change, search in the file for the placeholders **YOUR-PRODUCT** and **YOUR-IDP-ENTITY**. Every parameter comes with a brief explanation of what it is meant for.

4. 重启 spacewalk 服务以应用更改：

```
mgradm restart
```

访问 Uyuni URL 时，您将重定向到 SSO 的 IdP，需要在其中完成身份验证。身份验证成功后，您将重定向到 Uyuni Web UI，并以经过身份验证的用户身份登录。如果您在使用 SSO 登录时遇到问题，请查看 Uyuni 日志了解详细信息。

4.1.3. 示例 SSO 实现

In this example, SSO is implemented by exposing three endpoints with Uyuni, and using Keycloak 26.5.3 or later as the identity service provider (IdP).

To set up Single Sign-On (SSO) with Uyuni and Keycloak, you first configure the Keycloak identity provider, obtain its public SAML certificate, and set up the client and users. After that, you configure the Uyuni Server to trust Keycloak.

> 此示例仅用于说明目的。SUSE 不建议也不支持第三方身份服务提供者，并且与 Keycloak 不存在从属关系。有关 Keycloak 支持信息，请参见 https://www.keycloak.org/。

可以直接在您的计算机上安装 Keycloak，或者在容器中运行 Keycloak。此示例在 Podman 容器中运行 Keycloak。有关安装 Keycloak 的详细信息，请参见 https://www.keycloak.org/guides#getting-started 上的 Keycloak 文档。

4.1.3.1. Setting up the identity service provider (Keycloak)

First, run Keycloak, create a realm, obtain the SAML public certificate, and configure the client.

Procedure: Setting up Keycloak

1. 根据 Keycloak 文档所述在 Podman 容器中安装 Keycloak。
2. Run the container using the **-td** argument to ensure the process remains running:

```
podman run -td --name keycloak -p 8080:8080 -e KEYCLOAK_ADMIN=admin -e
KEYCLOAK_ADMIN_PASSWORD=admin quay.io/keycloak/keycloak:26.1.1 start-dev
```

3. Sign in to the Keycloak Admin Console as the **admin** user.
4. Create an authentication realm:
   - In the top-left corner, click the realm selector dropdown (showing **master**) and select **Manage realms**.
   - On the **Manage realms** page, click the **[Create realm]** button in the top-right corner.
   - In the **Realm name** field, enter a name (for example, **MLM**).
   - 单击 **[创建]**。

5. Obtain the SAML public certificate:
   - From the left navigation menu, click **Realm settings**.
   - Click the **Keys** tab.
   - Find the active **RSA** key (with provider **rsa-generated**) in the active keys table.
   - Click the **[Certificate]** button in that row.
   - Copy the entire Base64-encoded certificate from the modal that opens. Keep this certificate handy for the server configuration.
6. Add Uyuni as a Client:
   - From the left navigation menu, click **Clients**.
   - Click **[Create client]**.
   - In the **Client type** field, select **SAML**.
   - In the **Client ID** field, enter the Uyuni Metadata endpoint. For example: https://<FQDN_MLM>/rhn/manager/sso/metadata
   - Click **[Next]**.
   - In the **Valid redirect URIs** field, enter the Uyuni Assertion Consumer Service (ACS) endpoint. For example: https://<FQDN_MLM>/rhn/manager/sso/acs
   - 单击 **[保存]**。
7. Fine-tune client settings:
   - Under the **Settings** tab, scroll down to the **SAML capabilities** section:
     - Toggle **Sign assertions** to **On**.
     - In the **Signature algorithm** field, select **RSA_SHA1**.
     - In the **SAML Signature Key Name** field, select **Key ID**.
     - Click **Save**.
   - Under the **Keys** tab:
     - Toggle **Client signature required** to **Off**.
   - Under the **Advanced** tab, scroll down to the **Fine Grain SAML Endpoint Configuration** section:

- In both the **Assertion Consumer Service POST Binding URL** and **Assertion Consumer Service Redirect Binding URL** fields, enter the Uyuni ACS endpoint. For example: https://<FQDN_MLM>/rhn/manager/sso/acs
- In the **Logout Service POST Binding URL** field, enter the Uyuni SLS endpoint. For example: https://<FQDN_MLM>/rhn/manager/sso/sls
- In the **Logout Service Redirect Binding URL** field, enter the root URL of your Uyuni instance. For example: https://<FQDN_MLM>/
- Click **Save**.

8. Configure Client Scope and Mappers:
   - From the left navigation menu, click **Client scopes**.
   - Click **role_list** in the list.
   - Click the **Mappers** tab, and then click the **role list** mapper.
   - Toggle **Single Role Attribute** to **On**, and click **[Save]**.
   - Go back to **Clients**, click your Uyuni client ID (https://<FQDN_MLM>/rhn/manager/sso/metadata), and click the **Client scopes** tab.
   - Click the dedicated scope (for example, https://<FQDN_MLM>/rhn/manager/sso/metadata-dedicated).
   - Click **[Configure a new mapper]**.
   - Select **User Attribute** from the list, and configure the mapper with these details:
     - In the **Name** field, enter **uid**.
     - In the **Property** field, select **username**.
     - In the **SAML Attribute Name** field, enter **uid**.
   - 单击 **[保存]**。
9. Create Users:
   - From the left navigation menu, click **Users**.
   - Click **[Add user]**.
   - In the **Username** field, enter a username that matches an existing Uyuni user (for example, the administrative user).

- 单击 [**创建**]。
- Click the **Credentials** tab, click [**Set password**], set a password for the user, and toggle **Temporary** to **Off**, click [**Save**] and confirm by clicking [**Save password**].

4.1.3.2. Setting up the Uyuni server

After Keycloak is ready, configure the Uyuni Server to use single sign-on.

> Ensure that the users whom you intend to authenticate using SSO already exist in Uyuni. You can create users manually in the Uyuni Web UI, or use the command-line interface or API (using PAM-enabled or passwordless authentication).

Procedure: Setting up the Uyuni server

1. On the Uyuni Server, open the **/etc/rhn/rhn.conf** configuration file.
2. Add the following parameters. Replace **<FQDN_MLM>** with the fully qualified domain name of your Uyuni installation, **<FQDN_IDP>** with the FQDN of your Keycloak server, **<REALM>** with your realm name (for example, **MLM**), and **<CERTIFICATE>** with the SAML public certificate Base64-encoded string obtained from Keycloak in the previous steps:

```
java.sso = true

java.sso.onelogin.saml2.debug = true

java.sso.onelogin.saml2.sp.entityid                          =
https://<FQDN_MLM>/rhn/manager/sso/metadata
java.sso.onelogin.saml2.sp.assertion_consumer_service.url =
https://<FQDN_MLM>/rhn/manager/sso/acs
java.sso.onelogin.saml2.sp.single_logout_service.url      =
https://<FQDN_MLM>/rhn/manager/sso/sls

java.sso.onelogin.saml2.idp.entityid                      =
http://<FQDN_IDP>:8080/realms/<REALM>
java.sso.onelogin.saml2.idp.single_sign_on_service.url =
http://<FQDN_IDP>:8080/realms/<REALM>/protocol/saml
java.sso.onelogin.saml2.idp.single_logout_service.url  =
http://<FQDN_IDP>:8080/realms/<REALM>/protocol/saml

java.sso.onelogin.saml2.idp.x509cert = -----BEGIN CERTIFICATE-----
<CERTIFICATE> -----END CERTIFICATE-----

# Organization Metadata (Optional)
java.sso.onelogin.saml2.organization.name = SUSE Multi-Linux Manager
java.sso.onelogin.saml2.organization.displayname = SUSE Multi-Linux Manager
java.sso.onelogin.saml2.organization.url = https://<FQDN_MLM>
java.sso.onelogin.saml2.organization.lang =

# Contacts (Optional)
java.sso.onelogin.saml2.contacts.technical.given_name = SUSE Multi-Linux
Manager Admin
```

```
java.sso.onelogin.saml2.contacts.technical.email_address = admin@example.com
java.sso.onelogin.saml2.contacts.support.given_name = SUSE Multi-Linux Manager
Admin
java.sso.onelogin.saml2.contacts.support.email_address = admin@example.com
```

3. Save your changes and restart the spacewalk service to apply them:

```
mgradm restart
```

4. Navigate to the Uyuni Web UI. You should be redirected to the Keycloak login screen, and successfully log in back to Uyuni upon authentication.

4.1.3.3. 查错

If you encounter issues when authenticating with single sign-on, use these steps to diagnose:

- **Verify Logs:** Look for SSO errors in **/var/log/rhn/rhn_web_ui.log**. Detailed logging is enabled when **java.sso.onelogin.saml2.debug = true** is set in **/etc/rhn/rhn.conf**.
- **User Not Found:** If you see "User not found" or redirection fails after Keycloak authentication, ensure the username in Keycloak matches the username in Uyuni exactly. Users must be pre-created in Uyuni.
- **Signature and certificate errors:** If you see "Signature verification failed", verify that the certificate in **java.sso.onelogin.saml2.idp.x509cert** matches the active RSA certificate displayed under the **Keys** tab in Keycloak Realm Settings, and ensure that **Sign assertions** is toggled to **On** in the client settings.
- **Invalid Redirect URI:** If Keycloak displays "Invalid parameter: redirect_uri" on redirection, verify that **Valid redirect URIs** in client settings in Keycloak is set to https://<FQDN_MLM>/rhn/manager/sso/acs.

4.2. 使用 PAM 进行身份验证

Uyuni 支持使用可插入身份验证模块 (PAM) 和 SSSD 的基于网络的身份验证系统。PAM 是一个库套件，可用于将 Uyuni 与集中式身份验证机制相集成，让用户无需记住多个口令。Uyuni 支持 LDAP、Kerberos 和其他基于网络的身份验证协议。

4.2.1. SSSD 配置

过程：配置 SSSD

1. 在 Uyuni Web UI 中，导航到**用户 › 创建用户**，并允许新用户或现有用户使用 PAM 进行身份验证。

   > In usernames, additionally to alphanumeric characters, **-**, **_**, **.**, and **@** are allowed.

2. Check the **Pluggable Authentication Modules (PAM)** checkbox.
3. 在服务器容器中配置 SSSD。在 Uyuni 容器主机的命令提示符处，以 root 身份进入服务器容器：

```
mgrctl term
```

4. 在容器内部，执行以下步骤：
   a. Edit **/etc/sssd/sssd.conf** according to your configuration. For an example, see **Administration Guide › Auth Methods Pam › Auth Methods Pam Ad**.
   b. 完成后，退出容器：

   ```
   exit
   ```

5. 使用以下命令重启 Uyuni：

```
mgradm restart
```

在 Uyuni Web UI 中更改口令只会更改 Uyuni Server 上的本地口令。如果为该用户启用 PAM，则可能根本不会使用本地口令。例如，在上面的示例中，Kerberos 口令并未更改。使用网络服务的口令更改机制来更改这些用户的口令。

有关配置 PAM 的详细信息，请参见《SUSE Linux Enterprise Server 安全指南》，其中提供了一个通用示例，该示例同样适用于其他基于网络的身份验证方法。此外，它还介绍了如何配置 Active Directory (AD) 服务。有关详细信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/part-auth.html。

4.2.1.1. LDAP 与 Active Directory 集成的示例

要将 LDAP 与 Active Directory 集成，可以使用以下示例。

在代码段中，根据您的环境更改以下占位符：

$domain
: 您的域名

$ad_server
: FQDN of the AD server if it is not auto-detected from the **$domain** **$uyuni-hostname**: The name of the machine this AD client is supposed to be known. If not set, it will be **uyuni-server.mgr.internal**.

Example snippet for **/etc/sssd/sssd.conf**:

```
[sssd]
config_file_version = 2
services = nss, pam
domains = $domain

[nss]
```

```
[pam]

[domain/$domain]
id_provider = ad
chpass_provider = ad
access_provider = ad
auth_provider = ad

ad_domain = $domain
ad_server = $ad_server
ad_hostname = $uyuni-hostname

ad_gpo_map_network = +susemanager

krb5_keytab = FILE:/etc/rhn/krb5.conf.d/krb5.keytab
krb5_ccname_template = FILE:/tmp/krb5cc_%{uid}
```

章 5. 备份和恢复

This chapter contains information on the files you need to back up. With the built-in backup and restore solution (**mgradm backup**) you create Uyuni backups. Information about restoring from your backups in the case of a system failure completes this chapter.

由于 Uyuni 依赖数据库及已安装的程序和配置，因此备份安装的所有组件至关重要。请定期备份您安装的 Uyuni 系统，以防止数据丢失并确保快速恢复。

> 无论您使用哪种备份方法，可用空间必须是当前安装使用的空间量的至少三倍。空间不足可能导致备份失败，因此请经常检查可用空间。

5.1. 备份 Uyuni

The most comprehensive method for backing up your Uyuni installation is to use **mgradm backup create** command. This can save you time in administering your backup, and can be faster to reinstall and re-synchronize in the case of failure. However, this method requires significant disk space and could take a long time to perform the backup.

mgradm backup create command performs backup to a directory. This directory can be both local or mounted remote storage.

mgradm backup create command allows various customizations of the content of the backup. For all available options, see **mgradm backup create --help**.

5.1.1. Uyuni 的完整备份

Uyuni 的完整备份包含以下组件的备份：

- Uyuni 卷
- 数据库卷
- podman 网络配置
- podman 机密
- Uyuni systemd 服务
- Uyuni 容器映像

> 创建完整备份期间，Uyuni 服务将自动停止。此期间的服务中断时间可能较长。备份完成后，服务会自动重启。

Procedure: Creating full backup with **mgradm backup create**

1. 在容器主机上，以 root 身份执行以下命令创建备份：

```
mgradm backup create $path
```

Replace **$path** by the path to the backup location.

5.1.2. Uyuni 的部分备份

mgradm backup create tool allows creating partial backups. It is possible to skip individual or all volumes, skip database backup and images.

特别是，当跳过数据库备份时，系统会创建备份而不停止 Uyuni 服务，备份可以作为两阶段备份流程中的一个阶段运行。

> 部分备份仅针对部分数据执行，并未考量这些数据与其他未备份部分之间可能存在的依赖关系。因此，部分备份无法保障备份与恢复操作的一致性。

过程：创建部分备份（跳过数据库备份）

1. 在容器主机上，以 root 身份执行以下命令创建备份：

```
mgradm backup create --skipdatabase $path
```

Replace **$path** by the path to the backup location.

过程：创建部分备份（跳过卷）

1. 在容器主机上，以 root 身份执行以下命令创建备份：

```
mgradm backup create --skipvolumes $volumes $path
```

Replace **$path** by the path to the backup location.

Replace **$volumes** by the name of the volume name to be excluded from the backup, or by a comma separated list of volumes to be excluded.

Use **all** to skip all volumes, except database volumes.

5.1.3. 备份额外的卷

mgradm backup command uses internal list of Uyuni volumes. If additional volumes were configured during the installation, or additional volumes should be added to the backup, they need to be specified using **--extravolumes $volumes**.

过程：创建包含其他自定义卷的备份

1. 在容器主机上，以 root 身份执行以下命令创建备份：

```
mgradm backup create --extravolumes $volume $path
```

Replace **$path** by the path to the backup location.

Replace **$volumes** by the name of the volume name to be included in the backup. or by a comma separated list of volumes to be included.

5.1.4. 执行手动数据库备份

过程：执行手动数据库备份

1. 请为备份分配永久存储空间。
2. 在 Uyuni 容器主机的命令提示符处，以 root 身份执行以下命令：

```
mgradm backup create --skipvolumes all --skipconfig --skipimages $path
```

5.2. 从现有备份恢复 Uyuni

从现有备份恢复 Uyuni 时，系统将枚举卷、映像和配置的备份以进行恢复。与创建备份的场景不同，恢复操作不使用内部卷列表，而是自动检测备份中存在的所有卷或映像。

收集待恢复项目列表后，会执行存在性和完整性检查。存在性检查用于确保恢复备份时不会意外覆盖现有卷、映像或配置。完整性检查通过计算备份项目的校验和完成。

两项检查均通过后，才会执行实际的备份恢复操作。

恢复操作完成后，Uyuni 服务不会自动启动。

过程：从现有备份恢复

1. 在容器主机上以 root 身份运行以下命令，重新部署 Uyuni 服务器：

```
mgradm stop
mgradm backup restore $path
mgradm start
```

Replace **$path** by the path to the backup location.

Verification of the backup can be a time-consuming operation. If backup integrity is ensured by other means, verification can be skipped by using **--skipverify** option.

If for some reason it is needed to skip restoring a volume present in the backup, **--skipvolumes $volumes** option can be used.

5.2.1. 恢复备份后建议执行的步骤

过程：恢复 Uyuni 后建议执行的步骤

1. Re-synchronize your Uyuni repositories using either the Uyuni Web UI, or with the **mgr-sync** tool at the command prompt in the container. You can choose to re-register your product, or skip the registration and SSL certificate generation sections.
2. On the container host, check whether you need to restore **/var/lib/containers/storage/volumes/var-spacewalk/_data/packages/**. If **/var/lib/containers/storage/volumes/var-spacewalk/_data/packages/** was not in your backup, you need to restore it. If the source repository is available, you can restore `/var/lib/containers/storage/volumes/var-spacewalk/_data/packages/`u with a complete channel synchronization:

```
mgrctl exec -ti -- mgr-sync refresh --refresh-channels
```

3. Schedule the re-creation of search indexes next time the **rhn-search** service is started.

   此命令仅生成调试消息，而不生成错误消息。

   在容器主机上执行以下命令：

```
mgrctl exec -ti -- rhn-search cleanindex
```

5.3. Database Backup Management

This guide describes how to manage online database backups for the Uyuni server using **mgradm**. The backup system is based on PostgreSQL continuous archiving Write-Ahead Logging (WAL).

- To ensure the system can be fully recovered, perform regular backups of both the database and other system volumes.
- Regularly copy your backup volumes and WAL files to an external storage device or an off-site location for disaster recovery.

5.3.1. 概述

Continuous archiving reduces the risk of data loss by constantly backing up the transaction logs (WAL) to a dedicated volume.

The backup system utilizes a dedicated Podman volume named **var-pgsql-walbackup** to store:

- A base backup (a full snapshot of the database).
- WAL segments (incremental changes since the last base backup).

> Point-in-time recovery is currently not supported by **mgradm** and requires a manual workflow.
>
> Only full recovery is supported by **mgradm** tool. The restore procedure will restore everything from the backup location.

5.3.2. 先决条件

- **mgradm** tool installed on the host.
- Uyuni server running in a Podman environment.

5.3.3. Enabling backups

To enable continuous archiving, run the following command:

```
mgradm backup db enable
```

This command:

- Verifies that the backup is not already enabled.
- Configures PostgreSQL (**postgresql.conf**) to enable **archive_mode** and sets the **archive_command**.
- Configures the **var-pgsql-walbackup** volume for the database service.
- Restarts the database service to apply changes.
- Performs an initial base backup inside the container.

If the backup is already configured but you need to re-initialize it, use the **--force** flag.

5.3.4. Checking backup status

You can verify the current state of the database backup system at any time:

```
mgradm backup db status
```

The command will report one of the following statuses:

- **enabled**: Backup is correctly configured and active.
- **disabled**: Continuous archiving is explicitly turned off.
- **misconfigured**: There is a discrepancy between the configuration and the runtime state (for example, missing volume, incorrect archive command).

5.3.5. Maintaining backups (rebasing)

Over time, the number of WAL segments can grow significantly, consuming disk space and increasing restoration time. It is recommended to periodically "rebase" the backup by creating a new base backup and starting a new WAL chain.

```
mgradm backup db rebase
```

This command creates a fresh full snapshot of the database without requiring a service restart.

> Regularly rebase your database backups using command **mgradm backup db rebase** to minimize the time required for restore operations.

5.3.6. Restoring from backup

Restoring from a backup is a **destructive operation** that replaces the current database data with the content of the backup.

```
mgradm backup db restore
```

The restoration process:

- Stops the database service.
- Clears the current database data directory.
- Extracts the base backup from the backup volume.
- Configures PostgreSQL for recovery.
- Starts the database service.
- Replays the WAL segments to bring the database to the latest possible state.

The command will wait until the database has fully recovered and is ready to accept connections.

> Ensure you have a recent base backup and all necessary WAL segments in the **var-pgsql-walbackup** volume before performing a restore.

5.3.7. Disabling backups

To turn off continuous archiving:

```
mgradm backup db disable
```

By default, this command disables archiving in the configuration and restarts the database but preserves the backup volume.

To also remove the backup volume and all stored backups, use:

```
mgradm backup db disable --purge-volume
```

Regularly test your restore procedures to verify data integrity and ensure you are prepared for a recovery.

5.3.8. External documentation

For more information on PostgreSQL, see PostgreSQL 16: Continuous Archiving and Point-in-Time Recovery (PITR).

章 6. 通道管理

通道是将软件包分组的一种方法。

在 Uyuni 中，通道分为基础通道和子通道，基础通道按操作系统类型、版本和体系结构分组，子通道与其相关的基础通道兼容。将客户端指派到基础通道后，该系统只能安装相关的子通道。以这种方式组织通道可确保在每个系统上仅安装兼容的软件包。

软件通道使用储存库来提供软件包。通道镜像 Uyuni 中的储存库，软件包名称和其他数据存储在 Uyuni 数据库中。您可以使用与某个通道关联的任意数量的储存库。然后可以通过让客户端订阅相应的通道，将这些储存库中的软件安装在客户端上。

客户端只能指派到一个基础通道。然后，客户端可以安装或更新与该基础通道及其任何子通道关联的储存库中的软件包。

Uyuni 提供了许多供应商通道，这些通道为您提供了运行 Uyuni 所需的一切。Uyuni 管理员和通道管理员拥有通道管理权限，他们可以创建和管理自己的自定义通道。如果您想在环境中使用自己的软件包，可以创建自定义通道。自定义通道可以用作基础通道，或者您也可以将它们与供应商基础通道相关联。

有关创建自定义通道的详细信息，请参见 **Administration Guide › Custom Channels**。

6.1. 通道管理

默认情况下，任何用户都可以在系统中订阅通道。您可以使用 Web UI 对通道实施限制。

过程：限制订阅者对通道的访问权限

1. 在 Uyuni Web UI 中，导航到 **软件 › 通道列表**，并选择要编辑的通道。
2. Locate the **Per-User Subscription Restrictions** section and check **Only selected users within your organization may subscribe to this channel**. Click **[Update]** to save the changes.
3. Navigate to the **Subscribers** tab and select or deselect users as required.

6.2. 删除通道

可以从命令提示符删除供应商软件通道。

过程：删除供应商通道

1. 在 Uyuni Server 上的在命令提示符下，以 root 身份列出可用的供应商通道，并记下您要删除的通道：

```
mgr-sync list channels
```

2. 删除通道：

```
spacewalk-remove-channel -c <通道名称>
```

- 有关删除自定义通道的信息，请参见 **Administration Guide › Custom Channels**。

6.3. 自定义通道

自定义通道使您能够创建自己的软件包和储存库，然后可以使用这些软件包和储存库来更新您的客户端。它们还使您能够在环境中使用第三方供应商提供的软件。

本节提供有关如何创建、管理和删除自定义通道的更多细节。您必须拥有管理员特权才能创建和管理自定义通道。

6.3.1. 创建自定义通道和储存库

在创建自定义通道之前，请确定要将其关联到哪个基础通道，以及要为内容使用哪些储存库。

如果您需要将自定义软件包安装在客户端系统上，可以创建自定义子通道来管理它们。需要在 Uyuni Web UI 中创建通道，并为软件包创建储存库，然后将该通道指派到系统。

> 不要创建其中有软件包与客户端系统不兼容的子通道。

You can select a vendor channel as the base channel if you want to use packages provided by a vendor. Alternatively, select **none** to make your custom channel a base channel.

过程：创建自定义通道

1. 在 Uyuni Web UI 中，导航到**软件 › 管理 › 通道**，然后单击 **[创建通道]**。
2. On the **Create Software Channel** page, give your channel a name (for example, **My Tools SLES 15 SP1 x86_64**) and a label (for example, **my-tools-sles15sp1-x86_64**). 标签不能包含空格或大写字母。
3. In the **Parent Channel** drop down, choose the relevant base channel (for example, **SLE-Product-SLES15-SP1-Pool for x86_64**). 确保为软件包选择兼容的父通道。
4. In the **Architecture** drop down, choose the appropriate hardware architecture (for example, **x86_64**).
5. 根据环境的需要，在联系方法细节、通道访问控制和 GPG 字段中提供任何附加信息。
6. 单击 **[创建通道]**。

Custom channels sometimes require additional security settings. Many third party vendors secure packages with GPG. If you want to use GPG-protected packages in your custom channel, you need to trust the GPG key which has been used to sign the metadata. You can then check the **Has Signed Metadata?** check box to match the package metadata against the trusted GPG keys.

If remote channels and repositories are signed with GPG keys, you can import and trust these GPG keys. For example, execute the **spacewalk-repo-sync** from the command line on the Uyuni Server:

```
/usr/bin/spacewalk-repo-sync -c <通道标签名称> -t yum
```

此命令和过程仅可用于对 GPG 密钥进行临时同步。如需长期存储密钥，请参见后文相关章节。

The underlying **zypper** call will import the key, if it is available. The Web UI does not offer this feature.

但这仅在您要镜像的储存库满足以下条件时才生效：需要以特殊方式配置，且在储存库中与签名一同提供“密钥”。开放式构建服务（OBS）生成的所有储存库均符合此要求。对于其他储存库，则需执行特殊的准备步骤，具体说明如下。

By default, the **Enable GPG Check** field is checked when you create a new channel. If you would like to add custom packages and applications to your channel, make sure you uncheck this field to be able to install unsigned packages. Disabling the GPG check is a security risk if packages are from an untrusted source.

You can only add a repository to the Uyuni with the Web UI if it is a valid software repository. Check in advance that needed repository metadata are available. Tools such as **createrepo** and **reprepro** are useful in this regard. **mgrpush** can help with pushing a single RPM into a channel without creating a repository.

- For more information on **createrepo_c** see https://manpages.opensuse.org/Leap-15.6/createrepo_c/
- For more information on **reprepro** see https://manpages.opensuse.org/Leap-15.6/reprepro/

过程：添加软件储存库

1. 在 Uyuni Web UI 中，导航到**软件 › 管理 › 储存库**，然后单击 **[创建储存库]**。
2. On the **Create Repository** page, give your repository a label (for example, **my-tools-sles15sp1-x86_64-repo**).
3. In the **Repository URL** field, provide the path to the directory that contains the **repodata** file (for example, file:///opt/mytools/). You can use any valid addressing protocol in this field.
4. Uncheck the **Has Signed Metadata?** check box.
5. 可选：如果您的储存库需要客户端证书身份验证，请填写 SSL 字段。
6. 单击 **[创建储存库]**。

仅当您要镜像的储存库中在签名旁边提供了“key”时，才能正常完成上述过程。OBS 生成的所有储存库都是这种情况，但不是由 OBS 提供的操作系统储存库并非如此。

If the repository you want to use does not have a GPG key you can provide one yourself and import the GPG key into the keyring manually. If you import the key into the **/var/lib/spacewalk/gpgdir** keyring using the **gpg** command line tool it would be stored permanently. The key would also persist if the chroot environment would be cleaned.

过程：使用 GPG 密钥创建软件储存库

1. 在容器主机上，用于将密钥导入密钥环的命令是：

   ```
   mgradm gpg add /path/to/gpg.key
   ```

2. 有关详细信息，请参见 **Client Configuration Guide › Autoinst Owngpgkey**。

> Add Debian non-flat repositories using **uyuni_suite**, **uyuni_component**, and **uyuni_arch** query parameters.

uyuni_suite

is mandatory. In Debian documentation, this is also known as **distribution**. With this parameter you specify the apt source. Without this parameter the original approach is used. If the parameter ends with **/**, the repository is identified as flat.

uyuni_component

是可选的。此参数只能指定一个组件。无法列出组件。一个 apt 源项允许指定多个组件，但对于 Uyuni 无法做到这一点。对此，您必须为每个组件创建单独的储存库。

uyuni_arch

is optional. If omitted the architecture name is calculated with a SQL query for the channel from the database. Specify **uyuni_arch** explicitly if it does not match the architecture of the channel (sometimes architecture naming is ambiguous).

下面提供了一些示例：

表格 2. Debian 非平面储存库映射

Type	Source line / URL
apt source line	**deb** https://pkg.jenkins.io/debian-stable **binary/**
URL mapping	https://pkg.jenkins.io/debian-stable?uyuni_suite=binary/
apt source line	**deb** https://deb.debian.org/debian/ **stable main**

Type	Source line / URL
URL mapping	https://deb.debian.org/debian/?uyuni_suite=stable&uyuni_component=main
apt source line	**deb** http://archive.ubuntu.com/ubuntu/ **noble main**
URL mapping	http://archive.ubuntu.com/ubuntu/?uyuni_suite=noble&uyuni_component=main

以下有关 Debian 储存库定义格式的信息基于 https://wiki.debian.org/DebianRepository/Format#Overview。

储存库定义格式如下：

```
deb URI 套件 [组件 1] [组件 2] [...]
```

例如：

```
deb https://deb.debian.org/debian/ stable main
```

或

```
deb https://pkg.jenkins.io/debian-stable binary/
```

For each pair of **suite** and **component** the specification defines a distinct URL calculated on the base URL **+ suite + component**.

过程：将储存库分配给通道

1. 通过导航到**软件 › 管理 › 通道**，并单击新建的自定义通道的名称，将新储存库指派到自定义通道。
2. Navigate to the **Repositories** tab, and ensure the repository you want to assign to the channel is checked. Click **[Save Repositories]**.
3. 默认情况下，同步过程会立即开始。

有关通道同步的详细信息，请参见 **Administration Guide › Custom Channels › Custom Channel Sync**。

过程：向激活密钥添加自定义通道

1. 在 Uyuni Web UI 中，导航到**系统 › 激活密钥**，然后选择要将自定义通道添加到的密钥。
2. On the **Details** tab, in the **Child Channels** listing, select the channel to associate. 如果需要，您可以选择多个通道。
3. 单击 **[更新激活密钥]**。

6.3.2. 自定义通道同步

为了避免错过重要更新，SUSE 建议通过远程储存库更改来使您的自定义通道保持最新。

默认情况下，您创建的所有自定义通道将自动同步。具体而言，在以下情况下将发生同步：

- after adding a repository to a channel from the UI or by using **spacewalk-common-channels**
- as part of the daily task **mgr-sync-refresh-default**, which will synchronize all your custom and vendor channels.

To disable this default behaviour, set in **/etc/rhn/rhn.conf**:

```
java.unify_custom_channel_management = 0
```

关闭此属性后，将不会自动执行同步，要使自定义通道保持最新，您需要：

- synchronize it manually by navigating to the **Sync** tab and click **[Sync Now]**,
- set up an automated synchronization schedule from the **Repositories** tab.

该过程开始后，可通过多种方式检查通道是否已完成同步：

- In the Uyuni Web UI, navigate to **Admin › Setup Wizard** and select the **Products** tab. 产品同步时，此对话框将为每个产品显示一个完成栏。
- 在 Uyuni Web UI 中，导航到**软件 › 管理 › 通道**，然后单击与储存库关联的通道。 Navigate to the menu:[Repositories > Sync] tab. The **Sync Status** is shown next to the repository name.
- 在命令提示符下检查同步日志文件：

  ```
  tail -f /var/log/rhn/reposync/<channel-label>.log
  ```

  每个子通道在同步过程中会生成自身的日志。您需要检查所有基础通道和子通道日志文件，以确保同步完成。

可使用以下自定义通道同步选项：

保留已从储存库中去除的通道中的软件包

This turns off **strict** mode.

不同步勘误

不要同步补丁。

仅同步最新软件包

仅同步最新的软件包版本。

创建可无人值守安装树

此选项准备一个随时可用于自动安装 Kickstart 的目录树。

出现任何错误时终止

如果发生错误，则停止同步。

将为每个通道永久保存这些选项。在执行同步之前，单击[**立即同步**]按钮也会保存通道选项。

6.3.3. 将软件包和补丁添加到自定义通道

如果您创建新的自定义通道但未从现有通道克隆，则创建的通道不包含任何软件包或补丁。可以使用 Uyuni Web UI 添加所需的软件包和补丁。

自定义通道只能包含克隆的或者自定义的软件包或补丁，并且这些软件包或补丁必须与通道的基础体系结构相匹配。添加到自定义通道的补丁必须适用于通道中的软件包。

过程：将软件包添加到自定义通道

1. In the Uyuni Web UI, navigate to **Software › Manage › Channels**, and go to the **Packages** tab.
2. OPTIONAL: See all packages currently in the channel by navigating to the **List/Remove** tab.
3. Add new packages to the channel by navigating to the **Add** tab.
4. 选择要为其提供软件包的父通道，然后单击 [**查看软件包**] 以填充列表。
5. 检查要添加到自定义通道的软件包，然后单击 [**添加软件包**]。
6. 如果您觉得选择合适，请单击 [**确认添加**] 以将软件包添加到通道。
7. 可选：可以通过导航到**软件 › 管理 › 通道**，然后转到**软件包 › 比较**选项卡，将当前通道中的软件包与其他通道中的软件包进行比较。 要使两个通道相同，请单击 [**合并差异**] 按钮，然后解决任何冲突。

过程：将补丁添加到自定义通道

1. In the Uyuni Web UI, navigate to **Software › Manage › Channels**, and go to the **Patches** tab.
2. OPTIONAL: See all patches currently in the channel by navigating to the **List/Remove** tab.
3. Add new patches to the channel by navigating to the **Add** tab, and selecting what kind of patches you want to add.
4. 选择要为其提供补丁的父通道，然后单击 **[浏览关联补丁]** 以填充列表。
5. 检查要添加到自定义通道的补丁，然后单击 **[确认]**。
6. 如果您觉得选择合适，请单击 **[确认]** 以将补丁添加到通道。

6.3.4. 管理自定义通道

Uyuni 管理员和通道管理员可以更改或删除任何通道。

To grant other users rights to alter or delete a channel, navigate to **Software › Manage › Channels** and select the channel you want to edit. Navigate to the **Managers** tab, and check the user to grant permissions. Click **[Update]** to save the changes.

> 如果您删除某个已指派到一组客户端的通道，与已删除通道关联的任何客户端的通道状态将立即更新。这是为了确保在储存库文件中准确反映所做的更改。

无法使用 Web UI 删除 Uyuni 通道。只能删除自定义通道。

过程：删除自定义通道

1. 在 Uyuni Web UI 中，导航到**软件 › 管理 › 通道**，然后选择要删除的通道。
2. 单击 **[删除软件通道]**。
3. On the **Delete Channel** page, check the details of the channel you are deleting, and check the **Unsubscribe Systems** checkbox to remove the custom channel from any systems that might still be subscribed.
4. 单击 **[删除通道]**。

删除通道时，不会自动去除其中的软件包。您无法更新其通道已被删除的软件包。

可以在 Uyuni Web UI 中删除与通道没有关联的软件包。导航到**软件 › 管理 › 软件包**，选中要去除的软件包，然后单击 **[删除软件包]**。

6.4. 第三方通道

Uyuni 支持将第三方可选储存库的内容与部分可被其管理的产品进行同步。

部分第三方 GPG 密钥已默认包含在 Uyuni 数据库中，其余密钥则未包含。对于未包含的第三方密钥，在选择同步对应的第三方储存库时，您需要手动导入相关密钥。

如需导入 GPG 密钥，请在 Uyuni 主机服务器上执行以下命令：

```
mgradm gpg add <path-to-gpg-key-file-or-URL>
```

1. Example: Adding a GPG key from a file

   ```
   mgradm gpg add repomd.xml.key
   ```

2. Example: Adding a GPG key from a remote repository using a URL

   ```
   mgradm gpg add https://<3rd-party-domain>/path/to/repository/repodata/repomd.xml.key
   ```

如需列出当前存储在 Uyuni GPG 数据库中的密钥，请执行以下命令：

```
mgrctl exec -ti -- gpg --homedir /var/lib/spacewalk/gpgdir --list-keys
```

> 在客户端上安装软件包时，您可能还需要在客户端本地信任该 GPG 密钥，这要求对应的 GPG 密钥已存在于客户端系统中。
>
> 有关详细信息，请参见 **Client Configuration Guide › Gpg Keys**。

6.5. Configure Automatic Channels Synchronization

By default, channels are automatically synchronized with the upstream repositories to ensure that clients have access to the latest content and updates. However, in some cases, you may want to disable this automatic synchronization for specific channels.

This chapter explains how to disable automatic synchronization for a channel.

6.5.1. Verify channel synchronization status

To check if a channel is set to be automatically synchronized user can call an API.

To check the synchronization status of a channel using the API, use the following command. Replace **<Channel_Label>** with the actual label of the channel you want to check

```
spacecmd api -- --args "<Channel_Label>" channel.software.isAutoSync
INFO: Connected to http://localhost/rpc/api as admin
[
  true
```

```
]
```

The command will return **true** if automatic synchronization is enabled, or **false** if it is disabled.

This information is also available in the Web UI. To check the synchronization status, navigate to the channel details page and look for the field **Automatic Repository Synchronization**.

6.5.2. Set a channel to skip automatic synchronization

To enable or disable automatic synchronization for a channel, you can use the API call **channel.software.setAutoSync**.

Here is an example of how to set a channel to skip automatic synchronization using the API (replace **<Channel_Label>** with the actual label of the channel you want to update):

```
spacecmd api -- --args '["<Channel_Label>", false]' channel.software.setAutoSync
INFO: Connected to http://localhost/rpc/api as admin
[
  false
]
```

To re-enable automatic synchronization, set the second parameter to **true**:

```
spacecmd api -- --args '["<Channel_Label>", true]' channel.software.setAutoSync
INFO: Connected to http://localhost/rpc/api as admin
[
  true
]
```

6.5.3. Manually synchronize a channel

Even if you have disabled automatic synchronization for a channel, you can still manually synchronize it whenever needed.

Use **mgrctl term** before running steps inside the server container.

To manually synchronize a channel, you can use the command(replace **<Channel_Label>** with the actual label of the channel you want to synchronize):

```
spacewalk-repo-sync -c <Channel_Label>
```

章 7. 机密计算

借助Confidential Computing技术，可以使用基于硬件的可信执行环境（TEE）来保护使用中的数据，这种环境提供更高的安全级别来确保数据完整性、数据机密性和代码完整性。

7.1. Uyuni 的Confidential Computing

TEE 的可信度已通过认证流程进行检查。Uyuni 可用作其中已注册系统的认证服务器。它会针对此模式下运行的系统生成报告页面。需要定期对这些系统进行认证和检查。还可以存储并按请求提供以往的检查历史记录。

Confidential Computing认证取决于所用的硬件以及要认证的系统运行所在的环境。

Confidential Computing Attestation is only available when Uyuni is running on x86-64 architecture.

7.2. 要求

可以在具有以下特征的环境中设置Confidential Computing：

- Attested system (virtual machine) is bootstrapped to Uyuni and runs on one of the following systems:
  - SUSE Linux Enterprise 15 SP6
  - SUSE Linux Enterprise 15 SP7
  - SUSE Linux Enterprise 16
  - openSUSE Leap 15 SP6
  - openSUSE Leap 15 SP7
  - openSUSE Leap 16
- Hardware must have one of the following CPUs:
  - **AMD EPYC Milan**
  - **AMD EPYC Genoa**
  - **AMD EPYC Bergamo**
  - **AMD EPYC Siena**
  - **AMD EPYC Turin**
  - **IBM z16**
  - **IBM z17**
- 必须将 BIOS 配置为允许Confidential Computing认证
- Host OS and the virtualization software (KVM and libvirt) must support Confidential Computing
- For information about required URLs when performing confidential computing attestation for IBM, see see **Installation and Upgrade Guide › Required URLs**.

7.3. 限制

Secure boot is attested. However, currently KVM secure boot and AMD attestation tool **snpguest** are not working together.

7.4. 使用 Uyuni 中的Confidential Computing

有关在主机上设置和配置Confidential Computing的具体步骤，请参见操作系统供应商文档。

过程：在安装 Uyuni 期间启用认证容器

1. The attestation container is enabled during the installation of Uyuni with **mgradm install**.
2. Add the following to file **mgradm.yaml**.

```
coco:
    replicas: 1
```

过程：在安装 Uyuni 之后启用认证容器

1. To enable the attestation container after the installation, use the command line parameter **mgradm**.
2. 运行命令

```
mgradm scale uyuni-server-attestation --replicas 1
```

过程：在安装 Uyuni 后禁用认证容器

1. 要禁用已启用的认证容器，请运行以下命令：

```
mgradm scale uyuni-server-attestation --replicas 0
```

过程：启用认证

1. 对于选定的系统，转到选项卡**审计 › 机密计算 › 设置**。
2. 通过选择切换按钮来启用认证。
3. In the field **Environment Type** select the correct option from the drop-down list.
4. If the field **Environment Type** is set to one of the **IBM z-series** CPU values, two more settings are required, **Host Key Document** and **Secure Execution Header**.

Host Key Document is a public key certificate of the IBM hypervisor server instance where the attested system is going to run.

Secure Execution Header is a binary file needed to build the IBM secure VM image of the attested system.

Both files should be requested by the administrator creating the attested system VM running on the IBM hypervisor.

Host Key Document and **Secure Execution Header** are not security sensitive, therefore it is safe to store them on the Uyuni server.

5. 单击 [**保存**] 按钮保存更改。

过程：安排新认证

1. 对于选定的系统，转到选项卡**审计 › 机密计算 › 列出认证**。
2. 单击 [**安排认证**]。此时会打开新窗体。
3. In the field **Earliest** select the time of running the attestation.
4. If needed, add the newly created attestation to the action chain by selecting **Add to** option.
5. 单击 [**日程安排**] 保存新认证并安排其执行。

过程：从“系统细节”界面查看认证报告

1. 对于选定的系统，转到选项卡**审计 › 机密计算 › 列出认证**。
2. 查找并选择要查看的报告。
3. After clicking the selected attestation report tab **Overview** will open.
   a. If the selected CPU is **AMD** you will be shown additional tabs **SEV-SNP** and **Secure boot**.
   b. If the selected CPU is **IBM** you will be shown additional tab **IBM SELPVATTEST**.

过程：从“审计”界面查看认证报告

1. 从导航栏中选择**审计 › 机密计算**。
2. 所有认证的列表将显示在主面板中。
3. 查找并选择要查看的报告。

7.4.1. 报告状态

认证报告处于下列状态之一：

待处理

这是已安排的认证的默认状态。报告仍不可用，原因是认证过程尚未开始或完成。

成功

When the scheduled attestation creates a report which can be viewed, the status of the process is **Successful**.

失败

When the scheduled fails and does not create a report as a result, the status of the process is **Failed**.

7.5. 相关主题

有关 Confidential Computing 的详细信息，请参见此处。

章 8. 内容生命周期管理

内容生命周期管理允许您在更新生产客户端之前自定义和测试软件包。如果您需要在有限的维护时段内应用更新，此功能将特别有用。

内容生命周期管理允许您选择软件通道作为源、根据环境的需要调整软件通道，并在安装到生产客户端之前对其进行全面的测试。

虽然您无法直接修改供应商通道，但可以克隆它们，然后通过添加或去除软件包和自定义补丁来修改克隆版本。可以将这些克隆的通道指派到测试客户端，以确保它们按预期工作。

By default, cloned vendor channels match the original vendor channel and automatically select the dependencies. You can disable the automatic selection for cloned channels by adding the following option in **/etc/rhn/rhn.conf**:

```
java.cloned_channel_auto_selection = false
```

然后，当所有测试都通过后，您可以将克隆的通道升级为生产服务器。

此功能是通过一系列环境实现的，软件通道在其生命周期内可以在这些环境之间转移。大多数环境生命周期至少包括测试和生产环境，但您可以创建任意所需数量的环境。

本章介绍基本的内容生命周期过程以及可用的过滤器。有关更具体的示例，请参见 **Administration Guide › Content Lifecycle Examples**。

8.1. 创建内容生命周期项目

要设置内容生命周期，需要从一个项目开始。该项目定义软件通道源、用于查找软件包的过滤器，以及构建环境。

过程：创建内容生命周期项目

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 项目**并单击 **[创建项目]**。
2. In the **Label** field, enter a label for your project. The **Label** field only accepts lowercase letters, numbers, periods, hyphens, and underscores.
3. In the **Name** field, enter a descriptive name for your project.
4. 单击 **[创建]** 按钮以创建项目并返回项目页面。
5. 单击 **[关联源/解除源关联]**。
6. In the **Sources** dialog, select the source type, and select a base channel for your project. 此时会显示所选基础通道的可用子通道，包括有关通道是

必需通道还是建议通道的信息。

7. 选中所需的子通道，然后单击 [**保存**] 返回项目页面。 您选择的软件通道现在应会显示出来。
8. 单击 [**关联过滤器/解除过滤器关联**]。
9. In the **Filters** dialog, select the filters you want to attach to the project. 要创建新过滤器，请单击 [**创建新过滤器**]。
10. 单击 [**添加环境**]。
11. In the **Environment Lifecycle** dialog, give the first environment a name, a label, and a description, and click [**Save**]. The **Label** field only accepts lowercase letters, numbers, periods, hyphens, and underscores.
12. 继续创建环境，直到为整个生命周期创建了所有环境。 You can select the order of the environments in the lifecycle by selecting an environment in the **Insert before** field when you create it.

8.2. 过滤器类型

Uyuni 允许您创建各种类型的过滤器来控制用于构建项目的内容。使用过滤器可以选择要在构建中包含或排除哪些软件包。例如，可以排除所有内核软件包，或仅包含某些软件包的特定发行版。

支持的过滤器为：

- 软件包过滤
  - 按名称
  - 按名称、纪元、版本、发行版和体系结构
  - 按提供的名称
- 补丁过滤
  - 按建议名称
  - 按建议类型
  - 按摘要
  - 按关键字
  - 按日期
  - 按受影响的软件包
- 模块
  - 按流

在内容过滤期间不会解析软件包依赖项。

可以将多个匹配器与过滤器配合使用。哪些匹配器可用取决于所选的过滤器类型。

可用的匹配器为：

- 包含
- 匹配（必须采用正则表达式形式）
- 等于
- 大于
- 大于或等于
- 低于或等于
- 低于
- 高于或等于

8.2.1. Filter `rule` parameter

Each filter has a **rule** parameter that can be set to either **Allow** or **Deny**. The filters are processed like this:

- If a package or patch satisfies a **Deny** filter, it is excluded from the result.
- If a package or patch satisfies an **Allow** filter, it is included in the result (even if it was excluded by a **Deny** filter). **Allow** filters for packages only operate on package filters and **Allow** filters for patches only operate on patch filters. This means with a package filter you cannot add packages back in that were filtered out through patch filters or vice versa for patches that where filtered out through package filters.

This behavior is useful when you want to exclude large number of packages or patches using a general **Deny** filter and "cherry-pick" specific packages or patches with specific **Allow** filters.

内容过滤器在您的组织中是全局性的，并可以在项目之间共享。

如果您的项目已包含构建的源，当您添加环境时，该环境中会自动填充现有内容。内容是从周期中的前一个环境（如果有）提取的。如果前一个环境不存在，则将内容留空，直到再次构建了项目源。

8.3. 过滤器模板

为了帮助为某些常见方案创建过滤器，Uyuni 提供了过滤器模板。应用这些模板有助于提前创建一组针对特定用例定制的过滤器。

本节介绍可用的模板及其用法。

8.3.1. 根据 SUSE 产品进行实时修补

在包含实时修补的项目中，必须排除未来的常规内核软件包，以便仅将在线补丁软件包作为更新提供给客户端。另一方面，仍然必须包含已安装的常规内核软件包以保持系统完整性。

应用此模板会创建三个所需的过滤器来实现以下行为：

- Allow patches that contain **kernel-default** package equal to a base kernel version
- Deny patches that contain **reboot_suggested** keyword
- Deny patches that contain a package which provides the name **installhint(reboot-needed)**

有关如何设置实时修补项目的详细信息，请参见 **Administration Guide › Content Lifecycle Examples › Exclude Higher Kernel Version**。

过程：应用模板

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 **[创建过滤器]**。
2. 在对话框中单击 **[使用模板]**。输入将相应地更改。
3. In the **Prefix** field, type a name prefix. This value will be prepended to the name of every individual filter created by the template. If the template is being applied in the context of a project, this field will be prefilled with the project label.
4. In the **Template** field, select **Live patching based on a SUSE product**.
5. In the **Product** field, select the product you wish to set up live patching for.
6. In the **Kernel** field, select a kernel version from the list of versions available in the selected product. The filter to deny the later regular kernel patches will be based on this version.
7. 单击 **[保存]** 以创建过滤器。
8. 导航到**内容生命周期 › 项目**并选择您的项目。
9. 单击 **[关联过滤器/解除过滤器关联]**。
10. 选择具有指定前缀的三个过滤器，然后单击 **[保存]**。

8.3.2. 根据系统实时修补

如果您想要根据特定已注册系统中安装的内核版本设置实时修补项目，可以使用**根据系统实时修补**模板。

应用此模板会创建三个所需的过滤器来实现以下行为：

- Allow patches that contain **kernel-default** package equal to a base kernel version
- Deny patches that contain **reboot_suggested** keyword
- Deny patches that contain a package which provides the name **installhint(reboot-needed)**

有关如何设置实时修补项目的详细信息，请参见 **Administration Guide › Content Lifecycle Examples › Exclude Higher Kernel Version**。

过程：应用模板

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 **[创建过滤器]**。
2. 在对话框中单击 **[使用模板]**。输入将相应地更改。
3. In the **Prefix** field, type a name prefix. This value will be prepended to the name of every filter created by the template. If the template is being applied in the context of a project, this field will be prefilled with the project label.
4. In the **Template** field, select **Live patching based on a specific system**.
5. In the **System** field, select a system from the list, or start typing a system name to narrow down the options.
6. In the **Kernel** field, select a kernel version from the list of versions installed in the selected system. The filter to deny the later regular kernel patches will be based on this version.
7. 单击 **[保存]** 以创建过滤器。
8. 导航到**内容生命周期 › 项目**并选择您的项目。
9. 单击 **[关联过滤器/解除过滤器关联]**。
10. 选择具有指定前缀的三个过滤器，然后单击 **[保存]**。

8.3.3. 含默认值的 AppStream 模块

如果您想要在项目中包含模块化储存库中提供的所有可用模块，可以使用此过滤器模板自动添加这些模块。

应用后，此模板将为每个模块及其默认流创建一个 AppStream 过滤器。

如果从项目的页面完成此过程，则过滤器将自动添加到项目。否则，创建的过滤器可能会列在**内容生命周期 › 过滤器**中，并根据需要添加到任何项目。

可以编辑每个过滤器以选择不同的模块流，或去除所有这些过滤器以从目标储存库中排除该模块。

由于并非所有模块流都相互兼容，因此更改单个流可能会导致无法成功解析模块化依赖关

系。如果发生这种情况，项目细节页面中的过滤器窗格会显示描述问题的错误，并且在选择的所有模块都兼容之前，构建按钮将会禁用。

自 Red Hat Enterprise Linux 9 起，模块便不再具有任何定义的默认流。因此，对 Red Hat Enterprise Linux 9 源使用此模板将不会有任何效果。

有关如何使用内容生命周期管理设置 AppStream 储存库的详细信息，请参见 **Administration Guide › Content Lifecycle Examples › Appstream Filters**。

过程：应用模板

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 项目**，然后选择您的项目。
2. In the **Filters** section, click **[Attach/Detach Filters]**, and then click **[Create New Filter]**.
3. 在对话框中单击 **[使用模板]**。输入将相应地更改。
4. In the **Prefix** field, type a name prefix. This value will be prepended to the name of every filter created by the template. If the template is being applied in the context of a project, this field will be prefilled with the project label.
5. In the **Template** field, select **AppStream modules with defaults**.
6. In the **Channel** field, select a modular channel to get the modules from. In this dropdown, only the modular channels are displayed.
7. 单击 **[保存]** 以创建过滤器。
8. Scroll to the **Filters** section to see the newly attached AppStream filters.
9. 可以编辑/去除任何一个过滤器以根据您的需要定制项目。

8.4. 构建内容生命周期项目

创建项目、定义环境、关联源和过滤器后，可以首次构建项目。

构建过程会将过滤器应用于关联的源，并将过滤器克隆到项目中的第一个环境。

You can use the same vendor channels as sources for multiple content projects. In this case, Uyuni does not create new patch clones for each cloned channel. Instead, a single patch clone is shared between all of your cloned channels. This can cause problems if a vendor modifies a patch; for example, if the patch is retracted, or the packages within the patch are changed. When you build one of the content projects, all the channels that share the cloned patch are synchronized with the original by default, even if the channels are in other environments of your content project, or other content project channels in your organization. You can change this behavior by turning off automatic patch synchronization in your organization settings. To manually synchronize the patch later for all channels sharing the patch, navigate to **Software › Manage › Channels**, click the channel you want to synchronize and navigate to the **Sync** subtab. Even manual patch synchronization affects all organization channels sharing the patch.

过程：构建内容生命周期项目

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 项目**，然后选择您要构建的项目。

    在构建项目前，请确保环境可用。

2. 查看关联的源和过滤器，然后单击 **[构建]**。
3. 提供版本消息以描述此构建中的更改或更新。
4. You can monitor build progress in the **Environment Lifecycle** section.

构建完成后，环境版本号将会加 1，可将构建的源（例如软件通道）指派到您的客户端。

8.5. 升级环境

构建项目后，可将构建的源按顺序升级到环境。

过程：升级环境

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 项目**，然后选择您要处理的项目。
2. In the **Environment Lifecycle** section, locate the environment to promote to its successor, and click **[Promote]**.
3. You can monitor build progress in the **Environment Lifecycle** section.

8.6. 将客户端分配到环境

When you build and promote content lifecycle projects, Uyuni creates a tree of software channels. To add clients to the environment, assign the base and child software channels to your client using **Software › Software Channels** in the **System Details** page for the client.

新添加的克隆通道不会自动指派到客户端。如果您添加或升级了源，则需要手动检查并更新您的通道指派。

在将来的版本中，自动指派会添加到 Uyuni。

8.7. 内容生命周期管理示例

本节提供了一些常见示例来说明如何使用内容生命周期管理。请使用这些示例来构建您自己的个性化实现。

8.7.1. 为每月修补周期创建项目

每月修补周期的示例项目包括：

- 创建一个**按日期**过滤器
- 将过滤器添加到项目
- 将过滤器应用于新项目构建
- 从项目中排除补丁
- 在项目中包含补丁

8.7.1.1. 创建一个按日期过滤器

The **By Date** filter excludes all patches released after a specified date. This filter is useful for your content lifecycle projects that follow a monthly patch cycle.

Procedure: Creating the By Date filter

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 **[创建过滤器]**。
2. In the **Filter Name** field, type a name for your filter. For example, **Exclude patches by date**.
3. In the **Filter Type** field, select **Patch (Issue date)**.
4. In the **Matcher** field, **later or equal** is autoselected.
5. 选择日期和时间。
6. 单击 **[保存]**。

8.7.1.2. 将过滤器添加到项目

过程：将过滤器添加到项目

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 项目**并从列表中选择一个项目。
2. 单击 **[关联过滤器/解除过滤器关联]** 链接以查看所有可用的过滤器
3. Select the new **Exclude patches by date** filter.
4. 单击 **[保存]**。

8.7.1.3. 将过滤器应用于新项目构建

新过滤器已添加到过滤器列表，但仍然需要将其应用于项目。要应用过滤器，需要构建第一个环境。

过程：使用过滤器

1. 单击 **[构建]** 以构建第一个环境。
2. 可选：添加消息。 可以使用消息来帮助跟踪构建历史记录。
3. 在测试服务器上使用新通道来检查过滤器是否正常工作。
4. 单击 **[升级]** 将内容移到下一个环境。 如果您有大量过滤器，或者过滤器非常复杂，则构建过程需要较长时间。

8.7.1.4. 从项目中排除补丁

测试可以帮助您发现问题。如果发现问题，请排除**按日期**过滤器之前发布的有问题补丁。

过程：排除补丁

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 **[创建过滤器]**。
2. In the **Filter Name** field, enter a name for the filter. For example, **Exclude openjdk patch**.
3. In the **Filter Type** field, select **Patch (Advisory Name)**.
4. In the **Matcher** field, select **equals**.
5. In the **Advisory Name** field, type a name for the advisory. For example, **SUSE-15-2019-1807**.
6. 单击 **[保存]**。
7. 导航到**内容生命周期 › 项目**并选择您的项目。
8. Click **[Attach/Detach Filters]** link, select **Exclude openjdk patch**, and click **[Save]**.

When you rebuild the project with the **[Build]** button, the new filter is used together with the **by date** filter we added before.

8.7.1.5. 在项目中包含补丁

In this example, you have received a security alert. An important security patch was released several days after the first of the month you are currently working on. The name of the new patch is **SUSE-15-2019-2071**. You need to include this new patch into your environment.

> ! The **Allow** filters rule overrides the exclude function of the **Deny** filter rule. For more information, see **Administration Guide › Content Lifecycle**.

过程：在项目中包含补丁

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 [**创建过滤器**]。
2. In the **Filter Name** field, type a name for the filter. For example, **Include kernel security fix**.
3. In the **Filter Type** field, select **Patch (Advisory Name)**.
4. In the **Matcher** field, select **equals**.
5. In the **Advisory Name** field, type **SUSE-15-2019-2071**, and check **Allow**.
6. 单击 [**保存**] 以存储该过滤器。
7. 导航到**内容生命周期 › 项目**并从列表中选择您的项目。
8. Click [**Attach/Detach Filters**], and select **Include kernel security patch**.
9. 单击 [**保存**]。
10. 单击 [**构建**] 以重新构建环境。

8.7.2. 更新现有的每月修补周期

每月修补周期完成后，您可以更新下个月的修补周期。

过程：更新每月修补周期

1. In the **by date** field, change the date of the filter to the next month. 或者，创建一个新过滤器并更改项目的指派。
2. Check if the exclude filter for **SUSE-15-2019-1807** can be detached from the project. 可能有一个新补丁可以修复此问题。
3. Detach the **allow** filter you added previously. 默认已包含该补丁。
4. 重新构建项目以创建包含下个月补丁的新环境。

8.7.3. 通过实时修补增强项目

本节介绍如何设置过滤器以创建可实时修补的环境。

当您准备使用实时修补时，需要注意一些重要事项：

- 永远只在您的系统上使用一个内核版本。 实时修补软件包与特定内核一起安装。
- 实时修补更新作为一个补丁交付。

- Each kernel patch that begins a new series of live patching kernels displays the **required reboot** flag. 这些内核补丁附带实时修补工具。安装这些补丁后，需要在下一年开始之前至少重引导系统一次。
- 仅安装与已安装内核版本匹配的在线补丁更新。
- 在线补丁作为独立补丁提供。 您必须排除内核版本高于当前安装版本的所有常规内核补丁。

8.7.3.1. 排除内核版本较高的软件包

In this example you update your systems with the **SUSE-15-2019-1244** patch. This patch contains **kernel-default-4.12.14-150.17.1-x86_64**.

You need to exclude all patches which contain a higher version of **kernel-default** and **kernel-default-base**.

过程：排除内核版本较高的软件包

1. 在 Uyuni Web UI 中，导航到**内容生命周期 › 过滤器**并单击 **[创建过滤器]**。
2. In the **Filter Name** field, type a name for your filter. For example, **Exclude kernel greater than 4.12.14-150.17.1**.
3. In the **Filter Type** field, select **Patch (Contains Package)**.
4. In the **Matcher** field, select **version greater than**.
5. In the **Package Name** field, type **kernel-default**.
6. Leave the **Epoch** field empty.
7. In the **Version** field, type **4.12.14**.
8. In the **Release** field, type **150.17.1**.
9. 单击 **[保存]** 以存储该过滤器。
10. 导航到**内容生命周期 › 项目**并选择您的项目。
11. 单击 **[关联过滤器/解除过滤器关联]**。
12. Select **Exclude kernel greater than 4.12.14-150.17.1**, and click **[Save]**.

You need to repeat this procedure for the **kernel-default-base** package.

单击 **[构建]** 后，会创建一个新环境。新环境包含版本不高于所安装版本的所有内核补丁。

所有内核版本更高的内核补丁都将被去除。只要实时修补内核不是发行系列中的第一个，它们就仍然可用。

可以使用过滤器模板自动完成此过程。有关如何应用实时修补过滤器模板的详细信息，请参见 **Administration Guide › Content Lifecycle › Filter Templates**。

8.7.4. 切换到实时修补的新内核版本

特定内核版本的实时修补仅可使用一年。一年后，您必须更新系统上的内核。请执行以下环境更改：

过程：切换到新内核版本

1. 确定要升级到哪个内核版本。 例如：**4.12.14-150.32.1**
2. 创建新的内核版本过滤器。
3. 解除所有历史实时修补过滤器与旧内核的关联。
4. 关联新的内核版本过滤器。
5. 单击 **[构建]** 以重新构建环境。

    构建完成后，必须立即重新关联以下两个过滤器：

    - Deny patches that contain **reboot_suggested** keyword
    - Deny patches that contain a package which provides the name **installhint(reboot-needed)**

新环境包含版本不高于所选新内核版本的所有内核补丁。您需要在系统执行升级后重引导系统。新内核的有效期为一年。在当年安装的所有软件包与当前实时修补内核过滤器匹配。

8.7.5. AppStream 过滤器

在内容生命周期管理项目中，您可以使用 AppStream 过滤器将模块化储存库转换为常规储存库。为此，该过滤器会将软件包保留在储存库中，并除去模块元数据。可以在 Uyuni 中像使用常规储存库一样使用转换后的储存库。

> AppStream 储存库在整个 Web UI 中原生受到支持。
>
> 因此，要使用 AppStream 储存库，不一定非要完成此过程。

AppStream 过滤器将选择要包含在目标储存库中的单个模块流。您可以添加多个过滤器来选择多个模块流。

如果您不在 CLM 项目中使用 AppStream 过滤器，则模块化源中的模块元数据将保持不变，并且目标储存库包含相同的模块元数据。只要在 CLM 项目中至少启用一个 AppStream 过滤器，所有目标储存库就会转换为常规储存库。

In some cases, you might wish to build regular repositories without having to include packages from any module. To do so, add an AppStream filter using the matcher **none (disable modularity)**. This will disable all the modules in the target repository. This is especially useful for Red Hat Enterprise Linux 9 clients, where the default versions of most modules are already included in the AppStream repository as regular packages.

To use the AppStream filter, you need a CLM project with a modular repository such as **Red Hat Enterprise Linux AppStreams**. Ensure that you included the module you need as a source before you begin.

过程：使用 AppStream 过滤器

1. 在 Uyuni Web UI 中，导航到您的 Red Hat Enterprise Linux 8 或 9 CLM 项目。 确保已包含项目的 AppStream 通道。
2. 单击 **[创建过滤器]** 并使用以下参数：
   - In the **Filter Name** field, type a name for the new filter.
   - In the **Filter Type** field, select **Module (Stream)**.
   - In the **Matcher** field, select **equals**.
   - In the **Module Name** field, type a module name. For example, **postgresql**.
   - In the **Stream** field, type the name of the desired stream. For example, **10**. If you leave this field blank, the default stream for the module is selected.
3. 单击 **[保存]** 以创建新过滤器。
4. 导航到**内容生命周期 › 项目**并选择您的项目。
5. 单击 **[关联过滤器/解除过滤器关联]**，选择您的新 AppStream 过滤器，然后单击 **[保存]**。

You can use the browse function in the **Create/Edit Filter** form to select a module from a list of available module streams for a modular channel.

过程：浏览可用模块流

1. 在 Uyuni Web UI 中，导航到您的 Red Hat Enterprise Linux 8 或 9 CLM 项目。 确保已包含项目的 AppStream 通道。
2. 单击 **[创建过滤器]** 并使用以下参数：
   - In the **Filter Name** field, type a name for the new filter.
   - In the **Filter Type** field, select **Module (Stream)**.
   - In the **Matcher** field, select **equals**.

3. Click **Browse available modules** to see all modular channels.
4. 选择一个通道以浏览模块和流：
   - In the **Module Name** field, start typing a module name to search, or select from the list.
   - In the **Stream** field, start typing a stream name to search, or select from the list.

只能在浏览模块时选择通道。选定的通道不会与过滤器一起保存，并且不会对 CLM 过程产生任何影响。

您可以为要包含在目标储存库中的任何其他模块流创建额外的 AppStream 过滤器。会自动包含选定流所依赖的任何模块流。

请小心不要指定有冲突的、不兼容的或缺失的模块流。例如，从同一个模块中选择两个流是无效的。

过程：禁用模块化

1. 在 Uyuni Web UI 中，导航到您的 Red Hat Enterprise Linux 8 或 9 CLM 项目。 确保已包含项目的 AppStream 通道。
2. 单击 **[创建过滤器]** 并使用以下参数：
   - In the **Filter Name** field, type a name for the new filter.
   - In the **Filter Type** field, select **Module (Stream)**.
   - In the **Matcher** field, select **none (disable modularity)**.
3. 单击 **[保存]** 以创建新过滤器。
4. 导航到**内容生命周期 › 项目**并选择您的项目。
5. 单击 **[关联过滤器/解除过滤器关联]**，选择您的新 AppStream 过滤器，然后单击 **[保存]**。

这样会有效从目标储存库中去除模块元数据，并排除属于模块的所有软件包。

当您使用 Web UI 中的 **[构建]** 按钮构建 CLM 项目时，目标储存库是不包含任何模块的常规储存库，其中包含来自选定模块流的软件包。

在 Red Hat Enterprise Linux 8 项目中完全禁用模块化可能导致环境存在缺陷，因为在 Red Hat Enterprise Linux 8 中，某些模块对于系统的健康运行是不可或缺的。

章 9. 内容暂存

客户端使用暂存来预先下载软件包，然后再安装软件包。这样就可以在做好安排后立即开始安装软件包，从而减少维护时段占用的时间。

9.1. 启用内容暂存

You can manage content staging across your entire organization. In the Uyuni Web UI, navigate to **Admin › Organizations** to see a list of available organizations. Click the name of an organization, and check the **Enable Staging Contents** box to allow clients in this organization to stage package data.

> 您必须以 Uyuni 管理员身份登录才能创建和管理组织。

You can also enable staging at the command prompt by editing **/etc/sysconfig/rhn/up2date**, and adding or editing these lines:

```
stagingContent=1
stagingContentWindow=24
```

The **stagingContentWindow** parameter is a time value expressed in hours and determines when downloading starts. It is the number of hours before the scheduled installation or update time. In this example, content is downloaded 24 hours before the installation time. The start time for download depends on the selected contact method for a system.

下一次安排操作后，会自动下载但不安装软件包。将在安排的时间安装暂存的软件包。

9.2. 配置内容暂存

有两个参数用于配置内容暂存：

- **salt_content_staging_advance** is the advance time for the content staging window to open, in hours. 它是距离安装开始的小时数，经过这段时间后即可开始下载软件包。
- **salt_content_staging_window** is the duration of the content staging window, in hours. 这是在安装开始之前客户端必须将软件包暂存的一段时间。

For example, if **salt_content_staging_advance** is set to six hours, and **salt_content_staging_window** is set to two hours, the staging window opens six hours before the installation time, and remain open for two hours. No packages are downloaded in the four remaining hours until installation starts.

If you set the same value for both **salt_content_staging_advance** and **salt_content_staging_window** packages are able to be downloaded until installation begins.

Configure the content staging parameters in **/usr/share/rhn/config-defaults/rhn_java.conf**.

默认值：

- **salt_content_staging_advance: 8 hours**
- **salt_content_staging_window: 8 hours**

> 必须启用内容暂存才能让这些参数正常工作。

章 10. 断开连接的设置

无法将 Uyuni 连接到互联网时，您可以在断开连接的环境中使用它。

在 SUSE Linux Enterprise 15 和更高版本上可以使用储存库镜像工具 (RMT)。RMT 取代了订阅管理工具 (SMT)，后者在较旧的 SUSE Linux Enterprise 安装中可用。

在断开连接的 Uyuni 设置中，RMT 或 SMT 使用外部网络连接到 SUSE Customer Center。所有软件通道和储存库都会同步到可卸存储设备。之后，可以使用该存储设备更新断开连接的 Uyuni 安装。

此设置可让您的 Uyuni 安装保留在断开连接的脱机环境中。

> 您的 RMT 或 SMT 实例必须用于直接管理 Uyuni Server。它不能用于管理级联的另一个 RMT 或 SMT 实例。

有关 RMT 的详细信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/book-rmt.html。

10.1. 从 SCC 同步通道和储存库

10.1.1. 同步 RMT

可以在 SUSE Linux Enterprise 15 安装中使用 RMT 来管理运行 SUSE Linux Enterprise 12 或更高版本的客户端。

我们建议您为每个 Uyuni 安装设置一个专用 RMT 实例。

过程：设置 RMT

1. 在 RMT 实例上安装 RMT 软件包：

```
zypper in rmt-server
```

2. 使用 YaST 配置 RMT：

```
yast2 rmt
```

3. 按照提示完成安装。

有关设置 RMT 的详细信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/book-rmt.html。

过程：将 RMT 与 SCC 同步

1. 在 RMT 实例上，列出您的组织可用的所有产品和储存库：

```
rmt-cli products list --all
rmt-cli repos list --all
```

2. 同步您的组织可用的所有更新：

```
rmt-cli sync
```

还可以使用 systemd 将 RMT 配置为定期同步。

3. 启用所需的产品。 例如，要启用 SLES 15，请运行以下命令：

```
rmt-cli product enable sles/15/x86_64
```

4. 将已同步的数据导出到可卸存储。 In this example, the storage medium is mounted at **/mnt/usb**:

```
rmt-cli export data /mnt/usb
```

5. 将已启用的储存库导出到可卸存储：

```
rmt-cli export settings /mnt/usb
rmt-cli export repos /mnt/usb
```

Ensure that the external storage is mounted to a directory that is writeable by the RMT user. You can change RMT user settings in the **cli** section of **/etc/rmt.conf**.

10.1.2. 同步 SMT

SMT 已随附在 SUSE Linux Enterprise 12 中，可用于管理运行 SUSE Linux Enterprise 10 或更高版本的客户端。

SMT 要求您在 SMT 实例上创建一个本地镜像目录以同步储存库和软件包。

有关安装和配置 SMT 的更多细节，请参见 https://documentation.suse.com/sles/12-SP5/html/SLES-all/book-smt.html。

过程：将 SMT 与 SCC 同步

1. 在 SMT 实例上，创建一个数据库替换文件：

```
smt-sync --createdbreplacementfile /tmp/dbrepl.xml
```

2. Export the synchronized data to your removable storage. In this example, the storage medium is mounted at **/mnt/usb**:

```
smt-sync --todir /mnt/usb
smt-mirror --dbreplfile /tmp/dbrepl.xml --directory /mnt/usb \
```

```
         --fromlocalsmt -L /var/log/smt/smt-mirror-export.log
curl https://scc.suse.com/multi-linux-manager/product_tree.json -o
/mnt/usb/product_tree.json
```

Ensure that the external storage is mounted to a directory that is writeable by the RMT user. You can change SMT user settings in **/etc/smt.conf**.

10.2. 必需通道

需要启用相应的 Uyuni 客户端工具通道，才能使 Uyuni 能够同步给定的通道。如果未启用这些通道，Uyuni 可能无法检测到该产品。

运行以下命令启用这些必需的通道：

SLES 12 和基于它的产品，例如 SLES for SAP 或 SLE HPC

RMT: **rmt-cli products enable sle-manager-tools/12/x86_64**

SMT: **smt repos -p sle-manager-tools,12,x86_64**

SLES 15 和基于它的产品，例如 SLES for SAP 或 SLE HPC

RMT: **rmt-cli products enable sle-manager-tools/15/x86_64**

SMT: **smt repos -p sle-manager-tools,15,x86_64**

然后镜像通道并导出。

可以启用其他发行套件或体系结构。有关启用要镜像的产品通道或储存库的详细信息，请参见文档：

RMT

https://documentation.suse.com/sles/15-SP7/html/SLES-all/cha-rmt-mirroring.html#sec-rmt-mirroring-enable-disable

SMT

https://documentation.suse.com/sles/12-SP5/single-html/SLES-smt/index.html#smt-mirroring-manage-domirror

10.3. 不联网服务器

要将 Uyuni 设置为不联网服务器，请按照隔离部署的说明操作。

10.3.1. 部署

建议使用提供的映像将不联网服务器部署为虚拟机 (VM)。如需了解 Uyuni 服务器的隔离部署方式，请参见 **Installation and Upgrade Guide › Container Deployment › Server Air Gapped Deployment Mlm**。

Keep in mind to execute the final command with the **--mirror** option and replace **</media/disk>** with your mount point:

```
mgradm install --mirror </media/disk>
```

10.3.2. 同步

如果您有加载了 SUSE Customer Center 数据的可卸媒体，可以使用它来同步不联网服务器。

> 用于同步的可移动媒体必须始终在同一挂载点上可用。如果未挂载存储媒体，请不要触发同步，否则会导致数据损坏。

过程：同步不联网服务器

1. 重启 Tomcat 服务：

   ```
   mgrctl exec -ti -- systemctl restart tomcat
   ```

2. 刷新本地数据：

   ```
   mgrctl exec -ti -- mgr-sync refresh
   ```

3. 执行同步：

   ```
   mgrctl exec -ti -- mgr-sync list channels
   mgrctl exec -ti -- mgr-sync add channel channel-label
   ```

> Be aware that if **server.susemanager.fromdir** is set, Uyuni will not be able to check if SUSE Customer Center credentials are valid or not. Instead, a warning sign will be displayed and no SCC online check will be performed.

对于不联网设置，还有一种方法是使用服务器间同步 (ISS) 在服务器之间复制内容。有关详细信息，请参见 **Specialized Guides › Large Deployments › 服务器间同步 - 版本 2**。

章 11. 磁盘空间管理

磁盘空间不足可能会对 Uyuni 数据库和文件结构造成严重影响，在某些情况下，这种影响不可恢复。

Uyuni monitors some directories for free disk space. You can modify which directories are monitored, and the warnings that are created using environment variables in systemd service override files.

当某个受监控目录的可用空间低于警告阈值时，会向配置的电子邮件地址发送一条消息，并在登录页面顶部显示一条通知。

11.1. 受监控的目录

By default, the Uyuni Server container monitors these directories:

- **/var/spacewalk**
- **/var/cache**
- **/srv**

You can change which directories are monitored by setting the **DISKCHECKDIRS** environment variable in the server container systemd service. You can specify multiple directories by separating them with a space.

> The **DISKCHECKDIRS** environment variable applies only to the server container. The database container always monitors **/var/lib/pgsql/data** and this cannot be overridden.

Procedure: Configuring Monitored Directories

1. Create a systemd override directory for the server service:

```
mkdir -p /etc/systemd/system/uyuni-server.service.d/
```

2. Create a configuration file with the custom directories:

```
cat > /etc/systemd/system/uyuni-server.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKDIRS=/var/spacewalk /var/cache /srv"
EOF
```

3. Apply the configuration by restarting Uyuni:

```
mgradm restart
```

For more information about container volumes, see **Installation and Upgrade Guide › Container Management › Persistent Container Volumes**.

11.2. 阈值

By default, Uyuni creates a warning when a monitored directory reaches 90% disk usage. A critical alert is created when a monitored directory reaches 95% disk usage, and the container health check will fail.

You can change these alert thresholds by setting the **DISKCHECKALERT** and **DISKTHRESHOLD** environment variables.

Procedure: Configuring Disk Check Thresholds

1. Create systemd override directories:

```
mkdir -p /etc/systemd/system/uyuni-server.service.d/
mkdir -p /etc/systemd/system/uyuni-db.service.d/
```

2. Create configuration files for both server and database services:

```
cat > /etc/systemd/system/uyuni-server.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKALERT=90"
Environment="DISKTHRESHOLD=95"
EOF

cat > /etc/systemd/system/uyuni-db.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKALERT=90"
Environment="DISKTHRESHOLD=95"
EOF
```

3. Apply the configuration by restarting Uyuni:

```
mgradm restart
```

> For large deployments with multi-terabyte storage volumes, you may want to use higher thresholds to make more efficient use of disk space. For example, on a 10 TB volume, keeping 5% free means 500 GB unused. For tuning recommendations for large deployments, see **Specialized Guides › Large Deployments › Tuning › Diskcheck Thresholds**.

11.3. Database Disk Check

The database container has its own disk check that monitors only the PostgreSQL data directory (**/var/lib/pgsql/data**). This directory is always monitored and cannot be changed or overridden with the **DISKCHECKDIRS** environment variable.

The database disk check uses the **DISKCHECKALERT** and **DISKTHRESHOLD** environment variables configured for the **uyuni-db** service.

The database disk check runs as a PostgreSQL function and is invoked during container health checks. When the threshold is exceeded, the health check will fail and the container will be automatically stopped to prevent data corruption.

The database container only monitors **/var/lib/pgsql/data**. You cannot configure additional directories to monitor in the database container using **DISKCHECKDIRS**. To monitor other directories, use the server container configuration.

11.4. 禁用空间检查

The space checking tool is enabled by default and runs hourly via a Taskomatic scheduled task. Notification can be disabled by disabling diskcheck-task-queue-default Taskomatic job. To effectively disable disk space monitoring, set the threshold values to 99 (percent).

Procedure: Disabling Disk Space Monitoring

1. Edit the systemd override files to set very high thresholds:

```
cat > /etc/systemd/system/uyuni-server.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKALERT=99"
Environment="DISKTHRESHOLD=99"
EOF

cat > /etc/systemd/system/uyuni-db.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKALERT=99"
Environment="DISKTHRESHOLD=99"
EOF
```

2. Apply the configuration by restarting Uyuni:

```
mgradm restart
```

Disabling disk space monitoring is not recommended in production environments, as running out of disk space can cause severe and potentially unrecoverable damage to the database.

章 12. 映像构建和管理

12.1. 映像构建概述

Uyuni 允许系统管理员构建容器和操作系统映像，并将结果推送到映像存储区。

过程：构建和推送映像

1. 定义映像存储区。
2. 定义一个映像配置文件并将其与源（git 储存库或目录）相关联。
3. 构建映像。
4. 将映像推送到映像存储区。

Uyuni 支持两种构建类型：Dockerfile 和 Kiwi。Kiwi 构建类型用于构建系统映像、虚拟映像和其他映像。

The image store for the Kiwi build type is pre-defined as a file system directory in the **srv-www** volume.

The image files can be downloaded from https://UYUNI-HOST/os-images/ORGANIZATION-ID/FILE-NAME. The exact location can be determined from the image details page.

12.2. 容器映像

12.2.1. 要求

容器功能适用于运行 SUSE Linux Enterprise Server 12 或更高版本的 Salt 客户端。在开始之前，请确保您的环境满足以下要求：

- 一个包含 Dockerfile 和配置脚本的已发布 git 储存库。该储存库可以是公用或专用的，应托管在 GitHub、GitLab 或 BitBucket 上。
- 一个正确配置的映像存储区，例如 Docker 仓库。

有关容器的详细信息，请参见 https://documentation.suse.com/container/all/html/Container-guide/。

12.2.2. 创建构建主机

要使用 Uyuni 构建映像，您需要创建并配置一个构建主机。容器构建主机是运行 SUSE Linux Enterprise 12 或更高版本的 Salt 客户端。本节将指导您完成构建主机的初始配置。

> 构建主机上的操作系统必须与目标映像上的操作系统匹配。
>
> 例如，在运行 SUSE Linux Enterprise Server 15（SP2 或更高版本）操作系统版本的构建主机上构建基于 SUSE Linux Enterprise Server 15 的映像。在运行 SUSE Linux

> Enterprise Server 12 SP5 或 SUSE Linux Enterprise Server 12 SP4 操作系统版本的构建主机上构建基于 SUSE Linux Enterprise Server 12 的映像。
>
> 不支持跨体系结构构建。

在 Uyuni Web UI 中，执行以下步骤以配置构建主机：

过程：构建主机

1. 在**系统 › 系统**概览页面中选择要指定为构建主机的 Salt 客户端。
2. From the **System Details** page of the selected client assign the containers modules. Navigate to **Software › Software Channels** and enable the containers module (for example, **SLE-Module-Containers15-Pool** and **SLE-Module-Containers15-Updates**). Continue with [**Next**].
3. Schedule the ` Software Channel Change`, and click [**Confirm**].
4. From the **System Details** tab select **Properties** page, and enable **Container Build Host** from the **Add-on System Types** list. Confirm by clicking [**Update Properties**].
5. Install all required packages by applying **Highstate**. From the system details tab select **States › Highstate**, and click [**Apply Highstate**]. Alternatively, apply **Highstate** from the Uyuni Server command line:

```
salt '$your_client' state.highstate
```

12.2.3. 为容器创建激活密钥

通过 Uyuni 构建的容器器使用在构建映像时作为储存库关联到激活密钥的通道。本节将指导您为此目的创建一个临时激活密钥。

> 要构建容器，需要一个与通道关联的激活密钥，但该通道不能是 **SUSE Manager Default**。

Procedure: Creating an Activation Key

1. In the Uyuni Web UI, as an administrator, navigate to **Systems › Activation Keys**.
2. Click the [**Create Key**] button.
3. On the **Activation Key Details** page, in the **Description** field, enter a description of the activation key.

4. In the `Key` field, enter a name for the activation key. For example, `SLES15-SP7` for SUSE Linux Enterprise Server 15 SP7.

    - Do not use commas or double quotes in the `Key` field for any SUSE products. However, you **must** use commas for Red Hat Products.
    - All other characters are allowed, but `<> (){}` (this includes the space) are removed automatically.
    - If the field is left empty, a random string is generated.

5. In the `Usage` add number of times the key can be used, or leave it blank for unlimited usage.
6. In the `Base Channels` drop-down box, select the appropriate base software channel, and allow the relevant child channels to populate. For more information, see **Reference Guide › Admin › Setup Wizard** and **Administration Guide › Custom Channels**.
7. Select the `Child Channels` you need (for example, the mandatory SUSE Multi-Linux Manager tools and updates channels).
8. Check the `Add-On System Types` checkbox if you need to enable any of the options. For more information about system types, see **Client Configuration Guide › System Types**.
9. We recommend you leave the `Contact Method` set to `Default`. For more information about contact methods, see **Client Configuration Guide › Contact Methods Intro**.
10. We recommend you leave the `Universal Default` setting unchecked.
11. Click [Create Activation Key] to create the activation key.
12. After the activation key is created, the page shows additional checkbox `Configuration File Deployment` below the list of `Add-On System Types`. To enable deploying configuration files to systems on registration, select this checkbox.
13. Click [Update Activation Key] to save the change.

有关详细信息，请参见 **Client Configuration Guide › Activation Keys**。

12.2.4. 创建映像存储区

构建的所有映像都会推送到映像存储区。本节包含有关创建映像存储区的信息。映像存储区通常称为仓库。

过程：创建映像存储区

1. 选择**映像 › 存储区**。
2. Click **Create** to create a new store.
3. From the **Store Type** select the correct type.
4. Define a name for the image store in the **Label** field.
5. Provide the path to your image registry by filling in the **URI** field, as a fully qualified domain name (FQDN) for the container registry host (whether internal or external).

   ```
   registry.example.com
   ```

   仓库 URI 还可用于指定已被使用的仓库中的映像存储区。

   ```
   registry.example.com:5000/myregistry/myproject
   ```

6. 单击 **[创建]** 以添加新的映像存储区。

12.2.5. 创建映像配置文件

所有容器映像都是使用包含构建说明的映像配置文件构建的。本节包含有关使用 Uyuni Web UI 创建映像配置文件的信息。

过程：创建映像配置文件

1. 要创建映像配置文件，请选择**映像 › 配置文件**，然后单击 **[创建]**。
2. Provide a name for the image profile by filling in the **Label** field.

   > 如果您的容器映像标记采用类似于 **myproject/myimage** 的格式，请确保您的映像存储区仓库 URI 包含 **/myproject** 后缀。

3. Use **Dockerfile** as the **Image Type**.
4. Use the drop-down menu to select your registry from the **Target Image Store** field.

5. In the **Path** field, type a GitHub, GitLab, or BitBucket repository URL. The path can also be a local directory on the build host. The URL should be **http**, **https**, or a token authentication URL. For GitHub or GitLab, use one of these formats:

   GitHub 路径选项

   - GitHub 单用户项目储存库

     ```
     https://github.com/USER/project.git#branchname:folder
     ```

   - GitHub 组织项目储存库

     ```
     https://github.com/ORG/project.git#branchname:folder
     ```

   - GitHub 令牌身份验证

     如果您的 git 储存库是专用的，请修改配置文件的 URL 以包含身份验证。使用以下 URL 格式通过 GitHub 令牌进行身份验证：

     ```
     https://USER:<身份验证令牌>@github.com/USER/project.git#master:/container/
     ```

   - GitLab 单用户项目储存库

     ```
     https://gitlab.example.com/USER/project.git#master:/container/
     ```

   - GitLab 组项目储存库

     ```
     https://gitlab.example.com/GROUP/project.git#master:/container/
     ```

   - GitLab 令牌身份验证

     如果您的 git 储存库是专用的且不可公开访问，则需要修改配置文件的 git URL 以包含身份验证。使用以下 URL 格式通过 GitLab 令牌进行身份验证：

     ```
     https://gitlab-ci-token:<
     身份验证令牌>@gitlab.example.com/USER/project.git#master:/container/
     ```

     如果您未指定 git 分支，则默认会使用 **master** 分支。如果未

指定 **folder**，则映像源（Dockerfile 源）预期位于 GitHub 或 GitLab checkout 分支的根目录中。

6. 选择一个**激活密钥**。激活密钥可确保将使用配置文件的映像指派到正确的通道和软件包。

    将激活密钥与映像配置文件关联后，可以确保使用该配置文件的任何映像都会使用正确的软件通道以及该通道中的任何软件包。

7. 单击 **[创建]** 按钮。

12.2.5.1. 示例 Dockerfile 源

https://github.com/SUSE/manager-build-profiles 上发布了可重复使用的映像配置文件。

The **ARG** parameters ensure that the built image is associated with the desired repository served by Uyuni. The **ARG** parameters also allow you to build image versions of SUSE Linux Enterprise Server which may differ from the version of SUSE Linux Enterprise Server used by the build host itself.

For example: The **ARG repo** parameter and the **echo** command pointing to the repository file, creates and then injects the correct path into the repository file for the desired channel version.

储存库由您指派到映像配置文件的激活密钥决定。

```
FROM registry.example.com/sles12sp2
MAINTAINER Tux Administrator "tux@example.com"

### 开始：与 {productname} 配合使用时需要这些行

ARG repo
ARG cert

# 添加正确的证书
RUN echo "$cert" > /etc/pki/trust/anchors/RHN-ORG-TRUSTED-SSL-CERT.pem

# 更新证书信任存储
RUN update-ca-certificates

# 将储存库路径添加到映像
RUN echo "$repo" > /etc/zypp/repos.d/susemanager:dockerbuild.repo

### 结束：与 {productname} 配合使用时需要这些行

# 添加打包脚本
ADD add_packages.sh /root/add_packages.sh

# 运行打包脚本
RUN /root/add_packages.sh
```

```
# 构建后从映像中去除储存库路径
RUN rm -f /etc/zypp/repos.d/susemanager:dockerbuild.repo
```

12.2.5.2. 使用自定义信息键值对作为 Docker buildargs

您可以指派自定义信息键值对，以将信息关联到映像配置文件。此外，这些键值对将作为 **buildargs** 传递给 Docker 构建命令。

有关可用自定义信息键和创建其他信息键的详细信息，请参见 **Reference Guide › Systems › Custom System Info**。

12.2.6. 构建映像

可通过两种方式构建映像。第一种方式是从头开始创建。具体方法为：在左侧导航栏中选择**映像 › 构建**，或单击**映像 › 配置文件**列表中的构建图标，然后按照流程操作即可。

过程：构建映像

1. 选择**映像 › 构建**。
2. Add a different tag name if you want a version other than the default **latest** (only relevant to containers).
3. Select **Build Profile** and **Build Host**.

   > Notice the **Profile Summary** to the right of the build fields. When you have selected a build profile, detailed information about the selected profile is displayed in this area.

4. 要安排构建，请单击 **[构建]** 按钮。

12.2.7. 导入映像

The second way to get an image is to import and inspect arbitrary images. To do that, select **Images › Image List** from the left navigation bar. Complete the text boxes of the **Import** dialog. When it has processed, the imported image is listed on the **Image List** page.

过程：导入映像

1. From **Images › Image list** click **[Import]** to open the **Import Image** dialog.
2. In the **Import Image** dialog complete these fields:

   映像存储区

   要从中拉取映像进行检查的仓库。

映像名称

仓库中映像的名称。

映像版本

仓库中映像的版本。

构建主机

用于拉取和检查映像的构建主机。

激活密钥

激活密钥，它提供用于检查映像的软件通道的路径。

3. 单击 [导入] 以确认。

The entry for the image is created in the database, and an **Inspect Image** action on Uyuni is scheduled.

When it has been processed, you can find the imported image in the **Image List**. It has a different icon in the **Build** column, to indicate that the image is imported. The status icon for the imported image can also be seen on the **Overview** tab for the image.

12.2.8. 查错

12.2.8.1. 映像检查

基础容器映像 (BCI) 附带用于运行它的所有软件，但由于 BCI 为轻量级映像，它们可能不会附带您进行检查所需的所有工具和库。

在检查容器映像时，您可能会看到类似如下的错误消息：

```
libssl.so.1.1: cannot open shared object file: No such file or directory
```

BCI 适合用于除在容器构建主机上使用 Salt 捆绑包进行检查之外的其他场景，但如果您需要正常执行检查，则必须提前添加全部所需软件。

To avoid such issues you must add **libopenssl** to the image with **Dockerfile** and rebuild the image.

The same can happen with **libexpat**.

12.2.8.2. 一般问题

下面是处理映像时存在的一些已知问题：

- 用于访问仓库或 git 储存库的 HTTPS 证书应通过自定义状态文件部署到客户端。

- 目前不支持使用 Docker 进行 SSH git 访问。

12.3. 操作系统映像

操作系统映像由 Kiwi 构建系统构建。输出映像可自定义，可以是 PXE、QCOW2、LiveCD 或其他类型的映像。

有关 Kiwi 构建系统的详细信息，请参见 Kiwi 文档。

12.3.1. 要求

Kiwi 映像构建功能适用于运行 SUSE Linux Enterprise Server 12 和 SUSE Linux Enterprise Server 11 的 Salt 客户端。

必须可在以下位置之一访问 Kiwi 映像配置文件和配置脚本：

- Git 储存库
- HTTP 或 HTTPS 托管的 tar 归档
- 构建主机上的本地目录

有关 git 提供的完整 Kiwi 储存库的示例，请参见 https://github.com/SUSE/manager-build-profiles/tree/master/OSImage。

对于运行使用 Kiwi 构建的操作系统映像的主机，至少需要提供 1 GB RAM。具体所需磁盘空间取决于映像的实际大小。有关详细信息，请参见底层系统的文档。

12.3.2. Accessing Git repositories via an HTTP/HTTPS proxy when building images

When a build host needs to fetch sources from a git repository that is only reachable through an HTTP/HTTPS proxy, you may see git timeouts during the build because the git client invoked by the Salt state does not always pick up system-wide proxy settings (for example **/etc/sysconfig/proxy** or environment variables).

To make git use the proxy for non-interactive builds create **/etc/gitconfig** with an http.proxy entry:

```
# /etc/gitconfig
[http]
    proxy = http://proxy.example.com:3128
# or for HTTPS
[http]
    proxy = https://proxy.example.com:3128
```

If the proxy requires authentication, prefer a credential helper or use a credential store instead of embedding credentials in the URL.

12.3.2.1. Automating with Salt

You can manage **/etc/gitconfig** on build hosts through Configuration Management, the same way as on any other managed system. Create a Salt state file and assign it to the build host via a config channel, then apply a highstate from the Uyuni UI.

If you want to source the proxy address from a System Custom Info field, use the **custom_info:** pillar prefix:

```
/etc/gitconfig:
  file.managed:
    - user: root
    - group: root
    - mode: '0644'
    - contents: |
        [http]
            proxy = {{ salt['pillar.get']('custom_info:build_server:git_proxy',
'http://proxy.example.com:3128') }}
```

12.3.3. 基于容器的 Kiwi 映像构建支持

Uyuni 推出了基于容器的 Kiwi 映像构建系统，同时保留现有旧版 Kiwi 工具和 KiwiNG 工具。

12.3.3.1. 配置和覆盖设置

管理员可通过以下 pillar 或自定义值覆盖默认行为。如需进行相关配置，请在 Web UI 中导航至menu:[系统 > 自定义系统信息]，然后创建所需的键。

构建系统的选用取决于底层操作系统或特定 pillar 值：

- 对于 SLE 11 / SLE 12：旧版 Kiwi v7
- 对于 SLE 15：KiwiNG（v9 及容器化 Kiwi 10）

管理员可通过以下 pillar 或自定义值覆盖默认行为：

- **use_kiwi_ng**: force the use of Kiwi 9,
- **use_kiwi_container**: force the use of containerized Kiwi 10. To enable this, set the value to **1**.
- **use_bundle_build**: upload additional KIWI bundle build artifacts (like **.install.tar** which contains PXE files, and **.raw.xz**) to the server. To enable this, set the value to **true**.

12.3.3.2. 不同版本的配置

当为 SUSE Linux Enterprise 15 配置文件使用容器化构建主机时，需要进行特定配置。因为 SLES 15 配置文件依赖 Kiwi 9，而容器默认行为使用 Kiwi 10。

To ensure the correct version is used for SLES 15 profiles, you must define the **kiwi_image** custom info key with the following value:

- **Key**: kiwi_image

- **Value: registry.suse.com/bci/kiwi:9**

If this key is not set, the system defaults to the latest version (e.g., **registry.suse.com/bci/kiwi:10.2**), which may result in build issues for SLES 15 profiles.

12.3.4. 创建构建主机

要使用 Uyuni 构建各种映像，请创建并配置一个构建主机。操作系统映像构建主机是在 SUSE Linux Enterprise Server 15（SP2 或更高版本）或SUSE Linux Enterprise Server 12（SP4 或更高版本）上运行的 Salt 客户端。

此过程将指导您完成构建主机的初始配置。

> 构建主机上的操作系统必须与目标映像上的操作系统匹配。
>
> 例如，在运行 SUSE Linux Enterprise Server 15（SP2 或更高版本）操作系统版本的构建主机上构建基于 SUSE Linux Enterprise Server 15 的映像。在运行 SUSE Linux Enterprise Server 12 SP5 或 SUSE Linux Enterprise Server 12 SP4 操作系统版本的构建主机上构建基于 SUSE Linux Enterprise Server 12 的映像。
>
> 无法实现跨体系结构的构建。例如，必须在运行 SUSE Linux Enterprise Server 15 SP3 的 Raspberry PI（aarch64 体系结构）构建主机上构建 Raspberry PI SUSE Linux Enterprise Server 15 SP3 映像。

过程：在 Uyuni Web UI 中配置构建主机

1. 在**系统 › 概览**页面中选择要指定为构建主机的客户端。
2. Navigate to the **System Details › Properties** tab, and check the **Add-on System Type** > **OS Image Build Host** box.
3. 单击 **[更新属性]** 确认。
4. 导航到**系统细节 › 软件 › 软件通道**，根据构建主机版本启用所需的软件通道。
   - SUSE Linux Enterprise Server 12 build hosts require Uyuni Client tools (**SLE-Manager-Tools12-Pool** and **SLE-Manager-Tools12-Updates**).
   - SUSE Linux Enterprise Server 15 build hosts require SUSE Linux Enterprise Server modules **SLE-Module-DevTools15-SP4-Pool** and **SLE-Module-DevTools15-SP4-Updates**.
   - 配置日程安排，然后单击 **[确认]**。
5. 通过应用 **Highstate** 安装 Kiwi 和所有必需的软件包。在系统细节页面中选择**状态 › Highstate**，然后单击 **[应用 Highstate]**。或者，从 Uyuni Server 命令

行应用 Highstate：

```
salt '$your_client' state.highstate
```

12.3.4.1. Uyuni Web 服务器公共证书 RPM

构建主机置备将 Uyuni 证书 RPM 复制到构建主机。此证书用于访问 Uyuni 提供的储存库。

该证书由 **mgr-package-rpm-certificate-osimage** 打包脚本打包在 RPM 中。在全新安装 Uyuni 期间会自动调用该打包脚本。

当您升级 **spacewalk-certs-tools** 软件包时，升级方案会使用默认值调用该打包脚本。但是，如果证书路径已更改或不可用，请在升级过程完成后使用 **--ca-cert-full-path <证书路径>** 手动调用该打包脚本。

12.3.4.2. 封装脚本调用示例

```
/usr/sbin/mgr-package-rpm-certificate-osimage --ca-cert-full-path /root/ssl-build/RHN-ORG-
TRUSTED-SSL-CERT
```

包含证书的 RPM 软件包存储在 Salt 可访问的目录中，例如：

```
/usr/share/susemanager/salt/images/rhn-org-trusted-ssl-cert-osimage-1.0-1.noarch.rpm
```

包含证书的 RPM 软件包在本地构建主机储存库中提供：

```
/var/lib/Kiwi/repo
```

Specify the RPM package with the Uyuni SSL certificate in the build source, and make sure your Kiwi configuration contains **rhn-org-trusted-ssl-cert-osimage** as a required package in the **bootstrap** section.

列表 3. config.xml

```
...
  <packages type="bootstrap">
    ...
    <package name="rhn-org-trusted-ssl-cert-osimage" bootinclude=
"true"/>
  </packages>
...
```

12.3.5. 为操作系统映像创建激活密钥

创建与通道（在构建映像时操作系统映像可将其用作储存库）关联的激活密钥。

必须提供激活密钥才能构建操作系统映像。

要构建操作系统映像，需要拥有与通道关联的非**默认**激活密钥。

Procedure: Creating an Activation Key

1. In the Uyuni Web UI, as an administrator, navigate to **Systems › Activation Keys**.
2. Click the **[Create Key]** button.
3. On the **Activation Key Details** page, in the **Description** field, enter a description of the activation key.
4. In the **Key** field, enter a name for the activation key. For example, **SLES15-SP7** for SUSE Linux Enterprise Server 15 SP7.

    - Do not use commas or double quotes in the **Key** field for any SUSE products. However, you **must** use commas for Red Hat Products.
    - All other characters are allowed, but **<> (){}** (this includes the space) are removed automatically.
    - If the field is left empty, a random string is generated.

5. In the **Usage** add number of times the key can be used, or leave it blank for unlimited usage.
6. In the **Base Channels** drop-down box, select the appropriate base software channel, and allow the relevant child channels to populate. For more information, see **Reference Guide › Admin › Setup Wizard** and **Administration Guide › Custom Channels**.
7. Select the **Child Channels** you need (for example, the mandatory SUSE Multi-Linux Manager tools and updates channels).
8. Check the **Add-On System Types** checkbox if you need to enable any of the options. For more information about system types, see **Client Configuration Guide › System Types**.
9. We recommend you leave the **Contact Method** set to **Default**. For more information about contact methods, see **Client Configuration Guide › Contact Methods Intro**.
10. We recommend you leave the **Universal Default** setting unchecked.

11. Click **[Create Activation Key]** to create the activation key.
12. After the activation key is created, the page shows additional checkbox **Configuration File Deployment** below the list of **Add-On System Types**. To enable deploying configuration files to systems on registration, select this checkbox.
13. Click **[Update Activation Key]** to save the change.

有关详细信息，请参见 **Client Configuration Guide › Activation Keys**。

12.3.6. 创建映像存储区

OS Images can require a significant amount of storage space. By default, the image store is using the **srv-www** volume.

> 目前不支持 Kiwi 构建类型的用于构建系统映像、虚拟映像和其他映像的映像存储区。
>
> The image files can be downloaded from https://UYUNI-HOST/os-images/ORGANIZATION-ID/FILE-NAME. The exact location can be determined from the image details page.

12.3.7. 创建映像配置文件

使用 Web UI 管理映像配置文件。

过程：创建映像配置文件

1. 要创建映像配置文件，请在**映像 › 配置文件**中选择，然后单击 **[创建]**。
2. In the **Label** field, provide a name for the **Image Profile**.
3. Use **Kiwi** as the **Image Type**.
4. 系统会自动选择映像存储区。
5. Enter a **Config URL** to the directory containing the Kiwi configuration files. For example, a git URI such as https://github.com/SUSE/manager-build-profiles#master:OSImage/SLE-Micro54. Other options are a HTTP or HTTPS hosted tar archive or a local directory on the build host. For more information, see source format options at the end of this section.
6. Enter **Kiwi options** if needed. If the Kiwi configuration files specify multiple profiles, use **--profile <name>** to select the active one. For other options, see Kiwi documentation.

7. 选择一个**激活密钥**。激活密钥可确保将使用配置文件的映像指派到正确的通道和软件包。

   将激活密钥与映像配置文件相关联，以确保映像配置文件使用正确的软件通道和任何软件包。

8. 单击 **[创建]** 按钮确认。

源格式选项

* 指向储存库的 git/HTTP(S) URL

  指向公用或私用 git 储存库的 URL，该储存库包含要构建的映像的源代码。根据储存库的布局，该 URL 可能是：

  ```
  https://github.com/SUSE/manager-build-profiles
  ```

  可以在 URL 中的 **#** 字符后面指定一个分支。此示例使用了 **master** 分支：

  ```
  https://github.com/SUSE/manager-build-profiles#master
  ```

  可以在 **:** 字符后面指定包含映像源的目录。此示例使用了 **OSImage/POS_Image-JeOS6**：

  ```
  https://github.com/SUSE/manager-build-profiles#master:OSImage/POS_Image-JeOS6
  ```

* 指向 tar 归档的 HTTP(S) URL

  指向托管在 Web 服务器上的 tar 归档（压缩或未压缩）的 URL。

  ```
  https://myimagesourceserver.example.org/MyKiwiImage.tar.gz
  ```

* 构建主机上的目录的路径

  输入包含 Kiwi 构建系统源的目录的路径。此目录必须在选定的构建主机上存在。

  ```
  /var/lib/Kiwi/MyKiwiImage
  ```

12.3.7.1. Kiwi 源的示例

Kiwi 源至少包含 **config.xml**。通常其中还有 **config.sh** 和 **images.sh**。源还可以包含要安装在 **root** 子目录下的最终映像中的文件。

有关 Kiwi 构建系统的信息，请参见 Kiwi 文档。

SUSE 在 SUSE/manager-build-profiles 公共 GitHub 代码库中提供了功能齐备的映像源的示例。

列表 4. JeOS config.xml 示例

```
<?xml version="1.0" encoding="utf-8"?>

<image schemaversion="6.1" name="POS_Image_JeOS6">
    <description type="system">
        <author>Admin User</author>
        <contact>noemail@example.com</contact>
        <specification>SUSE Linux Enterprise 12 SP3 JeOS</specification>
    </description>
    <preferences>
        <version>6.0.0</version>
        <packagemanager>zypper</packagemanager>
        <bootsplash-theme>SLE</bootsplash-theme>
        <bootloader-theme>SLE</bootloader-theme>

        <locale>en_US</locale>
        <keytable>us.map.gz</keytable>
        <timezone>Europe/Berlin</timezone>
        <hwclock>utc</hwclock>

        <rpm-excludedocs>true</rpm-excludedocs>
        <type boot="saltboot/suse-SLES12" bootloader="grub2" checkprebuilt="true"
compressed="false" filesystem="ext3" fsmountoptions="acl" fsnocheck="true" image="pxe"
kernelcmdline="quiet"></type>
    </preferences>
    <!--    CUSTOM REPOSITORY
    <repository type="rpm-dir">
      <source path="this://repo"/>
    </repository>
    -->
    <packages type="image">
        <package name="patterns-sles-Minimal"/>
        <package name="aaa_base-extras"/> <!-- wouldn't be SUSE without that ;-) -->
        <package name="kernel-default"/>
        <package name="venv-salt-minion"/>
        ...
    </packages>
    <packages type="bootstrap">
        ...
        <package name="sles-release"/>
        <!-- this certificate package is required to access {productname} repositories
             and is provided by {productname} automatically -->
        <package name="rhn-org-trusted-ssl-cert-osimage" bootinclude="true"/>

    </packages>
    <packages type="delete">
        <package name="mtools"/>
        <package name="initviocons"/>
        ...
    </packages>
```

```
</image>
```

12.3.8. 构建映像

可以使用 Web UI 以两种方式构建或获取映像。选择**映像 › 构建**，或单击**映像 › 配置文件**列表中的构建图标。

过程：构建映像

1. 选择**映像 › 构建**。
2. Add a different tag name if you want a version other than the default **latest** (applies only to containers).
3. Select the **Image Profile** and a **Build Host**.

   A **Profile Summary** is displayed to the right of the build fields. When you have selected a build profile, detailed information about the selected profile is shown here.
4. 要安排构建，请单击 **[构建]** 按钮。

在映像构建过程中，构建服务器无法运行任何形式的自动挂载程序。如果适用，请确保不要以 root 身份运行 Gnome 会话。如果某个自动挂载程序正在运行，则映像构建可以成功完成，但映像的校验和将会不同，从而导致失败。

成功构建映像后，检查阶段随即开始。在检查阶段，SUSE Multi-Linux Manager 会收集有关映像的信息：

- 映像中安装的软件包列表
- 映像的校验和
- 映像类型和其他映像细节

如果构建的映像类型是 **PXE**，则还会生成一个 Salt pillar。映像 pillar 存储在数据库中，Salt 子系统可以访问有关生成的映像的细节。细节包括映像文件所在位置和提供映像文件的位置、映像校验和、网络引导所需的信息，等等。

生成的 pillar 可供所有连接的客户端使用。

12.3.9. 查错

构建映像需要完成几个相关的步骤。如果构建失败，调查 Salt 状态结果和构建日志可能有助于确定失败原因。构建失败时，可以执行以下检查：

- 构建主机是否可以访问构建源
- 构建主机和 Uyuni 服务器上是否为映像提供了足够的磁盘空间

- 激活密钥是否关联了正确的通道
- 使用的构建源是否有效
- 包含 Uyuni 公共证书的 RPM 软件包是否是最新的，并已在路径 **/usr/share/susemanager/salt/images/rhn-org-trusted-ssl-cert-osimage-1.0-1.noarch.rpm** 下提供。有关如何刷新公共证书 RPM 的详细信息，请参见 创建构建主机。

12.3.10. 限制

本节包含使用映像时存在的一些已知问题。

- 用于访问 HTTP 源或 git 储存库的 HTTPS 证书应通过自定义状态文件部署到客户端，或手动进行配置。
- 不支持导入基于 Kiwi 的映像。

12.4. 构建的映像列表

要列出可用的已构建映像，请选择**映像 › 映像列表**。此时会显示所有映像的列表。

Displayed data about images includes an image **Name**, its **Version**, **Revision**, and the build **Status**. You can also see the image update status with a listing of possible patch and package updates that are available for the image.

For OS Images, the **Name** and **Version** fields originate from Kiwi sources and are updated at the end of successful build. During building or after failed build these fields show a temporary name based on profile name.

Revision is automatically increased after each successful build. For OS Images, multiple revisions can co-exist in the store.

For Container Images the store holds only the latest revision. Information about previous revisions (packages, patches, etc.) are preserved and it is possible to list them with the **Show obsolete** checkbox.

单击映像上的 **[细节]** 按钮会显示一个详细视图。详细视图包含相关补丁的确切列表、映像中安装的所有软件包的列表和构建日志。

单击 **[删除]** 按钮会从列表中删除映像。同时还会删除操作系统映像存储区中关联的 pillar 和文件，以及过时的修订版。

> 仅当构建后的检查状态为成功时，才会显示补丁和软件包列表。

章 13. 基础架构维护任务

如果您使用安排的停机时段，可能发现很难记住在 Uyuni Server 发生这种事关重大的停机之前、期间和之后您要做的所有事情。Uyuni Server 相关系统（例如服务器间同步从属服务器或 Uyuni Proxy）也会受到影响，必须在日程安排中将其考虑在内。

SUSE 建议您始终将 Uyuni 基础结构保持更新状态。这包括服务器、代理和构建主机。如果不使 Uyuni Server 保持更新状态，可能无法在必要时更新环境的某些部分。

本章提供了一份停机时段核对清单，其中包含有关执行每个步骤的更多信息的链接。

13.1. 服务器

过程：服务器检查

1. 应用最新更新。
2. 根据需要升级到最新服务包。
3. Run **podman ps** and check whether all required services are up and running.

On SUSE Linux Enterprise Server 15 SP7, you can install updates using a package manager:

- 有关使用 YaST 的信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/cha-onlineupdate-you.html。
- 有关使用 zypper 的信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/cha-sw-cl.html#sec-zypper。

On SL Micro 6.2, you can install updates using the **transactional-update** command. For information on using **transactional-update**, see https://documentation.suse.com/sle-micro/6.2/html/Micro-transactional-updates/index.html.

默认会为 Uyuni Server 配置并启用多个更新通道。新软件包和更新的软件包会自动变为可用状态。

13.1.1. 客户端工具

When the server is updated consider updating some tools on the clients, too. Updating **venv-salt-minion**, **zypper**, and other related management package on clients is not a strict requirement, but it is a best practice in general. For example, a maintenance update on the server might introduce a major new Salt version. Then Salt clients continue to work but might experience problems later on. To avoid this SUSE makes sure that **venv-salt-minion** will always be updated safely.

13.2. 服务器间同步从属服务器

如果您使用服务器间同步从属服务器，请在 Uyuni 服务器更新完成后对其进行更新。

有关详细信息，请参见 **Specialized Guides › Large Deployments › 服务器间同步 - 版本 2**。

13.3. 监视服务器

如果您使用 Prometheus 监控服务器，请在 Uyuni Server 更新完成后对其进行更新。

有关监控的详细信息，请参见 **Administration Guide › Monitoring**。

13.4. 代理

Uyuni Server 更新完成后，代理应会立即更新。

一般情况下，不支持运行与其他版本上的服务器相连接的代理。唯一的例外情况是在更新期间，预期首先会更新服务器，因此代理可能暂时运行旧版本。

> 始终先升级服务器，然后再升级任何代理。

章 14. 使用 Uyuni 进行实时修补

执行内核更新通常需要重引导系统。应尽早应用公共漏洞和暴露（CVE）补丁，但如果您无法承受停机，可以使用实时修补来注入这些重要更新，这样就不需要重引导。

对于 SLES 12 和 SLES 15，设置实时修补的过程略有不同。本章将阐述适用于这两个版本的过程。

14.1. 设置实时修补通道

每次更新完整的内核软件包后都需要重引导。因此，必须确保使用实时修补的客户端在它们所指派到的通道中没有可用的更新内核。使用实时修补的客户端包含实时修补通道中正在运行的内核的更新。

可通过两种方式管理实时修补通道：

使用内容生命周期管理克隆产品树，并去除比正在运行的版本更新的内核版本。**Administration Guide › Content Lifecycle Examples › Enhance Project with Livepatching** 中介绍了此过程。这是建议的解决方法。

或者，可以使用 **spacewalk-manage-channel-lifecycle** 工具。此过程更偏向于手动操作，需要使用命令行工具以及 Web UI。本节针对 SLES 15 SP5 介绍了此过程，但内容同样适用于 SLE 12 SP4 或更高版本。

14.1.1. 为实时修补使用 spacewalk-manage-channel-lifecycle

spacewalk-manage-channel-lifecycle 已弃用，并将在后续版本中去除。建议用户改用功能更丰富且受官方支持的内容生命周期管理 (CLM) API。

Cloned vendor channels should be prefixed by **dev** for development, **testing**, or **prod** for production. In this procedure, you create a **dev** cloned channel and then promote the channel to **testing**.

过程：克隆实时修补通道

1. 在客户端上的命令提示符下，以 root 身份获取当前软件包通道树：

```
# spacewalk-manage-channel-lifecycle --list-channels
Spacewalk Username: admin
Spacewalk Password:
Channel tree:

 1. sles15-sp7-pool-x86_64
      \__ sle-live-patching15-pool-x86_64-sp7
      \__ sle-live-patching15-updates-x86_64-sp7
      \__ sle-manager-tools15-pool-x86_64-sp7
      \__ sle-manager-tools15-updates-x86_64-sp7
      \__ sles15-sp7-updates-x86_64
```

2. Use the **spacewalk-manage-channel** command with the **init** argument to automatically create a new development clone of the original vendor channel:

```
spacewalk-manage-channel-lifecycle --init -c sles15-sp7-pool-x86_64
```

3. Check that **dev-sles15-sp7-updates-x86_64** is available in your channel list.

Check the **dev** cloned channel you created, and remove any kernel updates that require a reboot.

过程：从克隆的通道中去除非在线内核补丁

1. Check the current kernel version by selecting the client from **Systems › System List**, and taking note of the version displayed in the **Kernel** field.
2. In the Uyuni Web UI, select the client from **Systems › Overview**, navigate to the **Software › Manage › Channels** tab, and select **dev-sles15-sp7-updates-x86_64**. Navigate to the **Patches** tab, and click **[List/Remove Patches]**.
3. In the search bar, type **kernel** and identify the kernel version that matches the kernel currently used by your client.
4. 去除所有比当前安装的内核更新的内核版本。

Your channel is now set up for live patching, and can be promoted to **testing**. In this procedure, you also add the live patching child channels to your client, ready to be applied.

过程：升级实时修补通道

1. At the command prompt on the client, as **root**, promote and clone the **dev-sles15-sp7-pool-x86_64** channel to a new **testing** channel:

```
# spacewalk-manage-channel-lifecycle --promote -c dev-sles15-sp7-pool-x86_64
```

2. 在 Uyuni Web UI 中，从**系统 › 概览**中选择客户端，然后导航到**软件 › 软件通道**选项卡。
3. Check the new **test-sles15-sp7-pool-x86_64** custom channel to change the base channel, and check both corresponding live patching child channels.
4. 单击 **[下一步]**，确认细节正确，然后单击 **[确认]** 以保存更改。

现在可以选择和查看可用的 CVE 补丁，并通过实时修补应用这些重要的内核更新。

14.2. SLES 15 上的实时修补

On SLES 15 systems and newer, live patching is managed by the **klp livepatch** tool.

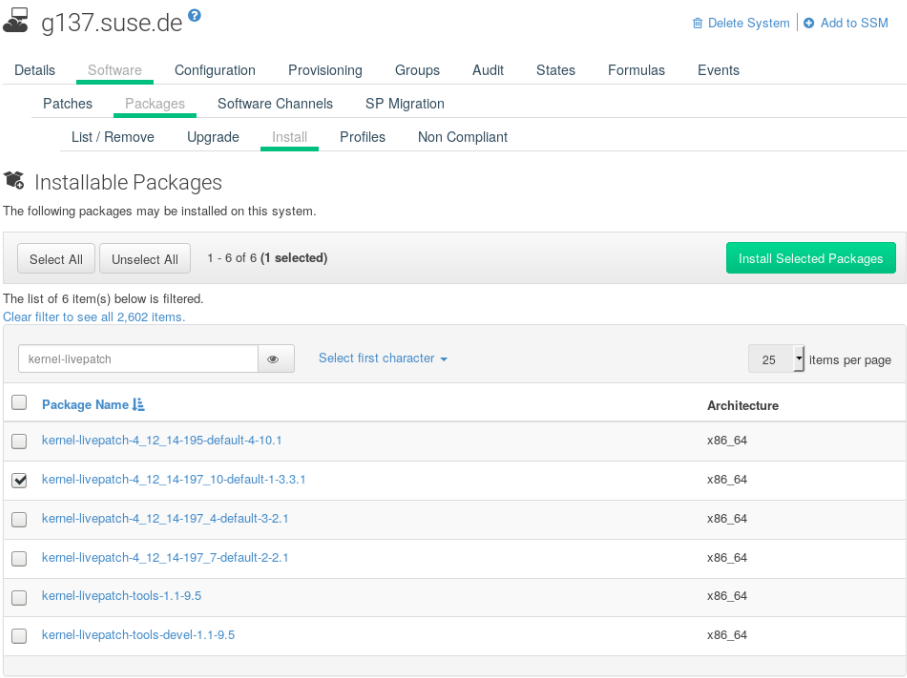
在开始之前，请确保：

- Uyuni 已完全更新。
- 您有一个或多个运行 SLES 15（SP1 或更高版本）的 Salt 客户端。
- 您的 SLES 15 Salt 客户端已注册到 Uyuni。
- 您有权访问适合您的体系结构的 SLES 15 通道，包括一个或多个实时修补子通道。
- 客户端已完全同步。

- 将客户端指派到为实时修补准备的克隆通道。 有关准备工作的详细信息，请参见 **Administration Guide › Live Patching Channel Setup**。

过程：设置实时修补

1. 从**系统 › 概览**中选择您要使用实时修补管理的客户端，然后导航到**软件 › 软件包 › 安装**选项卡。 Search for the **kernel-livepatch** package, and install it.



2. 应用 Highstate 以启用实时修补，然后重引导客户端。
3. 对您要使用实时修补管理的每个客户端重复上述过程。
4. To check that live patching has been enabled correctly, select the client from **Systems › System List**, and ensure that **Live Patch** appears in the **Kernel** field.

过程：将在线补丁应用于内核

1. 在 Uyuni Web UI 中，从**系统 › 概览**中选择客户端。 屏幕顶部的横幅显示了客户端可用的关键和非关键软件包数量。
2. 单击 **[关键]** 查看可用的关键补丁列表。
3. Select any patch with a synopsis reading **Important: Security update for the Linux kernel**. 安全 bug 还包含其 CVE 编号（如果适用）。
4. 可选：如果您知道要应用的补丁的 CVE 编号，可以在**审计 › CVE 审计**中搜索该补丁，并将它应用于任何需要它的客户端。

> - 并非所有内核补丁都是在线补丁。非在线内核补丁由**安全**盾牌图标旁边的**需要重引导**图标表示。这些补丁在应用后始终需要重引导。
> - 并非所有安全问题都可以通过应用在线补丁来解决。某些安全问题只能通过应用完整内核更新来解决，并需要重引导。针对这些问题指派的 CVE 编号不包含在在线补丁

中。CVE 审计会显示此要求。

14.3. SLES 12 上的实时修补

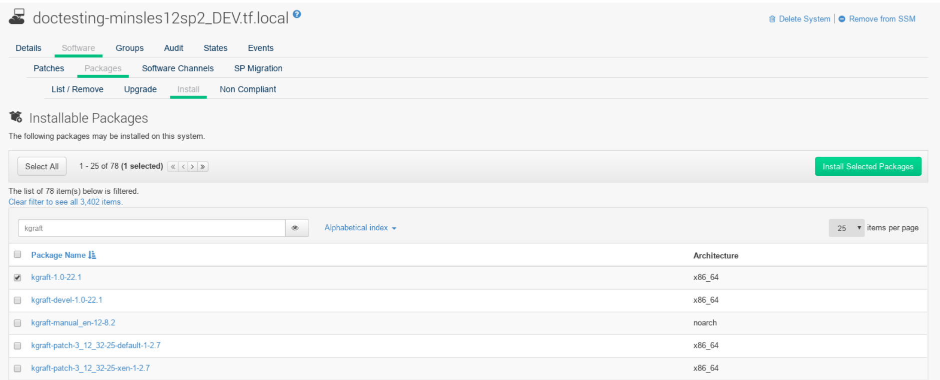
在 SLES 12 系统上，可以通过 kGraft 管理实时修补。有关 kGraft 用法等详细信息，请参见 https://documentation.suse.com/sles/12-SP5/html/SLES-all/cha-kgraft.html。

在开始之前，请确保：

- Uyuni 已完全更新。
- 您有一个或多个运行 SLES 12（SP1 或更高版本）的 Salt 客户端。
- 您的 SLES 12 Salt 客户端已注册到 Uyuni。
- 您有权访问适合您的体系结构的 SLES 12 通道，包括一个或多个实时修补子通道。
- 客户端已完全同步。
- 将客户端指派到为实时修补准备的克隆通道。 有关准备工作的详细信息，请参见 **Administration Guide › Live Patching Channel Setup**。

过程：设置实时修补

1. 从**系统 › 概览**中选择您要使用实时修补管理的客户端，然后在系统细节页面上导航到**软件 › 软件包 › 安装**选项卡。 Search for the **kgraft** package, and install it.



2. 应用 Highstate 以启用实时修补，然后重引导客户端。
3. 对您要使用实时修补管理的每个客户端重复上述过程。
4. To check that live patching has been enabled correctly, select the client from **Systems › System List**, and ensure that **Live Patching** appears in the **Kernel** field.

过程：将在线补丁应用于内核

1. 在 Uyuni Web UI 中，从**系统 › 概览**中选择客户端。 屏幕顶部的横幅显示了客户端可用的关键和非关键软件包数量。
2. 单击 **[关键]** 查看可用的关键补丁列表。

3. Select any patch with a synopsis reading **Important: Security update for the Linux kernel**. 安全 bug 还包含其 CVE 编号（如果适用）。

4. 可选：如果您知道要应用的补丁的 CVE 编号，可以在**审计 › CVE 审计**中搜索该补丁，并将它应用于任何需要它的客户端。

- 并非所有内核补丁都是在线补丁。非在线内核补丁由**安全**盾牌图标旁边的**需要重引导**图标表示。这些补丁在应用后始终需要重引导。
- 并非所有安全问题都可以通过应用在线补丁来解决。某些安全问题只能通过应用完整内核更新来解决，并需要重引导。针对这些问题指派的 CVE 编号不包含在在线补丁中。CVE 审计会显示此要求。

章 15. 维护时段

使用 Uyuni 中的维护时段功能，可以将操作安排在预定的维护时段执行。如果您创建了维护时段日程安排并将其应用于客户端，则无法在指定的时段以外执行某些操作。

> 维护时段的工作方式不同于系统锁定。系统锁可按需打开或关闭，而维护时段则是定义允许执行操作的时间段。此外，允许和受限的操作也不相同。有关系统锁的详细信息，请参见 **Client Configuration Guide › System Locking**。

Maintenance windows require both a calendar, and a schedule. The calendar defines the date and time of your maintenance window events, including recurring events, and must be in **ical** format. The schedule uses the events defined in the calendar to create the maintenance windows. You must create an **ical** file for upload, or link to an **ical** file to create the calendar, before you can create the schedule.

创建日程安排后，可将其指派给已注册到 Uyuni Server 的客户端。指派有维护日程安排的客户端在维护时段以外无法运行受限操作。

受限操作会对客户端进行重大修改，可能导致客户端停止运行。受限操作的一些示例包括：

- 软件包安装
- 客户端升级
- 产品迁移
- Highstate 应用

非受限制操作属于次要操作，一般认为它们是安全的，不太可能会在客户端上造成问题。非受限操作的一些示例包括：

- 软件包配置文件更新
- 硬件刷新
- 订阅软件通道

Before you begin, you must create an **ical** file for upload, or link to an **ical** file to create the calendar. You can create **ical** files in your preferred calendaring tool, such as Microsoft Outlook, Google Calendar, or KOrganizer.

过程：上载新的维护日历

1. 在 Uyuni Web UI 中，导航到**日程安排 › 维护时段 › 日历**，然后单击 **[创建]**。
2. In the **Calendar Name** section, type a name for your calendar.
3. Either provide a URL to your **ical** file, or upload the file directly.
4. 单击 **[创建日历]** 以保存您的日历。

过程：创建新的日程安排

1. 在 Uyuni Web UI 中，导航到**日程安排 › 维护时段 › 日程安排**，然后单击 **[创建]**。
2. In the **Schedule Name** section, type a name for your schedule.
3. OPTIONAL: If your **ical** file contains events that apply to more than one schedule, check **Multi**.
4. 选择要指派到此日程安排的日历。
5. 单击 **[创建日程安排]** 以保存您的日程安排。

过程：将日程安排指派到客户端

1. In the Uyuni Web UI, navigate to **Systems › Systems List**, select the client to be assigned to a schedule, locate the **System Properties** panel, and click **[Edit These Properties]**. 或者，可以通过系统集管理器指派客户端，方法是导航到**系统 › 系统集管理器**并使用**其他 › 维护时段**选项卡。
2. In the **Edit System Details** page, locate the **Maintenance Schedule** field, and select the name of the schedule to be assigned.
3. 单击 **[更新属性]** 以指派维护日程安排。

> 当您将新的维护日程安排指派到某个客户端时，可能已经为该客户端安排了一些受限操作，而这些操作现在可能与新的维护日程安排有冲突。如果发生这种情况，Web UI 会显示错误，并且您无法将该日程安排指派到该客户端。要解决此问题，请在指派日程安排时选中 **[取消受影响的操作]** 选项。这样就会取消以前安排的与新维护日程安排冲突的所有操作。

创建维护时段后，可以安排要在维护时段内执行的受限操作，例如软件包升级。

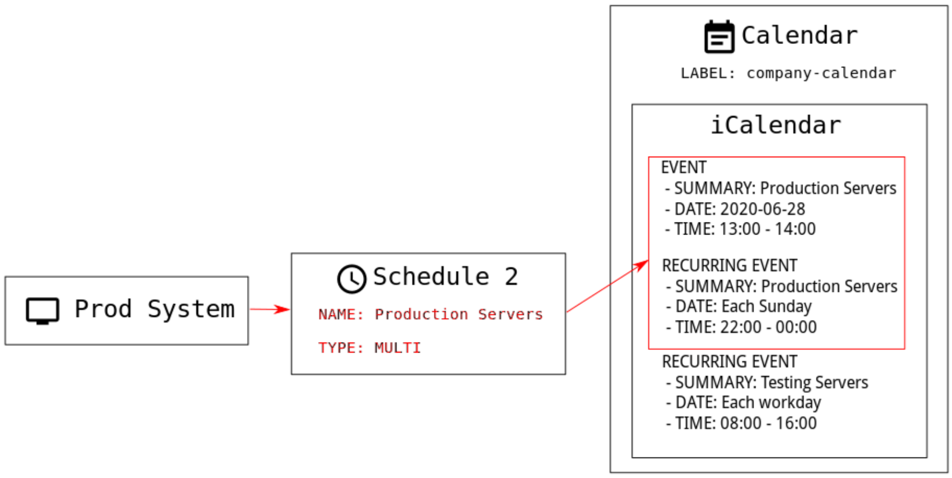
过程：安排软件包升级

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表**，选择要升级的客户端，然后转到**软件 › 软件包 › 升级**选项卡。
2. 从列表中选择要升级的软件包，然后单击 **[升级软件包]**。
3. In the **Maintenance Window** field, select which maintenance window the client should use to perform the upgrade.
4. 单击 **[确认]** 以安排软件包升级。

15.1. 维护日程安排类型

When you create a calendar, it contains a number of events, which can be either one-time events, or recurring events. Each event contains a **summary** field. If you want to create multiple maintenance schedules for one calendar, you can specify events for each using the **summary** field.

For example, you might like to create a schedule for production servers, and a different schedule for testing servers. In this case, you would specify **SUMMARY: Production Servers** on events for the production servers, and **SUMMARY: Testing Servers** on events for the testing servers.



There are two types of schedule: single, or multi. If your calendar contains events that apply to more than one schedule, then you must select **multi**, and ensure you name the schedule according to the **summary** field you used in the calendar file.

过程：创建多日程安排

1. 在 Uyuni Web UI 中，导航到**日程安排 › 维护时段 › 日程安排**，然后单击 **[创建]**。
2. In the **Schedule Name** section, type the name for your schedule. Ensure it matches the **summary** field of the calendar.
3. Check the **Multi** option.
4. 选择要指派到此日程安排的日历。
5. 单击 **[创建日程安排]** 以保存您的日程安排。
6. 要创建下一个日程安排，请单击 **[创建]**。
7. In the **Schedule Name** section, type the name for your second schedule. Ensure it matches the **summary** field of the second calendar.
8. Check the **Multi** option.
9. 单击 **[创建日程安排]** 以保存您的日程安排。
10. 对需要创建的每个日程安排重复上述过程。

15.2. 受限制和非受限操作

本节包含受限和非受限操作的完整列表。

受限操作会对客户端进行重大修改，可能导致客户端停止运行。受限操作只能在维护时段内运行。受限操作包括：

- 软件包操作（例如安装、更新或去除软件包）
- 补丁更新
- 重引导客户端

* 回滚事务
* 配置管理更改任务
* 应用 highstate
* 自动安装和重新安装
* 远程命令
* 产品迁移
* 群集操作

> 您可以随时导航到 **Salt › 远程命令**直接运行远程命令。无论客户端是否处于维护时段，均可执行此操作。有关远程命令的详细信息，请参见 **Administration Guide › Actions**。

Unrestricted actions are minor actions that are considered safe and are unlikely to cause problems on the client. If an action is not restricted it is, by definition, unrestricted, and can be run at any time.

章 16. Using mgr-sync

The **mgr-sync** tool is used at the command prompt. It provides functions for using Uyuni that are not always available in the Web UI. The primary use of **mgr-sync** is to connect to the SUSE Customer Center, retrieve product and package information, and prepare channels for synchronization with the Uyuni Server.

此工具应与 SUSE 支持订阅配合使用。openSUSE、CentOS 和 Ubuntu 等开源发行套件不需要此工具。

The available commands and arguments for **mgr-sync** are listed in this table. Use this syntax for **mgr-sync** commands:

```
mgr-sync [-h] [--version] [-v] [-s] [-d {1,2,3}] {list,add,refresh,delete}
```

表格 3. mgr-sync 命令

Command	Description	Example Use
list	List channels, organization credentials, or products	**mgr-sync list channels**
add	Add channels, organization credentials, or products	**mgr-sync add channel <channel_name>**
refresh	Refresh the local copy of products, channels, and subscriptions	**mgr-sync refresh**
delete	Delete existing SCC organization credentials from the local system	**mgr-sync delete credentials**
sync	Synchronize specified channel or ask for it when left blank	**mgr-sync sync channel <channel_name>**

要查看特定于某个命令的完整选项列表，请使用以下命令：

```
mgr-sync <命令> --help
```

表格 4. mgr-sync 可选参数

Option	Abbreviated option	Description	Example Use
help	**h**	Display the command usage and options	**mgr-sync --help**
version	N/A	Display the currently installed version of **mgr-sync**	**mgr-sync --version**

Option	Abbreviated option	Description	Example Use
verbose	**v**	Provide verbose output	**mgr-sync --verbose refresh**
store-credentials	**s**	Store credentials a local hidden file	**mgr-sync --store -credentials**
debug	**d**	Log additional debugging information. Requires a level of 1, 2, 3. 3 provides the highest amnount of debugging information	**mgr-sync -d 3 refresh**
no-sync	N/A	Use with the **add** command to add products or channels without beginning a synchronization	**mgr-sync --no-sync add <channel_name>**

Logs for **mgr-sync** are located in:

- **/var/lib/containers/storage/volumes/var-log/_data/rhn/mgr-sync.log**
- **/var/lib/containers/storage/volumes/var-log/_data/rhn/rhn_web_api.log**

章 17. 使用 Prometheus 和 Grafana 进行监控

您可以使用 Prometheus 和 Grafana 监控 Uyuni 环境。Uyuni Server 和 Proxy 能够提供自我运行状况监控指标。您还可以在 Salt 客户端上安装和管理一些 Prometheus 导出器。

17.1. 要求

Prometheus 和 Grafana 软件包包含在以下发行套件的 Uyuni 客户端工具中：

- SUSE Linux Enterprise 15
- SUSE Linux Enterprise 16
- openSUSE Leap 15.x
- openSUSE Leap 16.x

仅支持使用上述列出的客户端作为监控服务器。

您需要在一台与 Uyuni 服务器相互独立的计算机上安装 Prometheus 和 Grafana。我们建议使用受管 Salt SUSE 客户端作为监控服务器。

Prometheus 使用拉取机制提取指标，因此服务器必须能够与受监控客户端建立 TCP 连接。客户端上必须打开相应的端口，并且必须可以通过网络访问客户端。或者，您可以使用反向代理来建立连接。

17.2. Prometheus 和 Grafana

17.2.1. Prometheus

Prometheus 是一个开源监控工具，用于在时序数据库中记录实时指标。指标通过 HTTP 拉取，从而可以实现高性能和可伸缩性。

Prometheus 指标是时序数据，或者是属于同一个组或维度的带时间戳值。指标由其名称和一组标签唯一标识。

```
   指标名称            标签                 时间戳     值
┌─────────────┴─────────────┐   ┌──────────────┴──────────────┐
┌───────────┴───────────┐ ┌──┴──┐
http_requests_total{status="200", method="GET"}  @1557331801.111  42236
```

每个受监控的应用程序或系统必须通过代码检测或 Prometheus 导出器按上述格式公开指标。

17.2.2. Prometheus 导出器

导出器是帮助将第三方系统中的指标作为 Prometheus 指标导出的库。每当无法直接使用 Prometheus 指标检测给定的应用程序或系统时，导出器就很有作用。可以在受监控主机上运行多个导出器以导出本地指标。

Prometheus 社区提供了官方导出器的列表，还有其他一些作为社区贡献内容提供的导出器。有关详细信息和导出器的详细列表，请参见 https://prometheus.io/docs/instrumenting/exporters/。

17.2.3. Grafana

Grafana 是一个数据可视化、监控和分析工具。它用于创建仪表板，其中的面板代表设置时间段内的特定指标。Grafana 通常与 Prometheus 一起使用，但也支持 ElasticSearch、MySQL、PostgreSQL 和 Influx DB 等其他数据源。有关 Grafana 的详细信息，请参见 https://grafana.com/docs/。

17.3. 设置监控服务器

要设置监控服务器，需要安装 Prometheus 和 Grafana 并对其进行配置。

> 仅支持使用 SUSE 客户端作为监控服务器。如需完整列表，请参见 **Administration Guide › Monitoring › Monitoring Requirements**。

17.3.1. 安装 Prometheus

如果您的监控服务器是 Salt 客户端，则您可以使用 Uyuni Web UI 安装 Prometheus 软件包。否则，可以在监控服务器上手动下载并安装该软件包。Prometheus 软件也可用于 Uyuni Proxy 和 Uyuni for Retail Branch Server。

> - To access a shell inside the Uyuni Server container run **mgrctl term** on the container host, or to execute one command run **mgrctl exec <options> — <command>**.
> - To copy files from inside the container to the container host use **mgrctl cp**.

> Prometheus 需要使用 POSIX 文件系统来存储数据。不支持不符合 POSIX 标准的文件系统，因此不支持 NFS 文件系统。

过程：使用 Web UI 安装 Prometheus

1. In the Uyuni Web UI, open the details page of the system where Prometheus is to be installed, and navigate to the **Formulas** tab.
2. Check the **Prometheus** checkbox to enable monitoring formulas, and click [**Save**].
3. Navigate to the **Prometheus** tab in the top menu.
4. In the **Uyuni Server** section, enter valid Uyuni API credentials. 确保您输入的身份凭证允许访问您要监控的系统集。
5. 根据需要自定义任何其他配置选项。
6. 单击 [**保存公式**]。
7. 应用 Highstate 并确认它成功完成。
8. Check that the Prometheus interface loads correctly. In your browser, navigate to the URL of the server where Prometheus is installed, on

port 9090 (for example, http://example.com:9090).

有关监控公式的详细信息，请参见 **Specialized Guides › Salt › Salt Formula Monitoring**。

过程：手动安装和配置 Prometheus

1. On the monitoring server, install the **golang-github-prometheus-prometheus** package:

```
zypper in golang-github-prometheus-prometheus
```

2. 启用 Prometheus 服务：

```
systemctl enable --now prometheus
```

3. Check that the Prometheus interface loads correctly. In your browser, navigate to the URL of the server where Prometheus is installed, on port 9090 (for example, http://example.com:9090).

4. Open the configuration file at **/etc/prometheus/prometheus.yml** and add this configuration information. Replace **server.url** with your Uyuni server URL and adjust **username** and **password** fields to match your Uyuni credentials.

```
# {productname} self-health metrics
scrape_configs:
- job_name: 'mgr-server'
  static_configs:
    - targets:
      - 'server.url:9100'  # Node exporter
      - 'server.url:9187'  # PostgreSQL exporter
      - 'server.url:5556'  # JMX exporter (Tomcat)
      - 'server.url:5557'  # JMX exporter (Taskomatic)
      - 'server.url:9800'  # Taskomatic
    - targets:
      - 'server.url:80'    # Message queue
      labels:
        __metrics_path__: /rhn/metrics

# Managed systems metrics:
- job_name: 'mgr-clients'
  uyuni_sd_configs:
   - server: "http://server.url"
     username: "admin"
     password: "admin"
  relabel_configs:
   - source_labels: [__meta_uyuni_exporter]
     target_label: exporter
   - source_labels: [__address__]
     replacement: "No group"
     target_label: groups
   - source_labels: [__meta_uyuni_groups]
     regex: (.+)
```

```
        target_label: groups
      - source_labels: [__meta_uyuni_minion_hostname]
        target_label: hostname
      - source_labels: [__meta_uyuni_primary_fqdn]
        regex: (.+)
        target_label: hostname
      - source_labels: [hostname, __address__]
        regex: (.*);.*:(.*)
        replacement: ${1}:${2}
        target_label: __address__
      - source_labels: [__meta_uyuni_metrics_path]
        regex: (.+)
        target_label: __metrics_path__
      - source_labels: [__meta_uyuni_proxy_module]
        target_label: __param_module
      - source_labels: [__meta_uyuni_scheme]
        target_label: __scheme__
```

5. 保存该配置文件。
6. 重启 Prometheus 服务：

```
systemctl restart prometheus
```

有关 Prometheus 配置选项的详细信息，请参见 https://prometheus.io/docs/prometheus/latest/configuration/configuration/ 上的 Prometheus 官方文档。

17.3.2. 安装 Grafana

如果监视服务器是 Uyuni 管理的客户端，则您可以使用 Uyuni Web UI 安装 Grafana 软件包。否则，可以在监视服务器上手动下载并安装该软件包。

不可以在 Uyuni Proxy 上使用 Grafana。

过程：使用 Web UI 安装 Grafana

1. In the Uyuni Web UI, open the details page of the system where Grafana is to be installed, and navigate to the **Formulas** tab.
2. Check the **Grafana** checkbox to enable monitoring formulas, and click [**Save**].
3. Navigate to the **Grafana** tab in the top menu.
4. In the **Enable and configure Grafana** section, enter the admin credentials you want to use to log in Grafana.
5. On the **Datasources** section, make sure that the Prometheus URL field points to the system where Prometheus is running.

6. 根据需要自定义任何其他配置选项。
7. 单击 **[保存公式]**。
8. 应用 Highstate 并确认它成功完成。
9. Check that the Grafana interface is loading correctly. In your browser, navigate to the URL of the server where Grafana is installed, on port 3000 (for example, http://example.com:3000).
   - 请确保防火墙已开放 3000 端口，以顺利访问 Grafana

> Uyuni 为服务器自我运行状况监控指标、基本客户端监控等信息提供预构建的仪表板。您可以在公式配置页面中选择要置备的仪表板。

过程：手动安装 Grafana

1. Install the **grafana** package:

```
zypper in grafana
```

2. 启用 Grafana 服务：

```
systemctl enable --now grafana-server
```

3. In your browser, navigate to the URL of the server where Grafana is installed, on port 3000 (for example, http://example.com:3000).
   - 请确保防火墙已开放 3000 端口，以顺利访问 Grafana。
4. On the login page, enter **admin** for username and password.
5. 单击 **[登录]**。如果成功登录，您将会看到要求您更改口令的提示。
6. 针对提示单击 **[确定]**，然后更改您的口令。
7. 将光标移到边栏菜单中的齿轮图标上，配置选项即会显示。
8. 单击 **[数据源]**。
9. 单击 **[添加数据源]** 会看到支持的所有数据源列表。
10. 选择 Prometheus 数据源。
11. 务必指定正确的 Prometheus 服务器 URL。
12. 单击 **[保存并测试]**。

13. 要导入仪表板，请单击边栏菜单中的 **[+]** 图标，然后单击 **[导入]**。
14. For Uyuni server overview load the dashboard ID: **17569**.
15. For Uyuni clients overview load the dashboard ID: **17570**.

- 有关监控公式的详细信息，请参见 **Specialized Guides › Salt › Salt Formula Monitoring**。
- 有关如何手动安装和配置 Grafana 的详细信息，请参见 https://grafana.com/docs。

17.4. 配置 Uyuni 监控

在 Uyuni 4 和更高版本中，您可以让服务器公开 Prometheus 自我运行状况监控指标，并可以在客户端系统上安装和配置导出器。

17.4.1. 服务器自我监控

服务器自我运行状况监控指标涵盖了硬件、操作系统和 Uyuni 内部组件。这些指标是结合 Prometheus 导出器通过 Java 应用程序检测提供的。

Uyuni 服务器随附了以下导出器：

- Node exporter: **golang-github-prometheus-node_exporter**.
  - 请参见 https://github.com/prometheus/node_exporter。
- PostgreSQL exporter: **prometheus-postgres_exporter**.
  - 请参见 https://github.com/wrouesnel/postgres_exporter。
- JMX exporter: **prometheus-jmx_exporter**.
  - 请参见 https://github.com/prometheus/jmx_exporter。

Uyuni Proxy 随附了以下导出器软件包：

* Node exporter: **golang-github-prometheus-node_exporter**.
    * 请参见 https://github.com/prometheus/node_exporter。
* Squid exporter: **golang-github-boynux-squid_exporter**.
    * 请参见 https://github.com/boynux/squid-exporter。

导出器已预安装在 Uyuni 服务器和代理中，但其各自的 systemd 守护程序默认已禁用。

过程：启用自我监控

1. 在 Uyuni Web UI 中，导航到**管理 › 管理器配置 › 监控**。
2. 单击 **[启用服务]**。
3. 重启 Tomcat 和 Taskomatic。
4. Navigate to the URL of your Prometheus server, on port 9090 (for example, http://example.com:9090)
5. In the Prometheus UI, navigate to **Status › Targets** and confirm that all the endpoints on the **mgr-server** group are up.
6. 如果您还使用 Web UI 安装了 Grafana，则可以通过 Uyuni 服务器仪表板上的**管理 › 管理器配置 › 监视**查看服务器深入信息。

只能使用 Web UI 启用服务器自我运行状况监控。Prometheus 不会自动收集代理的指标。要在代理上启用自我运行状况监控，需要手动安装导出器并启用它们。

在 Uyuni 服务器上收集以下相关指标。

表格 5. 服务器统计数据（端口 80）

指标	类型	说明
uyuni_all_systems	计量器	所有系统数
uyuni_virtual_systems	计量器	虚拟系统数
uyuni_inactive_systems	计量器	非活动系统数
uyuni_outdated_systems	计量器	安装了过时软件包的系统数

表格 6. PostgreSQL 导出器（端口 9187）

指标	类型	说明
pg_stat_database_tup_fetched	计数器	查询提取的行数
pg_stat_database_tup_inserted	计数器	查询插入的行数

指标	类型	说明
pg_stat_database_tup_updated	计数器	查询更新的行数
pg_stat_database_tup_deleted	计数器	查询删除的行数
mgr_serveractions_completed	计量器	完成的操作数
mgr_serveractions_failed	计量器	失败的操作数
mgr_serveractions_picked_up	计量器	拾取的操作数
mgr_serveractions_queued	计量器	排队的操作数

表格 7. JMX 导出器（Tomcat 端口 5556、Taskomatic 端口 5557）

指标	类型	说明
java_lang_Threading_ThreadCount	计量器	活动线程数
java_lang_Memory_HeapMemoryUsage_used	计量器	当前堆内存使用量

表格 8. 服务器消息队列（端口 80）

指标	类型	说明
message_queue_thread_pool_threads	计数器	到目前为止创建的消息队列线程数
message_queue_thread_pool_threads_active	计量器	当前处于活动状态的消息队列线程数
message_queue_thread_pool_task_count	计数器	到目前为止提交的任务数
message_queue_thread_pool_completed_task_count	计数器	到目前为止完成的任务数

表格 9. Salt 队列（端口 80）

指标	类型	说明
salt_queue_thread_pool_size	计量器	每个 Salt 队列创建的线程数
salt_queue_thread_pool_active_threads	计量器	每个队列当前处于活动状态的 Salt 线程数
salt_queue_thread_pool_task_total	计数器	每个队列已提交的任务数
salt_queue_thread_pool_completed_task_total	计数器	每个队列已完成的任务数

每个 salt_queue 值都有一个名为 **queue** 值为队列编号的标签。

表格 10. Taskomatic 日程安排程序（端口 9800）

指标	类型	说明
taskomatic_scheduler_threads	计数器	到目前为止创建的日程安排程序线程数
taskomatic_scheduler_threads_active	计量器	当前处于活动状态的日程安排程序线程数
taskomatic_scheduler_completed_task_count	计数器	到目前为止完成的任务数

17.4.2. 监控受管系统

可以使用公式在 Salt 客户端上安装和配置 Prometheus 指标导出器。这些软件包可以从 Uyuni 客户端工具通道获得，并且可以直接在 Uyuni Web UI 中启用和配置。

可将这些导出器安装在受管系统上：

- Node exporter: **golang-github-prometheus-node_exporter**.
  - 请参见 https://github.com/prometheus/node_exporter。
- PostgreSQL exporter: **prometheus-postgres_exporter**.
  - 请参见 https://github.com/wrouesnel/postgres_exporter。
- Apache exporter: **golang-github-lusitaniae-apache_exporter**.
  - 请参见 https://github.com/Lusitaniae/apache_exporter。

> 在 SL Micro 上，只能使用 Node Exporter 和 Blackbox Exporter。

安装并配置导出器后，您可以开始使用 Prometheus 从受监视系统收集指标。如果您已使用 Web UI 配置了监视服务器，则会自动收集指标。

过程：在客户端上配置 Prometheus 导出器

1. 在 Uyuni Web UI 中，打开要监视的客户端的细节页面，并导航到**公式**选项卡。
2. Check the **Enabled** checkbox on the **Prometheus Exporters** formula.
3. 单击 **[保存]**。
4. 导航到**公式 › Prometheus 导出器**选项卡。
5. 选择要启用的导出器并根据需要自定义参数。 The **Address** field accepts either a port number preceded by a colon (**:9100**), or a fully resolvable address (**example:9100**).
6. 单击 **[保存公式]**。

7. 应用 Highstate。

防火墙需启用以下端口，且不得拦截这些端口的通信：

- 监控端：9090/tcp
- 被监控端：9100/tcp、9117/tcp、9187/tcp

还可为系统组配置监控公式，只需应用相应组中各个系统使用的同一种配置即可。

有关监控公式的详细信息，请参见 **Specialized Guides › Salt › Salt Formula Monitoring**。

17.4.3. 更改 Grafana 口令

要更改 Grafana 口令，请按照 Grafana 文档中所述的步骤操作：

- https://grafana.com/docs/grafana/latest/administration/user-management/user-preferences/#change-your-grafana-password

In case you have lost the Grafana administrator password you can reset it as **root** with the following command:

```
grafana-cli --configOverrides cfg:default.paths.data=/var/lib/grafana --homepath
/usr/share/grafana admin reset-admin-password <新口令>
```

17.5. 网络边界

Prometheus 使用拉取机制提取指标，因此服务器必须能够与受监控客户端建立 TCP 连接。Prometheus 默认使用以下端口：

- Node 导出器：9100
- PostgreSQL 导出器：9187
- Apache 导出器：9117

此外，如果您不是在运行 Prometheus 的同一台主机上运行警报管理器，则还需要在所在主机上打开端口 9093。警报管理器是 Prometheus 解决方案的一部分。它处理客户端应用程序（如 Prometheus 服务器实例）发送的警报。有关警报管理器的详细信息，请参见 https://prometheus.io/docs/alerting/latest/alertmanager/。

对于安装在云实例上的客户端，可以将所需的端口添加到有权访问监控服务器的安全组中。

或者，可以在导出器的本地网络中部署 Prometheus 实例，并配置联合。这样，主监控服务器就能从本地 Prometheus 实例中抓取时序。如果您使用此方法，只需要打开 Prometheus API 端口 (9090) 即可。

有关 Prometheus 联合的详细信息，请参见 https://prometheus.io/docs/prometheus/latest/federation/。

您还可以通过网络边界来中转请求。PushProx 等工具会在网络屏障的两端部署代理和客户端，并允许 Prometheus 跨网络拓扑（例如 NAT）工作。

有关 PushProx 的详细信息，请参见 https://github.com/RobustPerception/PushProx。

17.5.1. 反向代理设置

Prometheus 使用拉取机制提取指标，因此服务器必须能够与受监控客户端上的每个导出器建立 TCP 连接。为了简化防火墙配置，可为导出器使用反向代理，以便在单个端口上公开所有指标。

过程：安装使用反向代理的 Prometheus 导出器

1. In the Uyuni Web UI, open the details page of the system to be monitored, and navigate to the **Formulas** tab.
2. Check the **Prometheus Exporters** checkbox to enable the exporters formula, and click **[Save]**.
3. Navigate to the **Prometheus Exporters** tab in the top menu.
4. Check the **Enable reverse proxy** option, and enter a valid reverse proxy port number. For example, **9999**.
   - 请确保防火墙已开放 9999/tcp 端口。
5. 根据需要自定义其他导出器。
6. 单击 **[保存公式]**。
7. 应用 Highstate 并确认它成功完成。

有关监控公式的详细信息，请参见 **Specialized Guides › Salt › Salt Formula Monitoring**。

17.6. 安全

Prometheus 服务器和 Prometheus Node 导出器提供内置机制用于通过 TLS 加密和身份验证来保护其端点。Uyuni Web UI 简化了所有相关组件的配置。TLS 证书必须由用户提供和部署。在 Uyuni 中可以启用以下安全模型：

- Node 导出器：TLS 加密和基于客户端证书的身份验证
- Prometheus：TLS 加密和基本身份验证

有关配置所有可用选项的详细信息，请参见 **Specialized Guides › Salt › Salt Formula Monitoring**。

17.6.1. 生成 TLS 证书

默认情况下，Uyuni 不提供任何用于保护监控配置的证书。为了提供安全性，您可以生成或导入自定义证书，

包括自我签名证书或由第三方证书颁发机构 (CA) 签名的证书。

本节说明如何为使用 Uyuni CA 自我签名的 Prometheus 和 Node 导出器受控端生成客户端/服务器证书。

过程：创建服务器/客户端 TLS 证书

1. 在 Uyuni 容器主机的命令行中，以 root 身份运行以下命令：
   a. 运行以下命令生成证书文件。

      Ensure that the **set-cname** parameter is the fully qualified domain name (FQDN) of your Salt client. You can use the **set-cname** parameter multiple times if you require multiple aliases:

      ```
      mgrctl exec -ti -- mgr-ssl-tool --gen-server --dir="/root/ssl-build" --set
      -country="COUNTRY" \
        --set-state="STATE" --set-city="CITY" --set-org="ORGANIZATION" \
        --set-org-unit="ORGANIZATION UNIT" --set-email="name@example.com" \
        --set-hostname="minion.example.com" --set-cname="minion.example.com" --no
      -rpm
      ```

      命令执行的结果：

      ```
      Generating the web server's SSL private key: /root/ssl-
      build/minion/server.key
      Generating web server's SSL certificate request: /root/ssl-
      build/minion/server.csr
      Generating/signing web server's SSL certificate: server.crt
      ```

   b. Copy **server.crt** and **server.key** files from the server container to the host:

      ```
      mgrctl cp server:/root/ssl-build/minion/server.key server.key
      mgrctl cp server:/root/ssl-build/minion/server.crt server.crt
      ```

   c. Copy **server.crt** and **server.key** files from the host to the monitoring client:

      ```
      ssh minion.example.com 'mkdir /etc/ssl/mlm-server-certs'
      scp /root/server.* minion.example.com:/etc/ssl/mlm-server-certs
      ssh minion.example.com 'chmod go+r /etc/ssl/mlm-server-certs/server.*; ls
      -la /etc/ssl/mlm-server-certs'
      ```

2. 要配置 Salt 公式，请输入之前的步骤中指定的目录名称。
   a. formular server

```
Server Certificate /etc/ssl/mlm-server-certs/server.crt
Server Key /etc/ssl/mlm-server-certs/server.key
```

b. 公式 受控端

章 18. 组织

组织用于管理 Uyuni 中的用户访问权限和许可权限。

对于大多数环境而言，一个组织便已足够。但是，较复杂的环境可能需要多个组织。您可以为企业中的每个物理位置或不同的业务职能创建一个组织。

创建组织后，可以创建用户并将其指派到组织。然后，可以在组织级别指派权限，这些权限默认将应用于指派到组织的每个用户。

您还可以为新组织配置身份验证方法，包括 PAM 和单点登录。有关身份验证的详细信息，请参见 **Administration Guide › Auth Methods**。

> 您必须以 Uyuni 管理员身份登录才能创建和管理组织。

过程：创建新组织

1. 在 Uyuni Web UI 中，导航到**管理 › 组织**，然后单击 **[创建组织]**。
2. In the **Create Organization** dialog, complete these fields:
   - In the **Organization Name** field, type a name for your new organization. 该名称的长度应为 3 到 128 个字符。
   - In the **Desired Login** field, type the login name you want to use for the organization’s administrator. 这必须是一个新的管理员帐户，您无法使用现有的管理员帐户（包括当前使用的登录帐户）登录到新组织。
   - In the **Desired Password** field, type a password for the new organization’s administrator. Confirm the password by typing it again in the **Confirm Password** field. Password strength is indicated by the colored bar beneath the password fields.
   - In the **Email** field, type an email address for the new organization’s administrator.
   - In the **First Name** field, select a salutation, and type a given name for the new organization’s administrator.
   - In the **Last Name** field, type a surname for the new organization’s administrator.
3. 单击 **[创建组织]**。

18.1. 管理组织

在 Uyuni Web UI 中，导航到**管理 › 组织**可查看可用组织的列表。单击某个组织的名称可对其进行管理。

在**管理 › 组织**部分，可以访问相应的选项卡来管理组织的用户、信任、配置和状态。

> 组织只能由其管理员管理。要管理组织，请确保以您要更改的组织的正确管理员身份登录。

18.1.1. 组织用户

Navigate to the **Users** tab to view the list of all users associated with the organization, and their role. Clicking a username takes you to the **Users** menu to add, change, or delete users.

18.1.2. 可信组织

Navigate to the **Trusts** tab to add or remove trusted organizations. Establishing trust between organizations allow them to share content between them, and gives you the ability to transfer clients from one organization to another.

18.1.3. 配置组织

Navigate to the **Configuration** tab to manage the configuration of your organization. This includes the use of staged contents, and the use of SCAP files.

有关内容暂存的详细信息，请参见 **Administration Guide › Content Staging**。

有关 OpenSCAP 的详细信息，请参见 **Reference Guide › Audit › Audit Openscap Overview**。

18.2. 管理状态

Navigate to the **States** tab to manage Salt states for all clients in your organization. States allow you to define global security policies, or add a common admin user to all clients.

有关 Salt 状态的详细信息，请参见 **Specialized Guides › Salt › Salt States and Pillars**。

18.2.1. 管理配置通道

可以选择要在整个组织中应用哪些配置通道。可以在 Uyuni Web UI 中导航到**配置 › 通道**来创建配置通道。使用 Uyuni Web UI 将配置通道应用于您的组织。

过程：将配置通道应用于组织

1. 在 Uyuni Web UI 中，导航到**首页 › 我的组织 › 配置通道**。
2. 使用搜索功能按名称查找通道。
3. 选中要应用的通道，然后单击 **[保存更改]**。 这会将更改保存到数据库，但不会将其应用于通道。
4. 单击 **[应用]** 以应用更改。 随即会安排将更改应用于组织中的所有客户端的任务。

章 19. 补丁管理

本章包含有关补丁管理的各个主题。

19.1. 已收回补丁

在供应商发布某个新补丁后，该补丁在某种情况下可能会产生不利的负面影响（在安全性、稳定性方面），而这种情况并未通过测试识别出来。如果出现这种情况（非常罕见），供应商通常会发布新补丁，此过程可能需要数小时甚至数日时间，具体取决于该供应商实施的内部流程。

SUSE 引入了一个称为**已收回补丁**的新机制 (2021)，可以通过将此类补丁的建议状态设置为**已收回**（而不是**最终**或**稳定**）来近实时地撤消这些补丁。

> 如果某个补丁的建议状态属性设置为 **retracted**，则会**收回**该补丁。如果某个软件包属于**已收回**的补丁，则会**收回**该软件包。

A retracted patch or package cannot be installed on systems with Uyuni. The only way to install a retracted package, is to do it manually with **zypper install** and specifying the exact package version. For example:

```
zypper install vim-8.0.1568-5.14.1
```

Uyuni Web UI 中的 ⊗ 图标描绘了补丁和软件包的已收回状态。例如，查看：

- 通道中软件包的列表
- 通道中补丁的列表

收回安装在系统上的补丁或软件包时，⊗ 图标也会显示在该系统的已安装软件包列表中。Uyuni 不提供降级此类补丁或软件包的方法。

19.1.1. 通道克隆

使用克隆的通道时，必须小心将原始通道的已收回建议状态传播到克隆版本。

将供应商通道克隆到组织中时，也会克隆通道补丁。

当供应商收回通道中的补丁并且 Uyuni 同步此通道（例如，通过夜间作业）时，**retracted** 属性不会传播到克隆的补丁，并且订阅克隆通道的客户端看不到该属性。要将该属性传播到克隆的通道，请使用以下方法之一：

- 补丁同步（**软件 › 管理 › 克隆的通道 › 补丁 › 同步**）。使用此功能可使克隆通道中补丁的属性与其原始通道保持一致。
- 内容生命周期管理。有关内容生命周期管理上下文中克隆的通道的详细信息，请参见 **Client Configuration Guide › Channels**。

19.1.2. 补丁共享

当您在组织中创建多个供应商通道克隆版本时，补丁不会克隆多次，而是在克隆的通道之间共享。因此，当您同步克隆的补丁时（使用补丁同步功能或上面提到的内容生命周期管理），使用该克隆补丁的所有通道都会看到这种变化。

示例：

1. Consider two Content Lifecycle Management projects **prj1** and **prj2**
2. Both of these projects have 2 environments **dev** and **test**
3. 这两个项目都有一个设置为源通道的供应商通道
4. 此场景中的所有通道（总共四个克隆的通道）与供应商通道的最新状态相一致
5. 供应商收回源通道中的某个补丁，夜间作业将此补丁同步到 Uyuni
6. 四个通道都看不到这种变化，因为它们使用的是补丁克隆版本，而不是直接使用该补丁。
7. 在您同步补丁（构建这两个项目中的任何一个，或者对四个克隆通道中的任何一个使用补丁同步功能）后，由于补丁共享的原因，**所有**克隆的通道都会发现该补丁已收回。

章 20. 在 Uyuni 中使用 PTF

SUSE 为目前支持的所有解决方案提供直接交付给客户的临时修复。这些 PTF（程序临时修复）现在以储存库的形式提供，后者可在 Uyuni 中同步。

20.1. 了解 PTF 软件包

PTF 软件包通过代理软件包安装，命名为 **ptf-xxxxxx**。其中 xxxxxx 是软件包的编号和名称部分，而不是版本。

它们取决于已知包含软件中的修正的软件包正确版本。这种软件包：

- 不可能意外安装（即 zypper 更新绝不会建议安装它们），
- 不可能意外去除（即更新的软件包版本不会替换 PTF 软件包，除非用户在 zypper 命令行上明确指示替换），
- 仅会在已知有更新版本可解决该 PTF 之前所解决的特定问题时更新，
- 仅会在系统上已安装软件包时更新（也就是说，如果软件拆分成多个软件包，该 PTF 仅会替换系统上目前安装的那些软件包）。

软件包的正确 ID 将由 SUSE 支持团队在进行支持案例调查期间提供，同时还会提供有关如何部署/重启受影响服务的说明。

20.2. 安装 PTF 软件包

PTF 软件包目前仅受基于 SLE 12 和 SLE 15 的系统支持。其他版本或操作系统不提供此功能，因此未显示与其对应的页面。

如需通过 Uyuni 访问 PTF 通道，您需要向三级技术支持提交申请。

过程：使用命令行启用和同步 PTF 储存库

1. On the console enter **mgr-sync refresh**.
2. Enter **mgr-sync list channel** and look for channels starting with your SCC account name and **ptfs** in its name. For example, **a123456-sles-15.3-ptfs-x86_64**.
3. Enable the PTF channel with **mgr-sync add channel <label>**.

此通道现在便可供使用，并可添加到使用相同基础通道的每个系统。

您需要明确安装 PTF 软件包，因为在更新系统时它们不会自动被选中。SUSE 客户支持团队将提供用于修复特定问题的 PTF 编号。可以根据该编号来识别 PTF 列表中的代理软件包。在 Uyuni Web UI 中，会针对有可供安装的 PTF 的每个系统显示一个页面来列出这些 PTF。

过程：通过 Uyuni Web UI 启用和同步 PTF 储存库

1. 在 Uyuni Web UI 中，导航到**管理 › 安装向导 › 产品**，然后查找要为其启用 PTF 储存库的产品。

2. 单击产品同步状态旁边的 **[显示产品的通道]**。
3. 您应该会看到一个弹出窗口，其中会列出该产品的必需和可选通道。
4. 在可选通道列表中，查找以您的 SCC 帐户名称开头且名称中包含 **ptfs** 的通道。例如 **a123456-sles-15.3-ptfs-x86_64**。
5. 使用该通道名称旁边的复选框将其选中，然后单击 **[确认]** 安排同步。

请注意，必须安装该产品才能为其添加可选通道。

过程：安装 PTF 软件包

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表**，然后选择要安装 PTF 的客户端。
2. Navigate to the **Systems › Software › Packages › Software Channels** and select the **PTF channel**.
3. Click **[Next]**, and **Confirm Software Channel Change** with **[Confirm]**.
4. 要检查通道指派是否已完成，请导航到**系统 › 事件 › 历史记录**查看结果。
5. 导航到**系统 › 软件 › PTF › 安装**子选项卡。
6. 选择要安装的 PTF 软件包。
7. Click **[Install PTF]**, and **Confirm Program Temporary Fixes (PTFs) Installation** with **[Confirm]**.
8. 要查看 PTF 安装结果，请导航到**系统 › 事件 › 历史记录**。

In case a PTF should be installed using the API, the normal **system.schedulePackageInstall** API can be used with the proxy package name.

20.3. 安装 PTF 后

一旦确认使用某个 PTF 来解决报告的问题，在将更新的软件包作为更新储存库中的常规维护更新广泛分发之前，需对其进行跟踪，以便在将来的维护更新中纳入该软件包。

在发布包含修复的此常规更新时，还会将 PTF 的一个更新版本发布到特定于帐户的 PTF 储存库中。更新的 PTF 将会解除严格的依赖关系，并允许再次安装更新。

通过标准软件包更新或补丁安装可自动将 PTF 替换为包含该修复的维护更新。

20.4. 去除软件包的已修补版本

如果需要在系统上卸装某个 PTF 并安装软件包的未修补版本，仅执行软件包去除流程无法实现这个目标。在标准软件包列表页面中，无法选择该 PTF 软件包。

过程：去除 PTF 软件包

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表**，然后选择要去除 PTF 的客户端。
2. 导航到**系统 › 软件 › PTF › 列出/去除**子选项卡。

3. 选择要去除的 PTF 软件包。
4. Click [**Remove PTFs**], and on the **Confirm Program Temporary Fixes (PTFs) Removal** page click [**Confirm**].
5. 要查看结果，请导航到**系统 › 事件 › 历史记录**。

Removing PTFs requires a special version of **libzypp** and **zypper** to be installed on the client system. Check **zypper --help** to confirm whether **removeptf** is supported. The **List/Remove** tab is only visible if this condition is met.

In case the PTF should be removed using the API, the normal **system.schedulePackageRemove** API can be used with the proxy package name.

20.5. 在客户端上去除软件包的已修补版本

In case a PTF should be removed directly on the client using the console, it is required to use a special command **zypper removeptf**. All other ways result either in an error or can lead to unwanted behavior like removing important packages from the system and make the system unusable.

章 21. 生成报告

Uyuni 允许用户生成各种报告。这些报告有助于清点已订阅的系统、用户和组织。使用报告通常比从 Uyuni Web UI 手动收集信息更方便，尤其是要管理众多的系统时。

While the command line tool **spacewalk-report** can be used to generate pre-configured reports, with the introduction of the **Administration Guide › Reporting Database** it is also possible to generate fully customized reports. This can be achieved by connecting any reporting tool that supports the SQL language to the reporting database and extract the data directly. For more information about the data availability and structure, see the reporting database schema documentation.

21.1. Using spacewalk-report

> Use **mgrctl term** before running steps inside the server container.

To generate reports, you must have the **spacewalk-reports** package installed. The **spacewalk-report** command allows you to organize and display reports about content, systems, and user resources across Uyuni.

> Due to the introduction of the **Administration Guide › Reporting Database**, **spacewalk-report** now gathers by default the data from the reporting database. See spacewalk-report and the reporting database for more information.

可以针对以下各项生成报告：

系统清单

列出所有已注册到 Uyuni 的系统。

补丁

列出与已注册系统相关的所有补丁。可以按严重性以及适用于特定补丁的系统将补丁排序。

用户

列出所有已注册用户以及与特定用户关联的所有系统。

要获取 CSV 格式的报告，请在服务器上的命令提示符下运行以下命令：

```
spacewalk-report <报告名称>
```

21.2. spacewalk-report and the reporting database

spacewalk-report uses by default the new reporting database to extract the data. This means that the new generated reports will have some differences in the structure and the format of the data. The differences common to all reports are:

- 报告数据不会实时更改，而只能通过执行安排的任务来更新；

- data duplication has been removed and columns that were previously considered "multival" contain now multiple values separated by **;**. This also means that the command line options **--multival-on-rows** and **--multival-separator** are no longer applicable to the new reports, as their behavior is now the default;
- in all reports the new columns **mgm_id** and **synced_date** have been introduced to identify the management server in the hub scenario and the last time the information was updated from the application database;
- all boolean values are now represented by the **True**/**False** and not by a **1**/**0** values;
- the column **org_id** has been replaced by **organization**, which contains the organization name and not the numerical identifier;
- The term "server" has been replaced by "system." So, for example, the column **server_id** is now called **system_id**.

有关特定于报告的更改，请参见可用报告列表。

> If this changed behavior causes trouble, the new option **--legacy-report** can be used to fall back to the old report, which is executed against the application database.

For more information about the reporting database, see **Administration Guide › Reporting Database**. For hub-specific aggregation of data from multiple servers, see **Specialized Guides › Large Deployments › Hub Reporting**.

21.3. 可用报告列表

下表列出了可用报告：

表格 11. spacewalk-report Reports

Report	Invoked as	Description	Uses reporting database	Specific differences
Actions	**actions**	All actions.	Yes	The column **id** is now called **action_id**
Activation Keys	**activation-keys**	All activation keys, and the entitlements, channels, configuration channels, system groups, and packages associated with them.	No	

Report	Invoked as	Description	Uses reporting database	Specific differences
Activation Keys: Channels	**activation-keys-channels**	All activation keys and the entities associated with each key.	No	
Activation Keys: Configuration	**activation-keys-config**	All activation keys and the configuration channels associated with each key.	No	
Activation Keys: Server Groups	**activation-keys-groups**	All activation keys and the system groups associated with each key.	No	
Activation Keys: Packages	**activation-keys-packages**	All activation keys and the packages each key can deploy.	No	
Channel Packages	**channel-packages**	All packages in a channel.	Yes	
Channel Report	**channels**	Detailed report of a given channel.	Yes	
Cloned Channel Report	**cloned-channels**	Detailed report of cloned channels.	Yes	
Configuration Files	**config-files**	All configuration file revisions for all organizations, including file contents and file information.	No	
Latest Configuration Files	**config-files-latest**	The most recent configuration file revisions for all organizations, including file contents and file information.	No	

Report	Invoked as	Description	Uses reporting database	Specific differences
Custom Channels	**custom-channels**	Channel metadata for all channels owned by specific organizations.	Yes	The column **id** is now called **channel_id**
Custom Info	**custom-info**	Client custom information.	Yes	
Patches in Channels	**errata-channels**	All patches in channels.	Yes	
Patches Details	**errata-list**	All patches that affect registered clients.	Yes	
All patches	**errata-list-all**	All patches.	No	
Patches for Clients	**errata-systems**	Applicable patches and any registered clients that are affected.	Yes	
Host Guests	**host-guests**	Host and guests mapping.	Yes	
Inactive Clients	**inactive-systems**	Inactive clients.	Yes	The mandatory parameter is now called **threshold**.
System Inventory	**inventory**	Clients registered to the server, together with hardware and software information.	Yes	The column **osad_status** has been removed.
Kickstart Scripts	**kickstart-scripts**	All kickstart scripts, with details.	No	
Kickstart Trees	**kickstartable-trees**	Kickstartable trees.	No	"
All Upgradable Versions	**packages-updates-all**	All newer package versions that can be upgraded.	Yes	
Newest Upgradable Version	**packages-updates-newest**	Newest package versions that can be upgraded.	Yes	

Report	Invoked as	Description	Uses reporting database	Specific differences
Proxy Overview	**proxies-overview**	All proxies and the clients registered to each.	Yes	
Repositories	**repositories**	All repositories, with their associated SSL details, and any filters.	No	
Result of SCAP	**scap-scan**	Result of OpenSCAP **sccdf** evaluations.	Yes	
Result of SCAP	**scap-scan-results**	Result of OpenSCAP **sccdf** evaluations, in a different format.	Yes	
System Data	**splice-export**	Client data needed for splice integration.	No	
System Currency	**system-currency**	Number of available patches for each registered client.	No	
System Extra Packages	**system-extra-packages**	All packages installed on all clients that are not available from channels the client is subscribed to.	Yes	
System Groups	**system-groups**	System groups.	Yes	
Activation Keys for System Groups	**system-groups-keys**	Activation keys for system groups.	No	
Systems in System Groups	**system-groups-systems**	Clients in system groups.	Yes	
System Groups Users	**system-groups-users**	System groups and users that have permissions on them.	No	
System Hardware	**system-hardware**	System hardware information.	No	

Report	Invoked as	Description	Uses reporting database	Specific differences
History: System	**system-history**	Event history for each client.	Yes	
History: Channels	**system-history-channels**	Channel event history.	Yes	
History: Configuration	**system-history-configuration**	Configuration event history.	Yes	The column **created_date** has been removed.
History: Entitlements	**system-history-entitlements**	System entitlement event history.	Yes	
History: Errata	**system-history-errata**	Errata event history.	Yes	The column **created_date** has been removed.
History: Kickstart	**system-history-kickstart**	Kickstart event history.	Yes	The column **created_date** has been removed.
History: Packages	**system-history-packages**	Package event history.	Yes	The column **created_date** has been removed.
History: SCAP	**system-history-scap**	OpenSCAP event history.	Yes	The column **created_date** has been removed.
MD5 Certificates	**system-md5-certificates**	All registered clients using certificates with an MD5 checksum.	No	
Installed Packages	**system-packages-installed**	Packages installed on clients.	Yes	
System Profiles	**system-profiles**	All clients registered to the server, with software and system group information.	No	
Users	**users**	All users registered to Uyuni.	Yes	The column **organization_id** has been removed.

Report	Invoked as	Description	Uses reporting database	Specific differences
MD5 Users	**users-md5**	All users for all organizations using MD5 encrypted passwords, with their details and roles.	Yes	The column **organization_id** has been removed.
Systems administered	**users-systems**	Clients that individual users can administer.	Yes	The column **organization_id** has been removed.

For more information about an individual report, run **spacewalk-report** with the option **--info** or **--list-fields-info** and the report name. This shows the description and list of possible fields in the report.

For further information on program invocation and options, see the **spacewalk-report(8)** man page as well as the **--help** parameter of the **spacewalk-report** command.

21.4. Reporting Database

Uyuni includes a reporting database on every server. The reporting database is a separate PostgreSQL database that stores a de-normalized, read-optimized snapshot of the most relevant data from the main Uyuni database. External reporting tools can connect directly to this database to query and visualize data.

In a hub deployment, the hub aggregates the reporting databases of all connected peripheral servers into a single central reporting database. For hub-specific setup and aggregation, see **Specialized Guides › Large Deployments › Hub Reporting**.

21.4.1. Setup

The reporting database and schema are set up by default using the local PostgreSQL server. The reporting database is a separate database accessible via the network.

> In a hub deployment, the connection between hub and peripheral servers is established via hub online synchronization. For more information, see **Specialized Guides › Large Deployments › Hub Online Sync**.

21.4.1.1. Create a DB User for the Reporting Database

Before connecting an external reporting tool to the database, create a user with read-only permissions. Use the **uyuni-setup-reportdb-user** command for this:

```
usage: uyuni-setup-reportdb-user [options]

options:
  --help
          show this help message and exit
```

```
--non-interactive
          Switches to non-interactive mode
--dbuser=DBUSER
          Report DB User
--dbpassword=DBPASSWORD
          Report DB Password
--add
          Add the new user
--delete
          Delete the user
--modify
          Set a new password
```

21.4.2. Database Schema

The schema exports the most important tables from the main Uyuni database as a de-normalized variant containing only data relevant for reporting.

Ready-to-use reports are provided as views, aggregating data over multiple tables.

Every table has an extra column (**mgm_id**) specifying the Uyuni server which provided the data. On a single Uyuni server this column has the standard value **1**, representing **localhost**. On the Uyuni Hub it is replaced with the real server ID the managed server has in the hub database.

Another common additional field is **synced_date**, which represents the time when the data were exported from the main Uyuni server database.

Navigate to **Help › Report Database Schema** to find the schema documentation.

21.4.3. Data Generation and Update

Uyuni uses two taskomatic jobs to manage reporting data:

update-reporting-default

Generates and updates the local reporting database from the main Uyuni database. Runs by default daily at midnight on every server.

update-reporting-hub-default

Fetches reporting data from all registered peripheral servers and aggregates it into the hub reporting database. This schedule is visible on all servers, but only performs meaningful work on a Uyuni Hub. Runs by default daily at 1:30 AM.

All times are in the local timezone of the server.

For more information about hub aggregation and schedule alignment across timezones, see **Specialized Guides › Large Deployments › Hub Reporting**.

21.4.4. Tuning

21.4.4.1. report_db_batch_size

Description	The maximum number of rows fetched and written from one database to the other in one shot.
Tune when	Out of memory errors or on processing speed problems.
Value default	2000
Value recommendation	500 - 5000
Location	**/etc/rhn/rhn.conf**
Example	**report_db_batch_size = 4000**
After changing	Monitor memory usage closely before and after the change.

21.4.5. Visualizing Reporting Data with Grafana

The Uyuni Grafana formula can connect directly to the reporting database and deploy pre-built dashboards for fleet management and compliance reporting.

Prerequisites:

- A Grafana instance configured using the Monitoring Formula. See **Specialized Guides › Salt › Salt Formula Monitoring**.
- A reporting database user created with **uyuni-setup-reportdb-user** (only required when using external reporting tools, as the Grafana formula automatically creates and configures a dedicated read-only database user). See Create a DB User for the Reporting Database.

The reporting dashboards provided are:

- **Hub Overview** (Hub servers only) — high-level overview of the Hub setup, listing peripheral servers and their fleets.
- **Fleet Overview & Security** — fleet composition, security posture, CVE analysis, patch compliance, proxy infrastructure, and virtualization overview.
- **Reports & History** — channels, packages, errata, actions, system inventory, user management, system history, and SCAP compliance.
- **Executive Overview** — dynamic statistics including system counts and package queries.

For configuration steps, see **Specialized Guides › Salt › Salt Formula Monitoring**.

章 22. 安全

22.1. 审计

在 Uyuni 中，您可以通过一系列审计任务来跟踪客户端。您可以检查客户端上是否安装了所有最新的公共安全补丁 (CVE)，执行订阅匹配，并使用 OpenSCAP 检查合规性。

In the Uyuni Web UI, navigate to **Audit** to perform auditing tasks.

22.1.1. CVE audits

CVE（常见漏洞和披露）是对公开已知安全漏洞的修复方案。

只要有可用的 CVE，就必须在客户端上应用它们。

Each CVE contains an identification number, a description of the vulnerability, and links to further information. CVE identification numbers use the form **CVE-YEAR-XXXX**.

在 Uyuni Web UI 中，导航到**审计 › CVE 审计** 以查看所有客户端及其当前补丁状态的列表。

默认情况下，补丁数据在每天 23:00 更新。我们建议您在开始进行 CVE 审计之前刷新数据，以确保应用最新的补丁。

Procedure: Updating patch data

1. In the Uyuni Web UI, navigate to **Admin › Task Schedules** and select the **cve-server-channels-default** schedule.
2. 单击 **[cve-server-channels-bunch]**。
3. 单击 **[单次运行安排]** 以安排任务。等待该任务完成，然后继续进行 CVE 审计。

Procedure: Verifying patch status

1. 在 Uyuni Web UI 中，导航到**审计 › CVE 审计**。
2. To check the patch status for a particular CVE, type the CVE identifier in the **CVE Number** field.
3. 选择您要查看的补丁状态，或保持选中所有状态以查看所有补丁状态。
4. 单击 **[审计服务器]** 检查所有系统，或单击 **[审计映像]** 检查所有映像。

有关此页面上使用的补丁状态图标的详细信息，请参见 **Reference Guide › Audit › Audit Cve Audit**。

For each system, the **Actions** column provides information about what you need to do to address vulnerabilities. If applicable, a list of candidate channels or patches is also given. You can also assign systems to a **System Set** for further batch processing.

You can use the Uyuni API to verify the patch status of your clients. Use the **audit.listSystemsByPatchStatus** API method. For more information about this method, see the Uyuni API Guide.

22.1.2. OVAL

CVE 审计操作依赖于两个主要数据源：通道和 OVAL（开放漏洞与评估语言）。这两个数据源为 CVE 审计提供元数据，且各自具备独特作用。

通道

通道包含更新后的软件包（包括补丁），并提供针对漏洞修复所需关键补丁的洞察信息。

OVAL

与之相对，OVAL 数据提供漏洞本身的信息，以及导致系统容易因某个 CVE 而受到攻击的软件包。

尽管仅使用通道数据即可进行 CVE 审计，但同步 OVAL 数据可提升结果的准确性，尤其是在处理零日漏洞或部分修补的漏洞时。

OVAL 数据相比通道数据更轻量。例如，openSUSE Leap 15.4 的 OVAL 数据约有 50 MB。

仅同步 OVAL 数据时，您可执行 CVE 审计并检查系统是否容易因某个 CVE 而受到攻击，但无法应用补丁，因为补丁来自通道。

OVAL 功能的核心特性包括：

- **Enabled by default**: The feature is enabled by default and requires no additional configuration.
- **Reversible**: If needed, users can disable OVAL data and revert to the standard channel-based CVE audit. See Disabling OVAL Data Support below.
- OVAL 数据在每天 23:00 更新。我们建议您在开始进行 CVE 审计之前刷新数据，以确保获得最新的漏洞元数据。

The following procedures can be used to enable, disable, or update OVAL data.

Procedure: Enabling OVAL data support

1. Add or modify the following setting in file **/etc/rhn/rhn.conf** in the container:

```
java.cve_audit.enable_oval_metadata=true
```

2. 重启 Tomcat 和 Taskomatic 服务：

```
systemctl restart tomcat taskomatic
```

Alternatively, use the procedure for disabling OVAL data support.

Procedure: Disabling OVAL data support

1. Add or modify the following setting in **rhn.conf**:

   ```
   java.cve_audit.enable_oval_metadata=false
   ```

2. 重启 Tomcat 和 Taskomatic 服务：

   ```
   systemctl restart tomcat taskomatic
   ```

Procedure: Updating OVAL data

1. In the Uyuni Web UI, navigate to **Admin › Task Schedules** and select the **oval-data-sync-default** schedule.
2. 单击 [**oval-data-sync-bunch**]。
3. 单击 [**单次运行安排**] 以安排任务。

等待该任务完成，然后继续进行 CVE 审计。

22.1.2.1. 收集 CPE

To be able to accurately identify what vulnerabilities apply to a certain client, we need to identify the operating system product that client uses. To do that, we collect the CPE (Common Platform Enumeration) of the client as a Salt grain, then we save it to the database.

The CPE of newly registered clients will be automatically collected and saved to the database. However, for existing clients, it is necessary to execute the **Update Packages List** action at least once.

Procedure: Update packages list

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表 › 全部**，然后选择一个客户端。
2. Then go to the **Software** tab and select the **Packages** sub-tab.
3. 单击 [**更新软件包列表**]，更新软件包并收集客户端的 CPE。

22.1.2.2. OVAL 数据源

为确保 OVAL 数据的完整性和时效性，Uyuni 仅使用各产品官方维护者提供的 OVAL 数据。以下是 OVAL 数据源列表：

表格 12. OVAL 数据源

Product	Source URL	OVAL supported versions in Uyuni
openSUSE Leap	https://ftp.suse.com/pub/projects/security/oval	openSUSE Leap 16 openSUSE Leap 15.6
openSUSE Leap Micro	https://ftp.suse.com/pub/projects/security/oval	openSUSE Leap Micro 6.2 openSUSE Leap Micro 6.1 openSUSE Leap Micro 6.0
SUSE Linux Enterprise Server	https://ftp.suse.com/pub/projects/security/oval	SUSE Linux Enterprise Server 16 SUSE Linux Enterprise Server 15 SUSE Linux Enterprise Server 12
SUSE Linux Enterprise Desktop	https://ftp.suse.com/pub/projects/security/oval	SUSE Linux Enterprise Desktop 15
SUSE Linux Enterprise Micro	https://ftp.suse.com/pub/projects/security/oval	SUSE Linux Enterprise Micro 5.5 SUSE Linux Enterprise Micro 5.4 SUSE Linux Enterprise Micro 5.3 SUSE Linux Enterprise Micro 5.2
SUSE Linux Micro	https://ftp.suse.com/pub/projects/security/oval	SUSE Linux Micro 6.2 SUSE Linux Micro 6.1 SUSE Linux Micro 6.0
SUSE Liberty Linux	https://ftp.suse.com/pub/projects/security/oval	SUSE Liberty Linux 10 SUSE Liberty Linux 9 SUSE Liberty Linux 8 SUSE Liberty Linux 7
RedHat Enterprise Linux	https://www.redhat.com/security/data/oval/v2	RedHat Enterprise Linux 9 RedHat Enterprise Linux 8 RedHat Enterprise Linux 7
Debian	https://www.debian.org/security/oval	Debian 13 Raspberry Pi OS
Ubuntu	https://security-metadata.canonical.com/oval	Ubuntu 26.04 Ubuntu 24.04 Ubuntu 22.04

Product	Source URL	OVAL supported versions in Uyuni
AlmaLinux	https://security.almalinux.org/oval	AlmaLinux 10 AlmaLinux 9 AlmaLinux 8
Oracle Linux	https://linux.oracle.com/security/oval/	Oracle Linux 10 Oracle Linux 9 Oracle Linux 8 Oracle Linux 7

OVAL 元数据仅在部分客户端（即使用 openSUSE Leap、SUSE 企业产品、RHEL、Debian 或 Ubuntu 的客户端）的 CVE 审计中使用。这是由于其他产品缺乏 OVAL 漏洞定义元数据。

22.1.3. CVE 状态

The CVE status of clients is usually either **affected**, **not affected**, or **patched**. These statuses are based only on the information that is available to Uyuni.

在 Uyuni 中，以下定义适用：

受特定漏洞影响的系统

系统中安装的某个软件包版本低于标记为漏洞的相关补丁中相同软件包的版本。

不受特定漏洞影响的系统

同时包含在标记为漏洞的相关补丁中的软件包未安装在系统上。

针对某个漏洞进行了修补的系统

系统中安装的某个软件包版本等同于或高于标记为漏洞的相关补丁中相同软件包的版本。

相关补丁

Uyuni 在相关通道中已知的补丁。

相关通道

由 Uyuni 管理的通道，该通道被指派到系统、是指派到系统的克隆通道的原始通道、是链接到系统上安装的产品的通道，或者是系统的过去或将来的服务包通道。

由于 Uyuni 中使用的定义，CVE 审计结果在某些情况下可能不正确。例如，非受管通道、非受管软件包或不合规的系统可能会错误地报告结果。

22.2. 设置用于客户端到主控端验证的指纹

在高度安全的网络配置中，您可能希望确保您的 Salt 客户端连接特定的主控端。要设置从客户端到主控端的验证，首先请在 Salt 受控端配置文件中输入主控端的指纹：

* **/etc/salt/minion.d/custom.conf** in cases of using classic Salt minion in your client, or
* **/etc/venv-salt-minion/minion.d/custom.conf** in case of using Salt Bundle in your client

然后按以下过程进行操作：

> To access a shell inside the Server container run **mgrctl term** on the container host.

过程：将主控端的指纹添加到客户端

1. On the master, at the command prompt, as root, use this command to find the **master.pub** fingerprint:

   ```
   salt-key -F master
   ```

   On your client, open the **/etc/salt/minion.d/custom.conf** or **/etc/venv-salt-minion/minion.d/custom.conf** configuration file. Add this line to enter the master’ s fingerprint replacing the example fingerprint:

   ```
   master_finger: 'ba:30:65:2a:d6:9e:20:4f:d8:b2:f3:a7:d4:65:11:13'
   ```

2. 重启服务。对于 salt-minion，请运行：

   ```
   systemctl restart salt-minion
   ```

3. 对于 venv-salt-minion，请运行：

   ```
   systemctl restart venv-salt-minion
   ```

有关 Salt 捆绑包的详细信息，请参见 **Client Configuration Guide › Contact Methods Saltbundle**。

有关从客户端配置安全性的信息，请参见 https://docs.saltproject.io/en/latest/ref/configuration/minion.html。

22.3. 镜像源软件包

如果您在本地构建自己的软件包，或者出于法律原因需要提供软件包的源代码，可以在 Uyuni Server 上镜像源软件包。

> 镜像源软件包可能会消耗大量磁盘空间。

> Use **mgrctl term** before running steps inside the server container.

过程：镜像源软件包

1. Open the **/etc/rhn/rhn.conf** configuration file, and add this line:

```
server.sync_source_packages = 1
```

2. 重启 Spacewalk 服务以应用更改：

```
mgradm restart
```

目前，只能为所有储存库全局启用此功能。无法选择单个要镜像的储存库。

激活此功能后，源软件包会在下次储存库同步后显示在 Uyuni Web UI 中。它们显示为二进制软件包的源，可以直接从 Web UI 下载。无法使用 Web UI 在客户端上安装源软件包。

22.4. 使用 OpenSCAP 确保系统安全

Uyuni 使用 OpenSCAP 来审计客户端。它允许您为任何客户端安排合规性扫描并查看扫描结果。

> An enhanced SCAP auditing experience is available as a beta feature. It provides centralized content management, reusable scan policies, recurring scans, and built-in remediation from the Web UI. To get started, see **Administration Guide › Enhanced SCAP Auditing (Beta)**.

22.4.1. SCAP 简介

安全内容自动化协议 (SCAP) 是根据社区观点衍生出的一套综合性可互操作规范。它是由美国国家标准与技术研究院 (NIST) 维护的一系列规范，用于维持企业系统的系统安全性。

制定 SCAP 的目的是提供一种标准化方法来维持系统安全性，并且使用的标准会根据社区和企业的需求不断变化。新规范根据 NIST 的 SCAP 发布周期进行控制，以提供一致且可重复的修订工作流程。有关详细信息，请参见：

- https://csrc.nist.gov/projects/security-content-automation-protocol
- https://www.open-scap.org/features/standards/
- https://ncp.nist.gov/repository?scap

Uyuni 使用 OpenSCAP 来实现 SCAP 规范。OpenSCAP 是一个利用可扩展配置清单描述格式 (XCCDF) 的审计工具。XCCDF 是表达清单内容和定义安全清单的标准方式。它还结合了其他规范，例如通用平台枚举 (CPE)、通用配置枚举 (CCE) 及开放漏洞和评估语言 (OVAL)，以创建 SCAP 表达的、可由 SCAP 验证过的产品处理的清单。

OpenSCAP 使用 SUSE 安全团队生成的内容来验证补丁是否存在。OpenSCAP 使用基于标准和规范的规则来检查系统安全配置设置，并检查系统是否存在遭受入侵的迹象。有关 SUSE 安全团队的详细信息，请参见 https://www.suse.com/support/security。

22.4.2. 为 SCAP 扫描准备客户端

在开始之前，需要为客户端系统的 SCAP 扫描做好准备。

OpenSCAP 审计在使用 SSH 联系方法的 Salt 客户端上不可用。

Scanning clients can consume a lot of memory and compute power on the client being scanned. Ensure you have at least 2 GB of RAM available on each client to be scanned (especially for Red Hat-based and other modern clients).

在开始之前，请在客户端上安装 OpenSCAP 扫描程序和 SCAP 安全指南（内容）软件包。根据操作系统，这些软件包要么包含在基本操作系统中，要么包含在 Uyuni 客户端工具中。

下表列出了所需的软件包：

表格 13. OpenSCAP 软件包

Operating system	Version	Scanner	Content
SLES	12, 15	**openscap-utils**	**scap-security-guide**
openSUSE	Leap 15, Tumbleweed	**openscap-utils**	**scap-security-guide**
RHEL / CentOS / Oracle Linux / Rocky Linux / AlmaLinux	7, 8, 9, 10	**openscap-utils**	**scap-security-guide-redhat**
SUSE Liberty [1]	7, 8, 9, 10	**openscap-utils**	**scap-security-guide**
Debian	13	**openscap-scanner**	Not officially supported [2]
Ubuntu	22.04	**libopenscap8**	**scap-security-guide-ubuntu**
Ubuntu	24.04	**openscap-scanner**	Not officially supported [3]
Ubuntu	26.04	**openscap-scanner**	Not officially supported [4]

For Debian 13, Ubuntu 24.04, and Ubuntu 26.04, SUSE does not provide custom security guide packages. To perform scans on these clients, you must install the upstream packages (**ssg-debian** and **ssg-debderived** respectively) from their respective community repositories. Please note that these upstream guides are not officially supported by SUSE.

RHEL 7 and compatible systems provide a **scap-security-guide** package, which contains outdated contents. You are advised to use the **scap-security-guide-redhat** package you will find in the Uyuni Client Tools.

SUSE provides the **scap-security-guide** package for different openscap profiles. In the current version of **scap-security-guide**, SUSE supports the following profiles:

- 适用于 SUSE Linux Enterprise Server 12 和 15 的 DISA STIG 配置文件

- 适用于 SUSE Linux Enterprise Server 12 和 15 的 ANSSI-BP-028 配置文件
- 适用于 SUSE Linux Enterprise Server 12 和 15 的 PCI-DSS 配置文件
- 适用于 SUSE Linux Enterprise Server 15 的 HIPAA 配置文件
- SUSE Linux Enterprise Server for SAP Applications 15 的公有云映像强化
- SUSE Linux Enterprise 15 的公有云强化
- SLE 12 和 15 的标准系统安全性配置文件

对于非 SUSE 操作系统，包含的配置文件由社区提供。SUSE 不为其提供官方支持。

22.4.3. OpenSCAP 内容文件

OpenSCAP 使用 SCAP 内容文件来定义测试规则。这些内容文件是根据 XCCDF 或 OVAL 标准创建的。除了 SCAP 安全指南之外，您还可以下载公开的内容文件并根据要求对其进行自定义。可为默认内容文件模板安装 SCAP 安全指南软件包。或者，如果您熟悉 XCCDF 或 OVAL 的话，可以创建自己的内容文件。

我们建议您使用模板来创建自己的 SCAP 内容文件。如果您创建并使用自己的自定义内容文件，需要自负风险。如果您的系统因使用自定义内容文件而损坏，SUSE 可能不会为您提供支持。

When you have created your content files, you need to transfer the file to the client. You can do this in the same way as you move any other file, using physical storage media, or across a network with Salt (for example, salt-cp or the Salt File Server), **ftp** or **scp**.

If you use the **Administration Guide › Enhanced SCAP Auditing (Beta)** features, you do not need to manually transfer content files. The server automatically deploys the required DataStreams and tailoring files to the client at runtime.

我们建议您创建一个软件包以将内容文件分发到使用 Uyuni 管理的客户端。可将软件包签名并对其进行校验以确保其完整性。有关详细信息，请参见 **Administration Guide › Custom Channels**。

22.4.4. 查找 OpenSCAP 配置文件

不同的操作系统提供不同的 OpenSCAP 内容文件和配置文件。一个内容文件可以包含多个配置文件。

在基于 RPM 的操作系统上，可使用以下命令确定可用 SCAP 文件的位置：

```
rpm -ql <表中的 SCAP 安全指南软件包名称>
```

在基于 DEB 的操作系统上，可使用以下命令确定可用 SCAP 文件的位置：

```
dpkg -L <表中的 SCAP 安全指南软件包名称>
```

确定了一个符合您需求的 SCAP 内容文件后，列出客户端上可用的配置文件：

```
oscap info /usr/share/xml/scap/ssg/content/ssg-sle15-ds.xml
Document type: Source Data Stream
Imported: 2021-03-24T18:14:45

Stream: scap_org.open-scap_datastream_from_xccdf_ssg-sle15-xccdf-1.2.xml
Generated: (null)
Version: 1.2
Checklists:
        Ref-Id: scap_org.open-scap_cref_ssg-sle15-xccdf-1.2.xml
                Status: draft
                Generated: 2021-03-24
                Resolved: true
                Profiles:
                        Title: CIS SUSE Linux Enterprise 15 Benchmark
                                Id: xccdf_org.ssgproject.content_profile_cis
                        Title: Standard System Security Profile for SUSE Linux Enterprise
15
                                Id: xccdf_org.ssgproject.content_profile_standard
                        Title: DISA STIG for SUSE Linux Enterprise 15
                                Id: xccdf_org.ssgproject.content_profile_stig
                Referenced check files:
                        ssg-sle15-oval.xml
                                system: http://oval.mitre.org/XMLSchema/oval-definitions-5
                        ssg-sle15-ocil.xml
                                system: http://scap.nist.gov/schema/ocil/2

https://ftp.suse.com/pub/projects/security/oval/suse.linux.enterprise.15.xml
                                system: http://oval.mitre.org/XMLSchema/oval-definitions-5
Checks:
        Ref-Id: scap_org.open-scap_cref_ssg-sle15-oval.xml
        Ref-Id: scap_org.open-scap_cref_ssg-sle15-ocil.xml
        Ref-Id: scap_org.open-scap_cref_ssg-sle15-cpe-oval.xml
Dictionaries:
        Ref-Id: scap_org.open-scap_cref_ssg-sle15-cpe-dictionary.xml
```

记下用于执行扫描的文件路径和配置文件。

22.4.5. 执行审计扫描

安装或传输内容文件后，您可以执行审计扫描。可以使用 Uyuni Web UI 触发审计扫描。还可以使用 Uyuni API 来安排定期扫描。

过程：从 Web UI 运行审计扫描

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表**，然后选择要扫描的客户端。
2. Navigate to the **Audit** tab, and the **Schedule** subtab.
3. In the **Path to XCCDF Document** field, enter the parameters for the SCAP template and profile you want to use on the client. For example:
   - **Command**: /usr/bin/oscap xccdf eval
   - **Command-line arguments**: --profile xccdf_org.ssgproject.content_profile_stig

- **Path to XCCDF document**: /usr/share/xml/scap/ssg/content/ssg-sle15-ds.xml

  If you use **--fetch-remote-resources** parameter a lot of RAM is required. In addition, you may need to increase the value of **file_recv_max_size**.

4. 扫描将在客户端进行下一次安排的同步时运行。

XCCDF 内容文件在远程系统上运行之前会经过验证。如果内容文件包含无效参数，则测试将会失败。

过程：从 API 运行审计扫描

1. 在开始之前，请确保要扫描的客户端上已安装 Python 和 XML-RPC 库。
2. Choose an existing script or create a script for scheduling a system scan through **system.scap.scheduleXccdfScan**. For example:

```
#!/usr/bin/python3
import xmlrpc.client
client = xmlrpc.client.ServerProxy('https://server.example.com/rpc/api')
key = client.auth.login('username', 'password')
client.system.scap.scheduleXccdfScan(key, <1000010001>,
    '<path_to_xccdf_file.xml>',
    '--profile <profile_name>')
client.auth.logout(session_key)
```

   在此示例中：

   - **<1000010001>** is the system ID (sid).
   - **<path_to_xccdf_file.xml>** is the path to the content file location on the client. For example, **/usr/share/xml/scap/ssg/content/ssg-sle15-ds.xml**.
   - **<profile_name>** is an additional argument for the **oscap** command. For example, use **united_states_government_configuration_baseline** (USGCB).
3. 在命令提示符下，对您要扫描的客户端运行该脚本。

22.4.6. 扫描结果

Information about the scans you have run is in the Uyuni Web UI. Navigate to **Audit › OpenSCAP › All Scans** for a table of results. For more information about the data in this table, see **Reference Guide › Audit › Openscap All Scans**.

To ensure that detailed information about scans is available, you need to enable it on the client. In the

Uyuni Web UI, navigate to **Admin › Organizations** and click on the organization the client is a part of. Navigate to the **Configuration** tab, and check the **Enable Upload of Detailed SCAP Files** option. When enabled, this generates an additional HTML file on every scan, which contains extra information. The results show an extra line similar to this:

```
详细结果： xccdf-report.html xccdf-results.xml scap-yast2sec-oval.xml.result.xml
```

To retrieve scan information from the command line, use the **spacewalk-report** command:

```
spacewalk-report system-history-scap
spacewalk-report scap-scan
spacewalk-report scap-scan-results
```

You can also use the Uyuni API to view results, with the **system.scap** handler.

22.4.7. Removing SCAP Scan Results

Only SCAP scans that have passed the configured retention period can be deleted. You can configure the retention period in the **Organization Configuration** section.

Procedure: Configuring the SCAP results retention period

1. Navigate to **Home › My Organization › Configuration**.
2. Under **SCAP**, check **Allow Deletion of SCAP Results**.
3. Enter a number of days in **Allow Deletion After** (default is 90).

Procedure: Deleting SCAP scan results

1. Navigate to **Audit › OpenSCAP › All Scans**.
2. Select the checkbox for the scan or scans you want to delete.
3. 单击 **[确认]**。

22.4.8. 修复

Remediation Bash scripts and Ansible playbooks are provided in the same SCAP Security Guide packages to harden the client systems.

> If you have the beta SCAP features enabled, you can apply remediation directly from scan results in the Web UI, including custom per-rule remediation scripts. See **Administration Guide › Enhanced SCAP Auditing - Remediation**.

例如：

列表 5. bash 脚本

```
/usr/share/scap-security-guide/bash/sle15-script-cis.sh
/usr/share/scap-security-guide/bash/sle15-script-standard.sh
/usr/share/scap-security-guide/bash/sle15-script-stig.sh
```

列表 6. Ansible 剧本

```
/usr/share/scap-security-guide/ansible/sle15-playbook-cis.yml
/usr/share/scap-security-guide/ansible/sle15-playbook-standard.yml
/usr/share/scap-security-guide/ansible/sle15-playbook-stig.yml
```

在客户端系统中启用 Ansible 后，可以使用远程命令或 Ansible 运行这些脚本和剧本。

22.4.8.1. 使用 Bash 脚本运行修复

Install the **scap-security-guide** package on all your target systems. For more information, see **Administration Guide › Ansible Setup Control Node**.

用于每个操作系统和发行套件的软件包、通道和脚本都不同。修复 Bash 脚本示例 一节中列出了示例。

22.4.8.1.1. 在单个系统上将 Bash 脚本作为远程命令运行

在单个系统上将 Bash 脚本作为远程命令运行。

1. 在**系统 › 概览**选项卡中选择您的实例。然后在**细节 › 远程命令**中编写一个 Bash 脚本，例如：

   ```
   #!/bin/bash
   chmod +x -R /usr/share/scap-security-guide/bash
   /usr/share/scap-security-guide/bash/sle15-script-stig.sh
   ```

2. 单击 **[日程安排]**。

> 不同发行套件和版本中的文件夹和脚本名称有所不同。修复 Bash 脚本示例 一节列出了示例。

22.4.8.1.2. 在多个系统上使用系统集管理器运行 Bash 脚本

一次性在多个系统上将 Bash 脚本作为远程命令运行。

1. When a system group has been created click **System Groups**, select **Use in SSM** from the table.
2. From the **System Set Manager**, under **Misc › Remote Command**, write a Bash script such as:

   ```
   #!/bin/bash
   chmod +x -R /usr/share/scap-security-guide/bash
   /usr/share/scap-security-guide/bash/sle15-script-stig.sh
   ```

3. 单击 **[日程安排]**。

22.4.8.2. 修复 Bash 脚本示例

22.4.8.2.1. SUSE Linux Enterprise openSUSE 和变体

SUSE Linux Enterprise 和 openSUSE 脚本数据示例。

软件包

scap-security-guide

通道

- SLE12：SLES12 更新
- SLE15：SLES15 模块 Basesystem 更新

Bash 脚本目录

/usr/share/scap-security-guide/bash/

Bash 脚本

```
opensuse-script-standard.sh
sle12-script-standard.sh
sle12-script-stig.sh
sle15-script-cis.sh
sle15-script-standard.sh
sle15-script-stig.sh
```

22.4.8.2.2. Red Hat Enterprise Linux 和 CentOS Bash 脚本数据

Red Hat Enterprise Linux 和 CentOS 脚本数据示例。

scap-security-guide in centos7-updates only contains the Red Hat Enterprise Linux script.

软件包

scap-security-guide-redhat

通道

- SUSE Manager 工具

Bash 脚本目录

/usr/share/scap-security-guide/bash/

Bash 脚本

```
centos7-script-pci-dss.sh
centos7-script-standard.sh
centos8-script-pci-dss.sh
centos8-script-standard.sh
fedora-script-ospp.sh
fedora-script-pci-dss.sh
fedora-script-standard.sh
ol7-script-anssi_nt28_enhanced.sh
ol7-script-anssi_nt28_high.sh
```

```
ol7-script-anssi_nt28_intermediary.sh
ol7-script-anssi_nt28_minimal.sh
ol7-script-cjis.sh
ol7-script-cui.sh
ol7-script-e8.sh
ol7-script-hipaa.sh
ol7-script-ospp.sh
ol7-script-pci-dss.sh
ol7-script-sap.sh
ol7-script-standard.sh
ol7-script-stig.sh
ol8-script-anssi_bp28_enhanced.sh
ol8-script-anssi_bp28_high.sh
ol8-script-anssi_bp28_intermediary.sh
ol8-script-anssi_bp28_minimal.sh
ol8-script-cjis.sh
ol8-script-cui.sh
ol8-script-e8.sh
ol8-script-hipaa.sh
ol8-script-ospp.sh
ol8-script-pci-dss.sh
ol8-script-standard.sh
rhel7-script-anssi_nt28_enhanced.sh
rhel7-script-anssi_nt28_high.sh
rhel7-script-anssi_nt28_intermediary.sh
rhel7-script-anssi_nt28_minimal.sh
rhel7-script-C2S.sh
rhel7-script-cis.sh
rhel7-script-cjis.sh
rhel7-script-cui.sh
rhel7-script-e8.sh
rhel7-script-hipaa.sh
rhel7-script-ncp.sh
rhel7-script-ospp.sh
rhel7-script-pci-dss.sh
rhel7-script-rhelh-stig.sh
rhel7-script-rhelh-vpp.sh
rhel7-script-rht-ccp.sh
rhel7-script-standard.sh
rhel7-script-stig_gui.sh
rhel7-script-stig.sh
rhel8-script-anssi_bp28_enhanced.sh
rhel8-script-anssi_bp28_high.sh
rhel8-script-anssi_bp28_intermediary.sh
rhel8-script-anssi_bp28_minimal.sh
rhel8-script-cis.sh
rhel8-script-cjis.sh
rhel8-script-cui.sh
rhel8-script-e8.sh
rhel8-script-hipaa.sh
rhel8-script-ism_o.sh
rhel8-script-ospp.sh
rhel8-script-pci-dss.sh
rhel8-script-rhelh-stig.sh
rhel8-script-rhelh-vpp.sh
rhel8-script-rht-ccp.sh
rhel8-script-standard.sh
rhel8-script-stig_gui.sh
rhel8-script-stig.sh
rhel9-script-pci-dss.sh
rhosp10-script-cui.sh
rhosp10-script-stig.sh
rhosp13-script-stig.sh
rhv4-script-pci-dss.sh
rhv4-script-rhvh-stig.sh
```

```
rhv4-script-rhvh-vpp.sh
sl7-script-pci-dss.sh
sl7-script-standard.sh
```

22.4.8.2.3. Ubuntu Bash 脚本数据

Ubuntu 脚本数据示例。

软件包

scap-security-guide-ubuntu

通道

- SUSE Manager 工具

Bash 脚本目录

/usr/share/scap-security-guide/

Bash 脚本

```
ubuntu1804-script-anssi_np_nt28_average.sh
ubuntu1804-script-anssi_np_nt28_high.sh
ubuntu1804-script-anssi_np_nt28_minimal.sh
ubuntu1804-script-anssi_np_nt28_restrictive.sh
ubuntu1804-script-cis.sh
ubuntu1804-script-standard.sh
ubuntu2004-script-standard.sh
```

22.4.8.2.4. Debian Bash 脚本数据

Debian 脚本数据示例。

软件包

scap-security-guide-debian

通道

- SUSE Manager 工具

Bash 脚本目录

/usr/share/scap-security-guide/bash/

Bash 脚本

```
# Debian 12
debian12-script-anssi_np_nt28_average.sh
debian12-script-anssi_np_nt28_high.sh
debian12-script-anssi_np_nt28_minimal.sh
debian12-script-anssi_np_nt28_restrictive.sh
debian12-script-standard.sh
```

22.4.9. Enhanced SCAP Auditing (Beta)

22.4.9.1. 概述

Uyuni introduces a modernized approach to SCAP (Security Content Automation Protocol) auditing, available as a beta feature. The enhanced SCAP integration streamlines compliance scanning by centralizing content management, introducing reusable policies, supporting automated remediation, and eliminating the need to pre-stage SCAP files on managed systems.

This feature is currently in **beta**. You must enable the beta feature flag in your user preferences before using it.

22.4.9.2. Enable the beta feature

To access the enhanced SCAP auditing features:

Procedure: Enabling the beta feature

1. Navigate to **Home › My Preferences**.
2. Check the **[Enable Beta Features]** checkbox.
3. Click **[Save Preferences]**.

Once enabled, additional menu entries appear under **Audit › OpenSCAP**, and the scan scheduling interface is replaced with the new beta UI.

The beta feature flag is a per-user setting. Each user who wants to use the enhanced SCAP features must enable it individually.

22.4.9.3. Key differences from legacy SCAP integration

Legacy integration	Enhanced integration (beta)
SCAP content files must exist on the managed system beforehand	SCAP content is transferred automatically from the server to the managed system at scan time
No centralized content management	Upload and manage SCAP content centrally via the Web UI
No tailoring file management	Dedicated UI for uploading and managing tailoring files
Scans are configured individually each time	Reusable SCAP policies combine content, profiles, and tailoring files
No built-in remediation	Apply remediation directly from scan results, or define custom remediation scripts
No recurring scan support tied to policies	Schedule recurring scans linked to policies for ongoing compliance tracking

22.4.9.4. SCAP content management

SCAP content files (DataStream and XCCDF) can be uploaded and managed centrally on the Uyuni server.

22.4.9.4.1. Upload SCAP content

Procedure: Uploading SCAP content

1. Navigate to **Audit › OpenSCAP › SCAP Content**.
2. Click **[Upload]**.
3. Provide a **Name** and optional **Description**.
4. Upload the files:
    - **DataStream file**: The filename must end with `-ds.xml`.
    - **XCCDF file**: The filename must end with `-xccdf.xml`.
    - Both files must share the same base name (for example, `ssg-sle15-ds.xml` and `ssg-sle15-xccdf.xml`).
5. Click **[Upload]**.

Uploaded files are stored on the server at `/srv/susemanager/scap/ssg/content/`.

22.4.9.4.2. Edit and delete SCAP content

Procedure: Editing and deleting SCAP content

1. Go to **Audit › OpenSCAP › SCAP Content**.
2. Click a content entry to edit its name, description, or replace the uploaded files.
3. Click **[Delete]** to remove SCAP content.

Deleting SCAP content that is referenced by a policy will break that policy. Ensure no active policies depend on the content before deleting it.

22.4.9.4.3. OVAL files

Due to the large size of OVAL (Open Vulnerability and Assessment Language) files, they are not transferred automatically from the server to the managed system. If your SCAP evaluation requires OVAL files, you must ensure they are already present on the managed system before scheduling the scan.

You can specify OVAL file paths when creating a SCAP policy.

22.4.9.5. Tailoring file management

SCAP tailoring files allow you to customize the behavior of an SCAP profile without modifying the original content. The enhanced integration provides a dedicated interface for managing tailoring files.

Tailoring files are scoped per organization. Each organization manages its own set of tailoring files independently.

22.4.9.5.1. Upload a tailoring file

Procedure: Uploading a tailoring file

1. Navigate to **Audit › OpenSCAP › Tailoring Files**.
2. 单击 [**创建**]。
3. Provide a **Name** and optional **Description**.
4. Upload the tailoring file (XML format).
5. Click [**Upload**].

Tailoring files are stored on the server at **/srv/susemanager/scap/tailoring-files/**.

22.4.9.5.2. Editing and deleting tailoring files

Procedure: Edit and delete tailoring files

1. Navigate to **Audit › OpenSCAP › Tailoring Files** list.
2. Click a tailoring file entry to edit its name, description, or replace the file.
3. Select one or more entries and click [**Delete**] to remove them.

22.4.9.6. SCAP policies

SCAP policies define a reusable combination of SCAP content, a specific profile, and an optional tailoring file. Policies simplify the process of scheduling consistent compliance scans across your infrastructure.

22.4.9.6.1. Creating a policy

Procedure: Creating a policy

1. Navigate to **Audit › OpenSCAP › SCAP Policies**.
2. 单击 [**创建**]。
3. Fill in the policy details:

Policy Name

A unique name for the policy within your organization.

说明

An optional description of the policy’ s purpose.

SCAP Content

Select from the uploaded SCAP content.

XCCDF Profile

Select a profile from the chosen SCAP content. The list of available profiles is loaded dynamically from the content file.

Tailoring File

(Optional) Select a tailoring file to customize the selected profile.

Tailoring Profile

(Optional) If a tailoring file is selected, choose a profile from the tailoring file.

OVAL Files

(Optional) Comma-separated list of OVAL file paths that must exist on the managed system.

Advanced Arguments

(Optional) Additional `oscap` command-line arguments (for example, `--fetch-remote-resources`).

Fetch Remote Resources

(Optional) Enable to allow `oscap` to download remote XCCDF resources during evaluation.

4. 单击 [创建]。

22.4.9.6.2. Policy details and scan history

Click on a policy name in the list to view its details, including:

- The associated SCAP content, profile, and tailoring file.

- A scan history showing all scans that were executed using this policy, with compliance results.

22.4.9.6.3. Edit and delete policies

Procedure: Editing and deleting policies

1. Navigate to **Audit › OpenSCAP › SCAP Policies** list.
2. Click the edit icon to modify a policy.
3. Click **[Delete]** to remove a policy.

22.4.9.7. Schedule SCAP scans

22.4.9.7.1. Scheduling a scan for a single system

Procedure: Scheduling a scan for a single system

1. Navigate to **Systems › System List › All**.
2. Select a system then go to **Audit › OpenSCAP › Schedule**.
3. Select a **SCAP Policy** to use for the scan.
4. Choose the scan schedule:
   - **Run now**: Execute the scan immediately.
   - **Schedule for later**: Pick a specific date and time.
5. 单击 **[日程安排]**。

Package **openscap-utils** needs to be installed on the targeted minion. For more information see **Administration Guide › Prepare clients for an SCAP scan**.

22.4.9.7.2. Schedule scans for multiple systems (SSM)

Procedure: Scheduling scans for multiple systems (SSM)

1. Add the target systems to the System Set Manager (SSM).
2. Navigate to **Systems › System Set Manager › Audit › Schedule**.
3. Select a **SCAP Policy**.
4. Choose the scan schedule.
5. 单击 **[日程安排]**。

22.4.9.7.3. Recurring scans

SCAP policies can be scheduled as recurring actions, enabling automated compliance monitoring.

To set up a recurring scan, run the procedure:

Procedure: Setting up a recurring scan

1. Navigate to the recurring actions configuration for a system or system group.
2. Select **SCAP Policy Scan** as the action type.
3. Choose the SCAP policy to apply.
4. Configure the recurrence schedule (for example, daily, weekly).
5. Optionally enable **Test Mode** to simulate the scan without applying changes.

> Recurring scans are linked to the selected SCAP policy. If you update the policy (for example, change the profile), future recurring scans will use the updated configuration.

22.4.9.8. How scans work

When a scan is scheduled with the enhanced (beta) SCAP integration:

1. The Uyuni server transfers the required SCAP content files (DataStream and XCCDF) from the server to the managed system using Salt’ s file management.
2. If a tailoring file is associated with the policy, it is also transferred.
3. The **oscap** tool is executed on the managed system with the appropriate parameters (profile, rules, tailoring, etc.).
4. Scan results (**results.xml** and **report.html**) are collected from the managed system and stored on the server.
5. Results are available for review in the WebUI under the system’ s audit section.

Files are transferred to **/var/cache/salt/minion/scap/** on the managed system.

> OVAL files are **not** transferred automatically due to their large size. If your scan requires OVAL files, they must be present on the managed system before the scan runs.

22.4.9.9. 修复

The enhanced SCAP integration allows you to apply remediation actions directly from scan results to fix non-compliant rules.

22.4.9.9.1. Remediation from scan results

After reviewing scan results, you can apply remediation for individual rules that have failed.

Procedure: Applying remediation from scan results

1. Navigate to a completed scan's results.
2. Identify the non-compliant rule.
3. Click the remediation action to apply the fix.

Remediation can be applied in two ways:

Bash remediation
: A shell script is executed on the managed system as root. The script is derived from the SCAP content's built-in fix elements.

Salt remediation
: A Salt state is applied to the managed system. This uses the **scap_beta.remediation** Salt state with the remediation content passed as pillar data.

22.4.9.9.2. Custom remediation

If the built-in remediation from the SCAP content is insufficient or you need to tailor the fix for your environment, you can define custom remediation scripts.

Custom remediation is scoped per organization and per rule.

To save a custom remediation, follow these steps:

Procedure: Saving a custom remediation

1. Navigate to the rule's remediation view.
2. Choose the script type:
   - **Bash**: A custom shell script.
   - **Salt**: A custom Salt state definition.
3. Enter or modify the remediation script.
4. 单击 **[保存]**。

Custom remediation overrides the default remediation from the SCAP content for the specific rule within your organization.

> Custom remediation includes an audit trail. The system tracks which user created and last modified each custom remediation script.

22.4.9.9.3. Deleting custom remediation

You can delete a custom remediation to revert to the default remediation provided by the SCAP content.

Navigate to the custom remediation view and click **[Delete]** for the specific script type.

22.4.9.10. Navigation reference

When the beta feature is enabled, the following menu entries are available:

Menu path	Description
Audit › OpenSCAP › All Scans	View all SCAP scan results (available with or without beta).
Audit › OpenSCAP › SCAP Policies	Create and manage reusable SCAP policies. (beta only)
Audit › OpenSCAP › SCAP Content	Upload and manage SCAP DataStream and XCCDF files. (beta only)
Audit › OpenSCAP › Tailoring Files	Upload and manage SCAP tailoring files. (beta only)
Audit › OpenSCAP › XCCDF Diff	Compare two SCAP scan results (available with or without beta).
Audit › OpenSCAP › Advanced Search	Search across all SCAP scan results (available with or without beta).
Systems › System Details › Audit › OpenSCAP › Schedule	Schedule a SCAP scan for a single system (uses the new policy-based UI when beta is enabled).
Systems › SSM › Audit › Schedule	Schedule SCAP scans for multiple systems (uses the new policy-based UI when beta is enabled).

22.4.9.11. Workflow example

The following example demonstrates a typical workflow using the enhanced SCAP features:

1. **Enable beta features** in your user preferences.
2. **Upload SCAP content**:

   Navigate to **Audit › OpenSCAP › SCAP Content** and upload a SCAP Security Guide DataStream file (for example, **ssg-sle15-ds.xml** and **ssg-sle15-xccdf.xml**).

3. **Upload a tailoring file** (optional):

   Navigate to **Audit › OpenSCAP › Tailoring Files** and upload a tailoring file if you need to customize a profile.

4. **Create a SCAP policy**:

Navigate to **Audit › OpenSCAP › SCAP Policies** and create a policy that references the uploaded content, selects a profile (for example, **CIS Benchmark Level 1**), and optionally includes the tailoring file.

5. **Schedule a scan**:

   Navigate to a system’ s **Audit › OpenSCAP › Schedule** tab, select the policy, and schedule the scan.

6. **Review results**:

   After the scan completes, review the results under the system’ s **Audit › OpenSCAP › List Scans** tab.

7. **Apply remediation**:

   For any non-compliant rules, apply the built-in remediation or define custom remediation scripts tailored to your environment.

8. **Set up recurring scans**:

   Configure recurring scans using the SCAP policy to maintain ongoing compliance monitoring.

22.5. 储存库元数据

需有一个自定义 GPG 密钥才能为储存库元数据签名。

> To access a shell inside the Server container run **mgrctl term** on the container host.

Procedure: Generating a custom GPG key

1. As the root user, use the **gpg** command to generate a new key:

   ```
   mgrctl exec -it -- gpg --full-generate-key
   ```

2. At the prompts, select **RSA** as the key type, with a size of 2048 bits, and select an appropriate expiry date for your key. Check the details for your new key, and type **y** to confirm.
3. 根据提示输入要与该密钥关联的名称和电子邮件地址。 You can also add a comment to help you identify the key, if desired. When you are happy with the user identity, type **O** to confirm.
4. 根据提示输入通行口令以保护您的密钥。
5. 该密钥应自动添加到您的密钥环。　可以通过列出密钥环中的密钥进行检查：

```
mgrctl exec -- gpg --list-keys
```

6. Add the password for your keyring to the **/etc/rhn/signing.conf** configuration file, by opening the file in your text editor and adding this line:

```
GPGPASS="password"
```

有关如何续订 GPG 密钥，请参见 **Administration Guide › Troubleshooting › Tshoot Sync**。

You can manage metadata signing on the command line using the **mgr-sign-metadata-ctl** command.

Procedure: Enabling metadata signing

1. 您需要知道所要使用的密钥的短标识符。 可以简短格式列出可用的公共密钥：

```
mgrctl exec -- gpg --keyid-format short --list-keys
...
pub   rsa4096/3E7BFE0A 2019-04-02 [SC] [expires: 2029-04-01]
      A43F9EC645ED838ED3014B035CFA51BF3E7BFE0A
uid         [ultimate] SUSE Manager
sub   rsa4096/118DE7FF 2019-04-02 [E] [expires: 2029-04-01]
```

2. Enable metadata signing with the **mgr-sign-metadata-ctl** command:

```
mgrctl exec -- mgr-sign-metadata-ctl enable 3E7BFE0A
OK. Found key 3E7BFE0A in keyring.
DONE. Set key 3E7BFE0A in /etc/rhn/signing.conf.
DONE. Enabled metadata signing in /etc/rhn/rhn.conf.
DONE. Exported key 3E7BFE0A to /srv/susemanager/salt/gpg/mgr-keyring.gpg.
DONE. Exported key 3E7BFE0A to /var/spacewalk/gpg/<KEY_NAME>.key.
NOTE. For the changes to become effective run:
   mgr-sign-metadata-ctl regen-metadata
```

3. 可以使用以下命令检查您的配置是否正确：

```
mgrctl exec -- mgr-sign-metadata-ctl check-config
```

4. Restart the container for the configuration changes to be detected.

```
mgradm restart
```

5. Schedule metadata regeneration to replace all metadata with new signed versions.

```
mgrctl exec -- mgr-sign-metadata-ctl regen-metadata
```

You can also use the **mgr-sign-metadata-ctl** command to perform other tasks. Use **mgr-sign-metadata-ctl --help** to see the complete list.

储存库元数据签名是全局选项。启用后，将对服务器上的所有软件通道启用该选项。这意味着，连接到服务器的所有客户端需要信任新的 GPG 密钥才能安装或更新软件包。

Procedure: Importing GPG keys on clients

1. 将 GPG 密钥部署到客户端的过程适用于 Salt 状态。
2. 使用 Uyuni Web UI 应用 Highstate。

有关 GPG 密钥查错的详细信息，请参见 **Administration Guide › Troubleshooting › Tshoot Sync**。

[1] SUSE Liberty Linux includes a compatible **scap-security-guide** package. However, the version shipped with the client tools is not compatible, so it should not be used for SCAP content.

[2] This package is community-provided and not officially supported by SUSE. Upstream **ssg-debian** can be used at your own risk.

[3] This package is community-provided and not officially supported by SUSE. Upstream **ssg-debderived** can be used at your own risk.

[4] This package is community-provided and not officially supported by SUSE. Upstream **ssg-debderived** can be used at your own risk.

章 23. 基于角色的访问控制 (RBAC)

基于角色的访问控制（RBAC）是一种安全机制，它根据用户被分配的角色来限制授权用户对资源的访问权限。在 Uyuni 中，RBAC 可确保用户仅能执行其获得明确授权的操作以及访问相关资源，从而提升安全性并简化管理流程。

RBAC 的核心原则包括：

- **最小权限原则：** 仅授予用户执行其任务所需的必要访问权限。
- **精细化控制：** 针对特定功能提供精细的控制能力。
- **职责分离：** 防止单个用户对关键流程拥有过多控制权。
- **可审计性：** 能够清晰跟踪用户的操作及权限配置。

23.1. RBAC 核心概念

理解以下核心概念对于高效管理 RBAC 至关重要：

- **Role:** A collection of permissions defining a specific set of capabilities within Uyuni.

  角色分配给用户后，用户将获得该角色对应的所有聚合权限。

- **权限：** 原子级的授权项，允许用户在 Uyuni 中执行特定操作、访问特定网页或调用特定 API 端点。在 Uyuni 中，权限通过名称空间及其访问模式来表示。
- **用户：** 与 Uyuni 进行交互的个人帐户。用户可被分配一个或多个角色。
- **名称空间：** 粒度化的访问控制单元，以树形结构组织。大多数名称空间均设有独立的“查看”或“修改”模式。

23.2. 在 Uyuni 中使用角色

Uyuni 提供了预定义角色，并允许您定义额外的自定义角色，且可选择从多个其他角色的组合继承权限。

23.2.1. 预定义角色

有关预定义角色及其说明的完整列表，请参见 **Administration Guide › Users › Administrator Roles**。

23.2.2. 定义额外角色

要定义额外角色，您可以执行以下操作：

- 选择多个现有角色，从中继承权限。
- 指定额外的名称空间，以授予访问权限。

23.3. 用于精细化访问控制的名称空间

名称空间提供精细的访问控制，以树形结构组织。对于大多数名称空间，其内部访问权限可通过“查看”和“修改”模式进一步细化。

表格 14. 示例：映像管理相关名称空间和访问模式

Namespace	Access Mode	Description
cm.build	Modify	Build container or Kiwi images
cm.image.import	Modify	Import container images from a registered image store
cm.image.list	View	List all images
cm.image.list	Modify	Delete images
cm.image.overview	View	View image details, patches, packages, build log and cluster information
cm.image.overview	Modify	Inspect, rebuild, delete images
cm.profile.details	View	View details of an image profile
cm.profile.details	Modify	Create image profiles, edit profile details
cm.profile.list	View	List all image profiles
cm.profile.list	Modify	Delete image profiles
cm.store.details	View	View details of an image store
cm.store.details	Modify	Create image stores, edit store details
cm.store.list	View	List all image stores
cm.store.list	Modify	Delete image stores

A comprehensive list of namespaces and their descriptions can be retrieved by making a call to the **access.listNamespaces** API method. Refer to Uyuni API documentation for detailed information, including request and response formats.

23.4. RBAC 管理

Managing RBAC roles and permissions is possible either through the API or through the Web UI. To assign roles to users via the Web UI, refer to **Administration Guide › Users**.

23.4.1. Managing RBAC via Web UI

In the Web UI, you can manage custom RBAC roles and assign permissions by navigating to **Admin › Access Control**. From there, you can click **[Create Access Group]** to define a new role and configure its permissions.

> Predefined roles cannot be seen or managed via the Web UI. Only custom roles created within your organization are visible in the **Admin › Access Control** section.

23.4.2. 通过 API 管理 RBAC

Uyuni API 提供了以编程方式管理角色、权限及用户分配的相关方法。

23.4.2.1. The access API

以下 API 方法用于管理角色及其关联的访问权限：

- **listNamespaces**: Lists available namespaces, access modes and their descriptions in Uyuni.
- **listPermissions**: Lists permitted namespaces of a role.
- **listRoles**: Lists existing roles in Uyuni.
- **createRole**: Creates a new role, optionally copying permissions from existing roles.
- **deleteRole**: Deletes a role.
- **grantAccess**: Grants access to namespaces.
- **revokeAccess**: Revokes access to namespaces.

23.4.2.2. The user API

以下 API 方法用于管理用户与角色的分配关系：

- **listPermissions**: Lists effective permissions of a user.
- **listRoles**: Lists a user’ s assigned roles.
- **addRole**: Assigns a role to a user.
- **removeRole**: Removes a role from a user.

有关详细的 API 文档（包括请求和响应格式），请参见 Uyuni API 参考手册。

23.5. RBAC 最佳实践

遵循以下最佳实践有助于维护安全、高效且易于管理的 RBAC 环境：

- **最小权限原则：**始终仅授予用户完成其职责所需的最低权限，避免授予过于宽泛的权限。
- **定期审查：**定期审查用户分配到的角色和权限，确保其仍然适用且符合当前的安全策略。
- **记录角色：**为您创建的每个自定义角色清晰记录其用途和权限。
- **职责分离：**设计并应用能够强制执行职责分离的角色，防止单个用户对关键流程拥有过多控制权。

章 24. SSL 证书

Uyuni 使用 SSL 证书来确保客户端注册到正确的服务器。

每个使用 SSL 注册到 Uyuni Server 的客户端将通过验证服务器证书来检查它是否连接到正确的服务器。此过程称为 SSL 握手。

在 SSL 握手期间，客户端将检查服务器证书中的主机名是否与预期相符。客户端还需要检查服务器证书是否受信任。

证书颁发机构 (CA) 是用于为其他证书签名的证书。所有证书必须由证书颁发机构 (CA) 签名，只有这样，才会将它们视为有效，并且客户端才能成功地匹配它们。

为使 SSL 身份验证能够正常进行，客户端必须信任根 CA。这意味着必须在每个客户端上安装根 CA。

默认的 SSL 身份验证方法是让 Uyuni 使用自我签名证书。在这种情况下，Uyuni 已生成所有证书，并且根 CA 已直接为服务器证书签名。

另一种方法是使用中间 CA。在这种情况下，根 CA 为中间 CA 签名。然后中间 CA 可为任意数量的其他中间 CA 签名，而最后一个中间 CA 为服务器证书签名。这称为链式证书。

如果您在链式证书中使用中间 CA，则根 CA 将安装在客户端上，而服务器证书则安装在服务器上。在 SSL 握手期间，客户端必须能够校验根 CA 与服务器证书之间的整个中间证书链，因此它们必须能够访问所有中间证书。

可以通过两种主要方式实现此目的。在较旧版本的 Uyuni 中，所有中间 CA 默认都安装在客户端上。不过，您也可以在服务器上配置服务以将其提供给客户端。在这种情况下，在 SSL 握手期间，服务器会提供服务器证书以及所有中间 CA。此机制现已用作默认配置。

Uyuni 默认使用没有中间 CA 的自我签名证书。为了提高安全性，您可以安排一个第三方 CA 来为您的证书签名。第三方 CA 执行检查以确保证书中包含的信息正确。他们通常针对此项服务收取年费。使用第三方 CA 可以提高证书的伪造难度，并为安装提供附加的保护。如果您的证书已由第三方 CA 签名，您可以将其导入 Uyuni 安装中。

本手册分 2 步介绍 SSL 证书的用法：

1. 如何使用 Uyuni 工具创建自我签名证书
2. 如何在 Uyuni Server 或 Proxy 上部署证书

如果证书由第三方实例（例如自有或外部的 PKI）提供，则可以跳过步骤 1。

- 有关如何创建自我签名证书的详细信息，请参见 **Administration Guide › Ssl Certs Selfsigned**。
- 有关如何导入证书的详细信息，请参见 **Administration Guide › Ssl Certs Imported**。

24.1. 为 Uyuni 容器提供 SSL 证书

24.1.1. Podman

SSL 证书以 Podman 机密的形式存储并会分配给相应容器。Podman SSL 机密包括：

- CA 证书
  - uyuni-ca
  - uyuni-db-ca
- 服务器证书和密钥
  - uyuni-cert
  - uyuni-key
- 数据库证书和密钥
  - uyuni-db-cert
  - uyuni-db-key

24.2. 自我签名 SSL 证书

Uyuni 默认使用自我签名证书。在这种情况下，证书由 Uyuni 创建和签名。此方法不使用独立证书颁发机构来保证证书的细节正确无误。第三方 CA 将会执行检查，以确保证书中包含的信息正确无误。

- 有关第三方 CA 的详细信息，请参见 **Administration Guide › Ssl Certs Imported**。
- 有关如何替换证书的详细信息，请参见 **Administration Guide › Ssl Certs Imported › Ssl Certs Import Replace**。

本节介绍如何在新安装或现有安装中创建或重新创建自我签名证书。

SSL 密钥和证书的主机名必须与其部署到的计算机的完全限定主机名相匹配。

> CA certificates generated by older Uyuni versions can miss the **critical** flag on the **X509v3 Basic Constraints** extension. Strict certificate validators, including Python 3.13 and later, reject such CA certificates and can report **Basic Constraints of CA cert not marked critical**. If your self-signed CA is affected, create a new CA and new server certificates, then deploy the new root CA to all clients. For more information about checking the CA certificate, see **Administration Guide › Ssl Certs Imported › Ssl Certs Verify Ca Basic Constraints**. For more information about replacing certificates and deploying the root CA to clients, see **Administration Guide › Ssl Certs Imported › Ssl Certs Import Deploy Root Ca**.

24.2.1. 重新创建现有的服务器证书

如果您的现有证书已失效或出于任何原因而不再正常工作，您可以从现有 CA 生成新的服务器证书。

过程：重新创建现有服务器证书

1. 在 Uyuni 容器主机的命令提示符处，重新生成服务器证书：

```
mgrctl exec -ti -- 'rhn-ssl-tool --gen-server --dir="/root/ssl-build" --set
-country="COUNTRY" \
--set-state="STATE" --set-city="CITY" --set-org="ORGANIZATION" \
--set-org-unit="ORGANIZATION UNIT" --set-email="name@example.com" \
--set-hostname="susemanager.example.com" --set-cname="example.com"'
```

Ensure that the **set-cname** parameter is the fully qualified domain name of your Uyuni Server. You can use the **set-cname** parameter multiple times if you require multiple aliases.

The private key and the server certificate can be found inside the server container in the directory **/root/ssl-build/susemanager/** as **server.key** and **server.crt**. The name of the last directory depends on the hostname used with **--set-hostname** option.

请通过更新容器的主机 Podman 机密来部署或导入新证书和密钥。有关如何导入刚才生成的证书的详细信息，请参见 **Administration Guide › Ssl Certs Imported › Ssl Certs Import Replace**。

24.2.2. 创建新的 CA 证书和服务器证书

替换根 CA 时请小心。这可能会破坏服务器与客户端之间的信任链。如果发生这种情况，需要让某个管理用户登录到每个客户端并直接部署 CA。

过程：创建新证书

1. 在 Uyuni 容器主机的命令提示符处，将旧证书目录移到新位置：

```
mgrctl exec -- mv /root/ssl-build /root/old-ssl-build
```

2. 生成新的 CA 证书：

```
mgrctl exec -ti -- 'rhn-ssl-tool --gen-ca --dir="/root/ssl-build" --set
-country="COUNTRY" \
--set-state="STATE" --set-city="CITY" --set-org="ORGANIZATION" \
--set-org-unit="ORGANIZATION UNIT" --set-common-name="SUSE Manager CA Certificate" \
--set-email="name@example.com"'
```

3. 生成新的服务器证书：

```
mgrctl exec -ti -- 'rhn-ssl-tool --gen-server --dir="/root/ssl-build" --set
-country="COUNTRY" \
--set-state="STATE" --set-city="CITY" --set-org="ORGANIZATION" \
--set-org-unit="ORGANIZATION UNIT" --set-email="name@example.com" \
--set-hostname="susemanager.example.top" --set-cname="example.com"'
```

Ensure that the **set-cname** parameter is the fully qualified domain name of your Uyuni Server. You can use the **set-cname** parameter multiple times if you require multiple aliases.

还需要使用代理的主机名和 cname 为每个代理生成服务器证书。

24.3. 导入 SSL 证书

本节介绍如何为新的 Uyuni 安装配置 SSL 证书，以及如何替换现有证书。

在开始之前，请确保已准备好：

- 证书颁发机构 (CA) SSL 公共证书。如果您使用 CA 链，则所有中间 CA 也必须可用。
- 一个 SSL 服务器私用密钥
- 一个 SSL 服务器证书
- 一个 SSL 数据库私用密钥
- 一个 SSL 数据库证书

All files must be in PEM format. All CA certificates in the chain must mark **X509v3 Basic Constraints** as **critical** and include **CA:TRUE**. For more information, see **Administration Guide › Ssl Certs Imported › Ssl Certs Verify Ca Basic Constraints**.

The hostname of the SSL server certificate must match the fully qualified hostname of the machine you deploy them on. You can set the hostnames in the **X509v3 Subject Alternative Name** section of the certificate. You can also list multiple hostnames if your environment requires it. Supported Key types are **RSA** and **EC** (Elliptic Curve).

> In previous versions the database SSL certificates required **reportdb** and **db** as **Subject Alternative Name**. This is no longer needed.

Third-party authorities commonly use intermediate CAs to sign requested server certificates. In this case, all CAs in the chain are required to be available. The **mgradm** commands are taking care of ordering the certificates. Ideally, the root CA should be in its own file. The server certificate file should contain the server certificate first, followed by all intermediate CA certificates in order.

24.3.1. Verify CA Basic Constraints

Strict certificate validators reject CA certificates when the **X509v3 Basic Constraints** extension is not marked as **critical**. This includes Python 3.13 and later, which is used by SUSE Linux Enterprise Server 16 and openSUSE Leap 16.

Before importing certificates or upgrading an existing installation, verify every root and intermediate CA certificate used by Uyuni. For each CA certificate, run:

```
openssl x509 -in root-ca.pem -noout -text | grep -A1 "Basic Constraints"
```

The output must include **critical** and **CA:TRUE**:

```
X509v3 Basic Constraints: critical
```

```
CA:TRUE
```

If a CA certificate does not include both values, replace or regenerate the affected CA certificate before using it with Uyuni. Replacing only the server certificate does not fix a CA certificate that misses the **critical** flag. For self-signed certificates generated by Uyuni, see **Administration Guide › Ssl Certs Selfsigned › Ssl Certs Selfsigned Create Replace**. To replace a CA that is still trusted by the clients without interrupting their connection, see **Administration Guide › Ssl Certs Ca Migration**.

24.3.2. 为新安装导入证书

Uyuni 默认使用自我签名证书。您可以在安装期间导入第三方证书。

过程：在新的 Uyuni 服务器上导入证书

1. Deploy the Uyuni Server according to the instructions in **Installation and Upgrade Guide › Install Server**. Make sure to pass the correct files as parameters to **mgradm install**. The parameters are:

```
第三方 SSL 证书标志:
      --ssl-ca-intermediate 字符串   中间 CA 证书路径
      --ssl-ca-root 字符串            根 CA 证书路径
      --ssl-server-cert 字符串        服务器证书路径
      --ssl-server-key 字符串         服务器密钥路径
      --ssl-db-ca-intermediate 字符串 数据库的中间 CA
证书路径（如果不同于服务器的相应证书）
      --ssl-db-ca-root 字符串          数据库的根 CA
证书路径（如果不同于服务器的相应证书）
      --ssl-db-cert 字符串             数据库证书路径
      --ssl-db-key 字符串              数据库密钥路径
```

中间 CA 可以在使用 **--ssl-ca-root** 指定的文件中提供，也可以使用 **--ssl-ca-intermediate** 作为附加选项来指定。可以多次指定 **--ssl-ca-intermediate** 选项。

24.3.3. 为新代理安装导入证书

代理证书已嵌入生成的配置中。如要使用第三方证书，需要在配置过程中提供。

过程：在新的 Uyuni 代理上导入证书

1. 根据 **Installation and Upgrade Guide › Install Proxy** 中的说明安装 Uyuni 代理。
2. 按照提示完成设置。

> 请使用同一证书颁发机构（CA）为服务器和代理的所有证书签名。使用不同 CA 签名的证书将不匹配。

24.3.4. 替换证书

您可以将 Uyuni 安装环境中的现有有效证书替换为新证书。需要考虑两种场景：仅替换服务器或数据库证书；替换根 CA 证书。

替换根证书需要更多时间并进行更多规划，以避免服务中断，因为在服务器层面切换至新证书前，所有已注册的代理和系统都需要在其数据库中导入新的 CA 证书。

使用由中间 CA 签名的第三方证书时，需要将中间 CA 证书追加至服务器或数据库证书文件中。

证书顺序至关重要：首先是服务器证书，随后依次为对该证书签名的 CA 证书，直至由根 CA 签名的证书。根 CA 证书不应追加至服务器证书文件中。

过程：替换所有现有证书

1. The following considers that you have **root-ca.pem**, **intermediate-ca1.pem**, **intermediate-ca2.pem**, **server.pem** and **server.key** files. It may be different depending on the number of intermediate CAs in the server certificate signature chain.
2. Combine the intermediate CAs and server certificates. The order matters, the server must be first and the intermediate CAs in order. Do not add the root CA last into the chain as it will be passed separately to **uyuni-ca** and **uyuni-db-ca** secrets. If there is no intermediate CA, then you can use the **server.pem** instead of the combined file in the next steps.

```
cat server.pem intermediate-ca1.pem intermediate-ca2.pem >combined-server.pem
```

3. 在 Uyuni 容器主机的命令行中，重新创建用于传递文件路径的 Podman 证书机密：

```
podman secret create --replace uyuni-ca $path_to_ca_certificate
podman secret create --replace uyuni-cert $path_to_combined_server_certificate
podman secret create --replace uyuni-key $path_to_server_key

podman secret create --replace uyuni-db-ca $path_to_database_ca_certificate
podman secret create --replace uyuni-db-cert
$path_to_combined_database_certificate
podman secret create --replace uyuni-db-key $path_to_database_key
```

过程：重启服务器

1. 在容器主机上，重启服务以应用更改：

```
mgradm restart
```

如果您使用的是代理，则需要使用相关代理的主机名和 cname 为每个代理生成一个服务器证书 RPM。生成新的配置 tarball 并进行部署。

有关详细信息，请参见 **Installation and Upgrade Guide › Container Deployment › Proxy Deployment Uyuni › Proxy Setup Containers Generate Config**. proxy-deployment-uyuni.adoc[]

如果根 CA 已更改，则需要将其部署至所有连接到 Uyuni 的客户端。理想情况下应提前完成，以最大程度减少服务中断。

过程：在客户端上部署根 CA

1. 在 Uyuni Web UI 中，导航到**系统 › 概览**。
2. 选中所有客户端以将其添加到系统集管理器。
3. 导航到**系统 › 系统集管理器 › 概览**。
4. In the **States** field, click **[Apply]** to apply the system states.
5. In the **Highstate** page, click **[Apply Highstate]** to propagate the changes to the clients.

24.4. Migrate SSL Certificate Authorities

Use this procedure to replace the root certificate authority (CA) of a Uyuni server, for example to replace an older self-signed CA that does not mark the **X509v3 Basic Constraints** extension as **critical**. For more information, see **Administration Guide › Ssl Certs Imported › Ssl Certs Verify Ca Basic Constraints**.

> Python 3.13 and later (and probably other SSL clients) validates certificates strictly and rejects any CA certificate that does not mark the **X509v3 Basic Constraints** extension as **critical**.

On Podman installations, the **mgradm ssl** commands help perform the rotation. They generate or import the new CA, update the server and database secrets, and restart the services.

The migration is done in two phases:

1. With **mgradm ssl addca**, you add the new CA to the trusted bundle so that both the old and new CA are trusted, and distribute it to all clients and proxies.
2. After all clients and proxies trust the new CA, **mgradm ssl rotate** switches the server to a certificate signed by the new CA and drops the old one.

Also consider any machine that is not managed by the server but still accesses it through the API or a browser, as it needs the new CA too. These machines are not detected by the trust check in ssl-certs-ca-migration.pdf, so track and verify them yourself.

> Do not switch the server certificate before all clients and proxies trust the new CA. A client that does not trust the new CA can no longer connect, and you have to deploy

the CA to it manually.

The certificates are stored in Podman secrets, which the container reads at startup. The server CA bundle, certificate, and key are in **uyuni-ca**, **uyuni-cert**, and **uyuni-key**; the database equivalents are **uyuni-db-ca**, **uyuni-db-cert**, and **uyuni-db-key**. On Kubernetes, the root CAs are read from the **uyuni-ca** and **db-ca** config maps, while the certificates are read from the **uyuni-cert** and **db-cert** secrets. The **mgradm ssl** commands manage these secrets on Podman; on Kubernetes, you update the config maps and secrets manually as described below.

24.4.1. Add the New CA to the Trust Bundle

On Podman, add the new CA with **mgradm ssl addca**. The command generates a new self-signed root CA, or imports a third-party one, appends it to the **uyuni-ca** and **uyuni-db-ca** secrets without changing the served certificate, restarts the services, and prints the SHA-256 fingerprint of the new CA.

To generate a new self-signed CA, run the following command on the container host. Replace the subject values with the ones for your organization. The server’ s fully qualified domain name is detected automatically, or you can pass it as an argument:

```
mgradm ssl addca --ssl-password CA_PASSWORD \
  --ssl-country COUNTRY --ssl-state STATE --ssl-city CITY \
  --ssl-org ORGANIZATION --ssl-ou "ORGANIZATION UNIT"
```

If **--ssl-password** is omitted, the command prompts for it. To use a third-party root CA instead, pass its certificate and any intermediate certificates in place of the generation options. Repeat **--ssl-ca-intermediate** for each intermediate CA:

```
mgradm ssl addca --ssl-ca-root /root/new-ca.pem \
  --ssl-ca-intermediate /root/intermediate-ca.pem
```

On Kubernetes, append the new root CA certificate to the **uyuni-ca** and **db-ca** config maps manually:

```
kubectl get configmap -n uyuni-server uyuni-ca -o jsonpath='{.data.ca\.crt}' >
combined.crt && \
echo "" >> combined.crt && \
cat ca.crt >> combined.crt && \
kubectl create configmap uyuni-ca --from-file=ca.crt=combined.crt --dry-run=client -o json
| \
kubectl patch -n uyuni-server configmap uyuni-ca --patch-file /dev/stdin  -o yaml && \
rm combined.crt
```

You also need to run a similar command for **db-ca**.

Once the server trusts the new CA, distribute it to every client and proxy. Deploy the new CA to all Salt clients by applying the highstate, as described in **Administration Guide › Ssl Certs Imported › Ssl Certs Import Deploy Root Ca**.

On each Uyuni Proxy, make it trust the new CA without a full redeployment. The proxy reads its trusted CA from the **uyuni-ca** Podman secret, so update that secret directly. After **mgradm ssl addca**, the server publishes the combined CA bundle, which contains both the old and new root CAs, at http://SERVER_FQDN/pub/RHN-ORG-TRUSTED-SSL-CERT. Download it on the proxy host, replace the secret, and restart the proxy:

```
curl -o /tmp/proxy-ca.pem http://SERVER_FQDN/pub/RHN-ORG-TRUSTED-SSL-CERT
podman secret create --replace uyuni-ca /tmp/proxy-ca.pem
systemctl restart uyuni-proxy-pod
```

Also make sure the new CA is deployed to any other machine that is not a reachable Salt minion but still accesses the server, such as proxies running on Kubernetes or unmanaged systems.

24.4.2. Confirm the Clients Trust the New CA

Before switching, confirm that the clients trust the new CA. On Podman, run **mgradm ssl rotate --check-only** to query the Salt minions and report which ones already trust the new CA and which do not, without performing the rotation:

```
mgradm ssl rotate --check-only
```

This check is best-effort: it only covers Salt minions that respond at the time of the check. Offline minions and non-minion systems such as proxies running on Kubernetes or unmanaged machines are not covered and must be verified manually.

To verify one of those manually, compare fingerprints. The **mgradm ssl addca** command already reported the SHA-256 fingerprint of the new CA. You can also recompute it from the new CA certificate. For a self-signed CA, read the staged certificate from the server:

```
mgrctl exec -- openssl x509 -in /root/ssl-build-rotation/RHN-ORG-TRUSTED-SSL-CERT \
  -noout -fingerprint -sha256
```

For a third-party CA, run the same command against the root CA file you passed to **mgradm ssl addca**.

The client trust file **/etc/pki/trust/anchors/RHN-ORG-TRUSTED-SSL-CERT** contains several certificates. List the fingerprints of all of them on a client and confirm that the new CA fingerprint is present:

```
while openssl x509 -noout -fingerprint -sha256; do :; done < /etc/pki/trust/anchors/RHN-
ORG-TRUSTED-SSL-CERT
```

24.4.3. Switch to the New CA

Only continue after all clients and proxies trust the new CA.

On Podman, switch to the new CA with **mgradm ssl rotate**. The command promotes the CA staged by **mgradm ssl addca**, issues a new server certificate signed by it, replaces the server and database certificates, keys, and CA secrets, and restarts the services. Before switching, it checks over Salt that the clients trust the new CA and refuses to continue if some do not, unless you pass **--force**. Like **mgradm ssl rotate --check-only**, this check only covers Salt minions that respond, so an offline minion that does not report does not block the rotation; verify such minions manually beforehand.

For a self-signed CA, run the following command. Use the fully qualified domain name of your Uyuni server, or allow it to be detected automatically, and repeat **--ssl-cname** for additional aliases:

```
mgradm ssl rotate --ssl-password CA_PASSWORD
```

To install a third-party server certificate signed by the new CA, provide the certificate, key, and root CA instead. The database reuses the server values unless you pass the matching **--ssl-db-*** options:

```
mgradm ssl rotate --ssl-ca-root /root/new-ca.pem \
  --ssl-ca-intermediate /root/intermediate-ca.pem \
  --ssl-server-cert /root/server.crt \
  --ssl-server-key /root/server.key
```

The **mgradm ssl addca** and **mgradm ssl rotate** commands are only available on Podman. On Kubernetes, replace the **uyuni-cert** and **db-cert** secrets with the new server certificate and key, update the **uyuni-ca** and **db-ca** config maps to contain only the new CA, then restart the server.

24.4.3.1. Rotate a Compromised CA Immediately

If the current CA is compromised, you cannot wait for the overlap period in which both the old and new CA are trusted. Pass **--emergency** to **mgradm ssl rotate** to generate a new CA together with a matching server certificate and drop the old CA in a single step, without the two-phase overlap:

```
mgradm ssl rotate --emergency --ssl-password CA_PASSWORD
```

An emergency rotation drops the old CA immediately, so any client or proxy that does not yet trust the new CA can no longer connect until you deploy the new CA to it manually. After an emergency rotation, distribute the new CA to all clients and proxies as described in ssl-certs-ca-migration.pdf, and update the proxies as described below.

24.4.4. Update Proxies

In ssl-certs-ca-migration.pdf, you already made each proxy trust the new CA by updating its **uyuni-ca** Podman secret. Now that the server uses a certificate signed by the new CA, every proxy also needs its own certificate signed by it. For each proxy, regenerate the proxy configuration and redeploy the proxy:

Installation and Upgrade Guide › Container Deployment › Proxy Deployment.

24.5. HTTP 严格传输安全性

HTTP 严格传输安全性 (HSTS) 是帮助防范网站遭受中间人攻击（例如协议降级攻击和 Cookie 劫持）的策略机制。

在 Uyuni 中，HSTS 功能默认处于启用状态。如果您需要在服务器上禁用该功能，请按照以下过程操作：

过程：在服务器上禁用 HSTS

1. On the server container host, as root, execute the following command to create a new configuration file with setting **max-age=0**:

```
mgrctl exec -- \
  echo 'Header always set Strict-Transport-Security "max-age=0;
includeSubDomains"' \
  > /etc/apache2/conf.d/zz-spacewalk-www-hsts.conf
```

2. 使用以下命令重启 Apache：

```
mgrctl exec -- systemctl restart apache2
```

如果您需要在代理上禁用该功能，请按照以下过程操作：

过程：在代理上禁用 HSTS

1. On the server container host, as root, execute the following command to create a new configuration file with setting **max-age=0**:

```
echo 'Header always set Strict-Transport-Security "max-age=0;
includeSubDomains' \
  > /etc/uyuni/custom-httpd.conf
```

2. 运行以下命令：

```
mgrpxy install --tuning-httpd /etc/uyuni/custom-httpd.conf config.tar.gz
```

When naming the new config file **<filename>.conf**, make sure it is loaded at the right time. For example, to override something defined in **spacewalk-www.conf** the new file needs to be alphabetically after this file. For more information about how Apache loads files, see https://httpd.apache.org/docs.

在使用 Uyuni 生成的默认 SSL 证书或自我签名证书的情况下启用 HSTS 后，浏览器将拒绝通过 HTTPS 进行连接，除非用于为此类证书签名的 CA 受浏览器信任。如果您使用的是 Uyuni 生成的 SSL 证书，可以通过将 http://<服务器主机名>/pub/RHN-ORG-TRUSTED-SSL-CERT 中的文件导入到所有用户的浏览器来信任该证书。

章 25. 订阅匹配

您的 SUSE 产品需要订阅，订阅由 SUSE Customer Center (SCC) 管理。Uyuni 会运行夜间报告，以检查您的 SCC 帐户下所有已注册客户端的订阅状态。该报告提供有关哪些客户端使用哪些订阅、您的剩余订阅数和可用订阅数，以及哪些客户端目前没有订阅的信息。

导航到**审计 › 订阅匹配**以查看报告。

The **Subscriptions Report** tab gives information about current and expiring subscriptions.

The **Unmatched Products Report** tab gives a list of clients that do not have a current subscription. This includes clients that could not be matched, or that are not currently registered with Uyuni. The report includes product names and the number of systems that remain unmatched.

The **Pins** tab allows you to associate individual clients to the relevant subscription. This is especially useful if the subscription manager is not automatically associating clients to subscriptions successfully.

The **Messages** tab shows all the messages generated by subscription matcher during the matching process. They provide information to help understand the results and to improve the matching.

You can also download the reports in .csv format, or access them from that command prompt in the /var/lib/spacewalk/subscription-matcher/ directory.

By default, the subscription matcher runs daily, at midnight. To change this, navigate to **Admin › Task Schedules** and click **gatherer-matcher-default**. Change the schedule as required, and click [**Update Schedule**].

由于报告只能将当前客户端与当前订阅相匹配，因此您可能会发现匹配结果随时间而变化。同一个客户端不一定总与同一个订阅匹配。原因可能是新客户端正在注册或未注册，或者添加了订阅或订阅失效。

订阅匹配器会自动尝试减少不匹配的产品数量，具体数量受您帐户中订阅的条款和条件限制。但是，如果硬件信息不完整、虚拟机主机指派未知，或者客户端在未知公有云中运行，则匹配器可能会显示您没有足够的可用订阅。请务必提供有关 Uyuni 中您的客户端的完整数据，以帮助确保信息准确。

订阅匹配器不一定总能准确匹配客户端和订阅。它并非旨在替代审计功能。

25.1. 将客户端关联到订阅

如果订阅匹配器无法自动将特定客户端与正确的订阅相匹配，您可以手动关联它们。创建关联后，订阅匹配器倾向于将特定订阅与给定的系统或系统组相匹配。

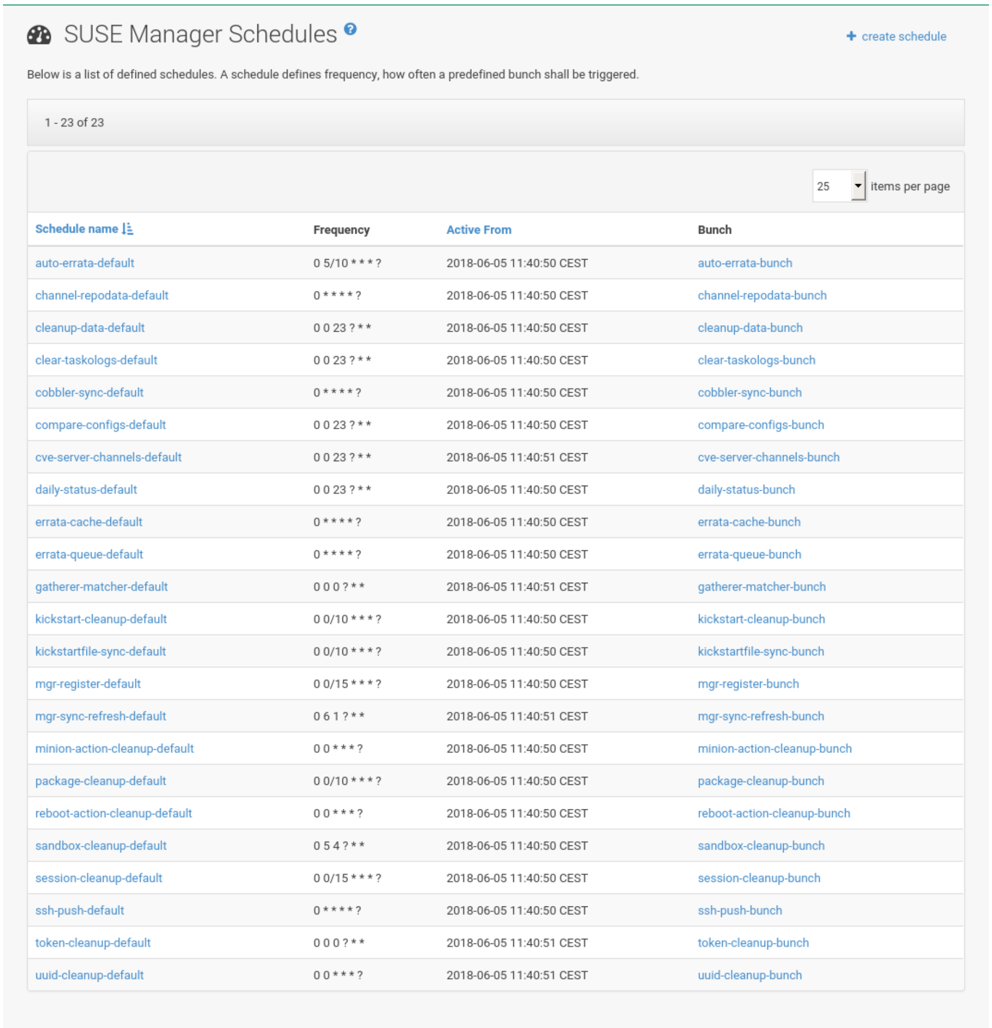
但是，匹配器并非始终遵守关联。这取决于订阅是否可用，以及订阅是否可以应用于该客户端。此外，如果关联导致匹配违反订阅的条款和条件，或者如果匹配器检测到在忽略关联的情况下匹配结果更准确，则就会忽略关联。

要添加新关联，请单击 [**添加关联**]，然后选择要关联的客户端。

我们不建议经常性使用关联或者对大量客户端使用关联。对于大多数安装而言，订阅匹配器工具通常已足够准确。

章 26. 任务日程安排

管理 › 任务日程安排下会列出所有预定义的任务组。


SUSE Manager Schedules

+ create schedule

Below is a list of defined schedules. A schedule defines frequency, how often a predefined bunch shall be triggered.

1 - 23 of 23

25 items per page

Schedule name	Frequency	Active From	Bunch
auto-errata-default	0 5/10 * * * ?	2018-06-05 11:40:50 CEST	auto-errata-bunch
channel-repodata-default	0 * * * * ?	2018-06-05 11:40:50 CEST	channel-repodata-bunch
cleanup-data-default	0 0 23 ? * *	2018-06-05 11:40:50 CEST	cleanup-data-bunch
clear-taskologs-default	0 0 23 ? * *	2018-06-05 11:40:50 CEST	clear-taskologs-bunch
cobbler-sync-default	0 * * * * ?	2018-06-05 11:40:50 CEST	cobbler-sync-bunch
compare-configs-default	0 0 23 ? * *	2018-06-05 11:40:50 CEST	compare-configs-bunch
cve-server-channels-default	0 0 23 ? * *	2018-06-05 11:40:51 CEST	cve-server-channels-bunch
daily-status-default	0 0 23 ? * *	2018-06-05 11:40:50 CEST	daily-status-bunch
errata-cache-default	0 * * * * ?	2018-06-05 11:40:50 CEST	errata-cache-bunch
errata-queue-default	0 * * * * ?	2018-06-05 11:40:50 CEST	errata-queue-bunch
gatherer-matcher-default	0 0 0 ? * *	2018-06-05 11:40:51 CEST	gatherer-matcher-bunch
kickstart-cleanup-default	0 0/10 * * * ?	2018-06-05 11:40:50 CEST	kickstart-cleanup-bunch
kickstartfile-sync-default	0 0/10 * * * ?	2018-06-05 11:40:50 CEST	kickstartfile-sync-bunch
mgr-register-default	0 0/15 * * * ?	2018-06-05 11:40:50 CEST	mgr-register-bunch
mgr-sync-refresh-default	0 6 1 ? * *	2018-06-05 11:40:51 CEST	mgr-sync-refresh-bunch
minion-action-cleanup-default	0 0 * * * ?	2018-06-05 11:40:50 CEST	minion-action-cleanup-bunch
package-cleanup-default	0 0/10 * * * ?	2018-06-05 11:40:50 CEST	package-cleanup-bunch
reboot-action-cleanup-default	0 0 * * * ?	2018-06-05 11:40:50 CEST	reboot-action-cleanup-bunch
sandbox-cleanup-default	0 5 4 ? * *	2018-06-05 11:40:50 CEST	sandbox-cleanup-bunch
session-cleanup-default	0 0/15 * * * ?	2018-06-05 11:40:50 CEST	session-cleanup-bunch
ssh-push-default	0 * * * * ?	2018-06-05 11:40:50 CEST	ssh-push-bunch
token-cleanup-default	0 0 0 ? * *	2018-06-05 11:40:51 CEST	token-cleanup-bunch
uuid-cleanup-default	0 0 * * * ?	2018-06-05 11:40:51 CEST	uuid-cleanup-bunch

单击 **SCC 日程安排 › 日程安排名称**打开**日程安排名称 › 日程安排基本细节**，在其中可以禁用该日程安排或更改其频率。

单击 **[编辑日程安排]** 可使用您的设置更新该日程安排。

要禁用日程安排，请单击右上角的 **[禁用日程安排]**。

> 由于日程安排对于 Uyuni 正常运行至关重要，因此请仅在绝对必要的情况下才禁用日程安排。

任务被禁用后仍会显示在列表中。单击**Uyuni 日程安排 › 日程安排名称**后，可通过单击 **[激活日程安排]** 重新激活该作业。

如果您单击某个组名称，将显示该组类型的运行列表和运行状态。

单击开始时间链接会返回**日程安排名称 › 日程安排基本细节**。

26.1. 预定义的任务组

系统默认会安排以下预定义的任务组，您可对其进行配置：

auto-errata-default

根据需要安排自动勘误更新。

channel-repodata-default

（重新）生成储存库元数据文件。

cleanup-data-default

从数据库中清理已过时的软件包更改日志和监控时序数据。

clear-taskologs-default

根据作业类型，从数据库中清除超过指定天数的任务引擎 (taskomatic) 历史数据。

cobbler-sync-default

将 Uyuni 中的分布数据和配置文件数据同步到 Cobbler。有关由 Cobbler 提供支持的自动安装的详细信息，请参见 **Client Configuration Guide › Autoinst Intro**。

compare-configs-default

将存储在配置通道中的配置文件与存储在所有已启用配置的服务器上的文件进行比较。要查看比较结果，请单击**系统**选项卡并选择相关系统。转到**配置 › 比较文件**。有关详细信息，请参见 **Reference Guide › Systems › Sd Configuration › Sd Config Compare Files**。

cve-server-channels-default

更新用于在**审计 › CVE 审计**页面上显示结果的内部预计算 CVE 数据。**审计 › CVE 审计**页面中的搜索结果将根据此日程安排的上次运行情况更新。有关详细信息，请参见 **Reference Guide › Audit › Audit Cve Audit**。

daily-status-default

将每日报告电子邮件发送到相关地址。有关如何为特定用户配置通知的详细信息，请参见 **Reference Guide › Users › User Details**。

errata-advisory-map-sync-default

更新内部的 SUSE 补丁供应商建议数据库表。如果有相关内容，每个补丁详细信息的“供应商建议”部分会显示 SUSE 提供的原始建议。

errata-cache-default

更新内部补丁缓存数据库表，这些表用于查找每个服务器的需要更新的软件包。此任务组还向可能对特定补丁感兴趣的用户发送通知电子邮件。有关补丁的详细信息，请参见 **Reference Guide › Patches › Patches Menu**。

errata-queue-default

为配置为接收自动更新（补丁）的服务器将这些更新（补丁）排队。

gatherer-matcher-default

通过运行在虚拟主机管理器中配置的虚拟主机收集器，来收集虚拟主机数据。当有更新的数据后，订阅匹配器作业即会运行。

kickstart-cleanup-default

清理过时的 Kickstart 会话数据。

kickstartfile-sync-default

生成与配置向导创建的 Kickstart 配置文件对应的 Cobbler 文件。

mgr-forward-registration-default

Synchronizes client registration data with SUSE Customer Center. By default, new, changed, or deleted client data are forwarded. To disable synchronization set in **/etc/rhn/rhn.conf**, run:

```
server.susemanager.forward_registration = 0
```

禁用与 SCC 的数据同步将导致 RMT、SMT、Uyuni 以及直接注册到 SCC 的客户端之间，对受管客户端的可见性降低。

通过同步数据，您可以确保所有注册客户端具有统一的视图。

告知我们您选择停用的原因，从而帮助改进我们的服务

mgr-sync-refresh-default

Synchronizes with SUSE Customer Center (**mgr-sync-refresh**). By default, all custom channels are also synchronized as part of this task. For more information about custom channel synchronization, see **Administration Guide › Custom Channels › Custom Channel Synchronization**.

minion-action-chain-cleanup-default

清理过时的操作链数据。

minion-action-cleanup-default

从文件系统中删除过时的客户端操作数据。首先，此任务组会尝试通过查找存储在 Salt 作业缓存中的相应结果，来完成任何可能未完成的操作。如果服务器遗漏了操作结果，就可能发生操作未完成的情况。对于成功完成的操作，此任务组会去除已执行的脚本文件等项目。

minion-checkin-default

在客户端上执行常规的签入。

notifications-cleanup-default

清理失效的通知消息。

oval-data-sync-default

生成所需的 OVAL 数据，以提高 CVE 审计查询的准确度。

package-cleanup-default

从文件系统中删除过时的软件包文件。

reboot-action-cleanup-default

任何超过六个小时未处理的重引导操作都将标记为失败，数据库中的关联数据将被清理。有关安排重引导操作的详细信息，请参见 **Reference Guide › Systems › Sd Provisioning › Sd Power Management**。

session-cleanup-default

清理过时的 Web 界面会话，这通常是用户登录之后，在注销之前关闭浏览器时临时存储的数据。

ssh-service-default

如果为客户端配置了 **SSH 推送**联系方法，则提示客户端通过 SSH 来与 Uyuni 通信。另外，在重引导后继续执行操作链。

system-overview-update-queue-default

更新系统概览数据。

system-profile-refresh-default

在所有系统上执行硬件刷新。此任务组每月只执行一次，可能会增加 Uyuni 服务器上的负荷。该作业会使用 **Specialized Guides › Salt › Salt 率限制**。如果要调整批次大小，请参见 **Specialized Guides › Large Deployments › Tuning › Java Salt Batch Size**。

token-cleanup-default

删除 Salt 客户端用来下载软件包和元数据的已失效储存库令牌。

update-payg-default

从配置的 PAYG 云实例收集身份验证数据。

update-reporting-default

更新本地报告数据库。

update-reporting-hub-default

从外围 Uyuni 服务器收集所有报告数据，并更新中心报告数据库。

update-system-overview-default

定期检查系统概览数据是否保持最新。

uuid-cleanup-default

清理过时的 UUID 记录。

章 27. 微调更改日志

某些软件包的更改日志项列表很长。默认会下载这些数据，但日志中保留的信息不一定有用。为了限制下载的更改日志元数据量并节省磁盘空间，您可以对磁盘上保留的项数施加限制。

> Use **mgrctl term** before running steps inside the server container.

This configuration option is in the **/etc/rhn/rhn.conf** configuration file. The parameter defaults to **20**. Changing this value to **0** will provide an unlimited number of entries.

```
java.max_changelog_entries = 20
```

如果您设置此参数，它只在同步新软件包后对其生效。

After changing this parameter, restart services with **mgradm restart**.

您可能想要删除再重新生成缓存的数据，以去除旧数据。

> 删除再重新生成缓存的数据可能需要很长时间。根据您的通道数量和要删除的数据量，此过程可能需要几个小时。该任务由 Taskomatic 在后台运行，因此您可以在操作完成时继续使用 Uyuni，但应该预料到性能会有所下降。

可以从命令行删除缓存的数据并请求重新生成数据：

```
spacewalk-sql -i
```

然后在 SQL 数据库提示符下输入：

```
DELETE FROM rhnPackageRepodata;
INSERT INTO rhnRepoRegenQueue (id, CHANNEL_LABEL, REASON, FORCE)
(SELECT sequence_nextval('rhn_repo_regen_queue_id_seq'),
        C.label,
        'cached data regeneration',
        'Y'
        FROM rhnChannel C);
\q
```

章 28. 用户

Uyuni 管理员可以添加新用户、授予权限以及停用或删除用户。如果您正在管理大量的用户，可以将用户指派到系统组，以便在组级别管理权限。还可以更改 Web UI 的系统默认值，包括语言和主题默认值。

> The **Users** menu is only available if you are logged in with the Uyuni administrator account.

To manage Uyuni users, navigate to **Users › User List › All** to see all users in your Uyuni Server. Each user in the list shows the username, real name, assigned roles, the date the user last signed in, and the current status of the user. Click **[Create User]** to create a new user account. Click the username to go to the **User Details** page.

要将新用户添加到您的组织，请单击 **[创建用户]**，填写新用户的细节，然后单击 **[创建登录名]**。

28.1. 口令要求

Uyuni 出厂时已预设一系列默认值。

为确保所有新用户口令均符合组织的安全标准，Uyuni 管理员可选择强制执行口令创建规则。

在 Web UI 中，导航至**管理 › 管理器配置 › 口令策略** 以定义口令要求。可组合使用以下字段：

口令长度下限

此字段用于定义口令的最小字符数。

口令长度上限

此字段用于定义口令的最大字符数。

需要包含数字

此字段用于指定口令是否必须包含数字 (0-9)。

需要包含小写字母

此字段用于指定口令是否必须包含小写字母 (a-z)。

需要包含大写字母

此字段用于指定口令是否必须包含大写字母 (A-Z)。

限制连续的字符

此字段用于指定是否限制使用连续的字符。

需要包含特殊字符

此字段用于指定口令是否必须包含特殊字符。

允许使用的特殊字符

This field is enable only of the **Require Special Characters** is selected. Use it to specify which special characters are allowed, for example **!@#$%&***, and others.

限制重复的字符

此字段用于指定禁止的字符重复次数。

重复字符数上限

此字段用于指定字符的最大重复次数。

单击 **[保存]** 可保存口令设置的任何更改。

单击 **[重置]** 可将所有设置恢复为默认值。

> Uyuni 口令策略的默认值如下：
>
> - 口令长度下限：4
> - 口令长度上限：32
> - 需要包含大写字母：已选中

28.2. 停用和删除帐户

您可以停用或删除不再需要的用户帐户。已停用的用户帐户随时可以重新激活。已删除的用户帐户将不可见，且不可检索。

用户可以停用自己的帐户。但是，如果用户具有管理员角色，则必须先去除该角色，然后才能停用帐户。

已停用的用户无法登录到 Uyuni Web UI 或安排任何操作。用户在停用之前安排的操作将保留在操作队列中。Uyuni 管理员可以重新激活已停用的用户。

28.3. 用户角色

用户可被分配多个角色，且任一角色在任意时刻均可由多个用户拥有。系统中必须始终至少存在一个活跃的 Uyuni 管理员。

要更改用户的角色（Uyuni 管理员角色除外），请导航到**用户 › 用户列表 › 所有**，选择要更改的用户，然后根据需要选中或取消选中管理员角色。

To change a user’s Uyuni Administrator role, navigate to **Admin › Users** and check or uncheck **Uyuni Admin?** as required.

表格 15. 用户角色权限

角色名称	说明
Uyuni 管理员	可以执行所有功能，包括更改其他用户的权限。
组织管理员	管理激活密钥、配置、通道和系统组。
激活密钥管理员	管理激活密钥。
映像管理员	管理映像配置文件、build 和存储区。
配置管理员	管理系统配置。
通道管理员	管理软件通道，包括使通道全局可订阅，以及创建新通道。
系统组管理员	管理系统组，包括创建和删除系统组、将客户端添加到现有组，以及管理用户对系统组的访问权限。
普通用户	提供标准的访问权限级别。新创建的用户会自动被分配此角色。

28.4. 创建额外角色

With Role-Based Access Control in Uyuni, you can create custom roles (referred to as Access Groups in the Web UI) to fine-tune user permissions.

To manage and create custom roles, navigate to **Admin › Access Control** in the Web UI and click **[Create Access Group]**.

Refer to **Administration Guide › Role Based Access Control** for more detailed information about these roles.

28.5. 用户权限和系统

如果您已创建系统组来管理客户端，可以将组指派到用户，让他们进行管理。

To assign a user to a system group, navigate to **Users › User List**, click the username to edit, and go to the **System Groups** tab. Check the groups to assign, and click **[Update Defaults]**.

您还可以为用户选择一个或多个默认系统组。当该用户注册新客户端时，默认会将其指派到所选的系统组。这样，该用户就能立即访问新注册的客户端。

To manage external groups, navigate to **Users › System Group Configuration**, and go to the **External Authentication** tab. Click **[Create External Group]** to create a new external group. Give the group a name, and assign it to the appropriate system group.

有关系统组的详细信息，请参见 **Reference Guide › Systems › System Groups**。

To see the individual clients a user can administer, navigate to **Users › User List**, click the username to edit, and go to the **Systems** tab. To carry out bulk tasks, you can select clients from the list to add them to the system set manager.

有关系统集管理器的详细信息，请参见 **Client Configuration Guide › System Set Manager**。

28.6. 用户和通道权限

可以将用户作为使用通道内容的订阅者，或者作为可以自行管理通道的管理员，指派到组织中的软件通道。

要为用户订阅通道，请导航到**用户 › 用户列表**，单击要编辑的用户名，然后转到**通道权限 › 订阅**选项卡。选中要指派的通道，然后单击 **[更新权限]**。

要为用户授予通道管理权限，请导航到**用户 › 用户列表**，单击要编辑的用户名，然后转到**通道权限 › 管理**选项卡。选中要指派的通道，然后单击 **[更新权限]**。

列表中的某些通道可能无法订阅。原因通常与用户的管理员状态或通道全局设置有关。

28.7. 用户默认语言

创建新用户时，您可以选择 Web UI 使用的语言。创建用户后，可以通过导航到**首页 › 我的首选项**来更改语言。

The default language is set in the **rhn.conf** configuration file. To change the default language, open the **/etc/rhn/rhn.conf** file and add or edit this line:

```
web.locale = <语言代码>
```

If the parameter is not set, the default language is **en_US**.

在 Uyuni 中可以使用以下语言：

表格 16. 可用语言代码

Language code	Language	Dialect
bn_IN	Bangla	India
ca	Catalan	
de	German	
en_US	English	United States
es	Spanish	
fr	French	
gu	Gujarati	
hi	Hindi	
it	Italian	

Language code	Language	Dialect
ja	Japanese	
ko	Korean	
pa	Punjabi	
pt	Portuguese	
pt_BR	Portuguese	Brazil
ru	Russian	
ta	Tamil	
zh_CN	Chinese	Mainland, Simplified
zh_TW	Chinese	Taiwan, Traditional

Translations in Uyuni are provided by the community, and could be incorrect or incomplete. Where a translation is not available, the Web UI defaults to English (**en_US**).

28.7.1. 用户默认界面主题

Uyuni Web UI 默认使用适用于您所安装的产品的主题。您可以更改主题以反映 Uyuni 或 SUSE Multi-Linux Manager 颜色。SUSE Multi-Linux Manager 主题还提供深色选项。

You can change the default theme in the **rhn.conf** configuration file. To change the default theme, open the **/etc/rhn/rhn.conf** file and add or edit this line:

```
web.theme_default = <主题>
```

表格 17. 可用的 WebUI 主题

Theme Name	Colors	Style
suse-light	SUSE Multi-Linux Manager	Light
suse-dark	SUSE Multi-Linux Manager	Dark
uyuni	Uyuni	Light

章 29. 支持

When you manage systems in Uyuni where you are entitled for support from SUSE, you can get the support data like **supportconfig** or **sosreport**. You will get the data from the managed client and upload it directly to SUSE Global Technical Support.

29.1. 生成服务请求编号

在向全球技术支持平台提交支持数据前，您需要先生成一个服务请求编号。

如需创建服务请求，请访问网址 https://scc.suse.com/support/requests，并按照页面提示完成操作，同时记录生成的服务请求编号。

> **隐私声明**
>
> SUSE 将系统报告视为机密数据。有关 SUSE 隐私政策承诺的详细内容，请访问 https://www.suse.com/company/policies/privacy/。

29.2. 从 Uyuni 中收集支持数据并上载到 SUSE

过程：通过 Uyuni Web UI 收集并上载支持数据

1. 在 Uyuni Web UI 中，导航到**系统 › 系统列表 › 所有**，选择要将其支持数据添加到案例中的客户端。
2. Then navigate to the **Details** tab and select the **Support** sub-tab.
3. In the field **Support Case Number** enter the service request number created above.
4. In the field **Upload Region** select option **EU** or **US** depending on the server where you want to upload your data.
5. In the field **Command-line Arguments** you can enter options for tool which is used to collect the data from the target system. Which tool will be used can be found in the tip below this field.
6. In the field **Earliest** select the time of running this action.
7. 单击 **[日程安排]** 按钮。该操作即进入日程安排队列，并在设定时间自动执行，完成客户端数据收集后直接将数据上载到目标服务器。

> 支持收集支持数据的产品：
>
> - Uyuni 代理

- 在中心辐射型架构场景中注册为外围服务器的 Uyuni 服务器
- 所有 SUSE Linux Enterprise 和 openSUSE 客户端
- SUSE Liberty 及兼容客户端
- Ubuntu 客户端
- Debian 客户端

To upload the support data from the main Uyuni Server please still use **mgradm support config** from the container host.

章 30. 查错

本章介绍了您在使用 Uyuni 时可能会遇到的一些常见问题及其解决方法。

30.1. 自动安装查错

如果使用普通的 Salt (salt-minion) 实现进行通信，您必须自行确保所需的所有软件（软件包）在配置的客户端通道中均可用。不建议将此实现与 Uyuni 一起使用。

使用默认的 Salt 捆绑包 (venv-salt-minion) 实现，您只需要一个客户端工具通道作为子通道，该通道与您的客户端基础通道兼容。所有必需的软件包都将包含在 Salt 捆绑包中。

- 检查是否为您的组织和用户提供了与自动安装配置文件中的基础通道相关的客户端工具软件通道。
- 检查是否为您的 Uyuni 提供了工具通道作为子通道。
- 仅当使用普通的 Salt（salt-minion）实现时，需要额外检查关联的通道中是否提供了必需的软件包和任何依赖项。

30.2. 对生命周期已结束产品的引导储存库进行查错

同步受支持的产品时，会自动在 Uyuni Server 上创建及重新生成引导储存库。当产品到达生命周期结束日期且不再受支持时，如果您要继续使用此产品，就必须手动创建引导储存库。

有关引导储存库的详细信息，请参见 **Client Configuration Guide › Bootstrap Repository**。

过程：创建生命周期已结束产品的引导储存库

1. 在 Uyuni 容器主机的命令提示符处，以 root 身份进入服务器容器：

```
mgrctl term
```

2. 在容器内部，执行以下步骤：

    a. List the available unsupported bootstrap repositories with the **--force** option, for example:

    ```
    mgr-create-bootstrap-repo --list --force
    1. SLE-12-SP2-x86_64
    2. SLE-12-SP3-x86_64
    ```

    b. 使用适当的储存库名称作为产品标签，创建引导储存库：

    ```
    mgr-create-bootstrap-repo --create SLE-12-SP2-x86_64 --force
    ```

如果您不想手动创建引导储存库，可以检查您需要的产品和引导储存库是否有 LTSS。

30.3. 对克隆的 Salt 客户端进行查错

如果您曾经使用过超级管理程序克隆实用程序，并尝试注册克隆的 Salt 客户端，您可能会收到以下错误：

```
抱歉，找不到该系统。
```

发生该错误的原因是新的克隆系统与现有的已注册系统具有相同的计算机 ID。您可以手动调整此数据以修复该错误，然后便可成功注册克隆的系统。

有关详细信息和说明，请参见 **Administration Guide › Troubleshooting › Tshoot Registerclones**。

30.4. 对容器全盘空间用尽事件进行查错

如果挂接为容器永久性存储媒体的专用磁盘的存储空间用尽，您需要采取紧急措施。

Proceed as follows to solve the problem with resizing the storage medium. Run all listed commands as **root** on the container host.

过程：调整存储媒体大小

1. 增加磁盘大小。需要采取什么措施取决于具体的安装情境。
2. If the disk is partitioned (for example, there is a **/dev/vdb1** for disk **/dev/vdb**), run the following commands:

   运行以下命令：

   a. **parted /dev/vdb**
   b. **(parted) print**
   c. **(parted) resizepart NUMBER 100%** where **NUMBER** is the partition number shown by **print** command (for example, **1** if **/dev/vdb1**)
   d. **(parted) quit**
3. 调整文件系统的大小。例如，对于 XFS 文件系统，请运行以下命令：

   ```
   xfs_growfs /dev/vdb1
   ```

完成该过程后，XFS 文件系统应该会使用该磁盘上的所有可用空间。

30.5. 对损坏的储存库进行查错

储存库元数据文件中的信息可能会损坏或过时。这可能会在更新客户端时造成问题。您可以通过去除再重新生成这些文件来解决此问题。生成新的储存库数据文件后，更新应该可以按预期进行。

过程：解决储存库数据损坏的问题

1. Remove all files from **/var/cache/rhn/repodata/<channel-label>**. 如果您不知道通道标签是什么，可以在 Uyuni Web UI 中导航到**软件 › 通道 › 通道标签**找到它。
2. 在容器主机上的命令行中执行以下命令，以在容器中重新生成文件：

```
mgrctl exec -ti -- spacecmd softwarechannel_regenerateyumcache <channel-label>
```

30.6. 对包含有冲突软件包的自定义通道进行查错

设置包含有冲突软件包的自定义通道时，某些功能（例如创建引导储存库）可能会导致未定义的行为，并使客户端注册失败。

For example, conflicting packages with higher version numbers could be included into the bootstrap repository. Such packages (for example, **python3-zmq** or **zeromq**) may corrupt the creation of the bootstrap repository or cause issues during bootstrap of the client.

When the custom channel (for example, an EPEL channel) is added below the parent vendor channel, issues with conflicting packages cannot be solved directly. The way how to solve this is to separate the custom channel from the vendor channel. The custom channel needs to be created in a separate tree. In case that the custom channel needs to be delivered as a child, such environment can be created using Content Lifecycle Management (CLM). Sources in a CLM project can be added there from different trees. Using such an approach, the custom channel stays below the parent within the built environment. However, the vendor channel tree stays without the custom channel and the bootstrap repository. Then registering clients works correctly.

将包含有冲突软件包的自定义通道（salt、zeromq 等）创建为子通道时，以下步骤可能有助于避免该问题：

过程：避免自定义通道中包含有冲突的软件包

1. 从父通道中去除作为子通道的自定义通道。有关详细信息，请参见 **Administration Guide › Custom Channels › Manage Custom Channels**。
2. 在单独的树中创建自定义通道。有关详细信息，请参见 **Administration Guide › Custom Channels › Creating Custom Channels and Repositories**。 要在内容生命周期管理 (CLM) 中添加作为子通道的自定义通道，请执行以下操作：
   - In the Uyuni Web UI, navigate to **Content Lifecycle**, and click **[Create Project]**. Enter **Name** and **Label**.
   - 将源关联到项目。使用所需的供应商通道和自定义通道。（分享使用 CentOS8 的示例）
   - 将环境添加到项目中。例如，使用 CentOS8。
   - 要构建环境，请单击 **[构建]** 按钮。这会创建一个包含供应商通道和自定义通道的环境，这些通道可与激活密钥相关联并用于引导客户端。
3. 重要说明：在 CLM 项目中，建议添加一个过滤器用于排除有问题或有冲突的软件包。否则会在客户端更新期间安装版本号较高的有冲突软件包。有关过滤的详细信息，请参见 **Administration Guide › Content Lifecycle Examples › Exclude Higher Kernel Version**。
4. 要将最新补丁添加到 CLM 环境（包含供应商通道和自定义通道），请在项目中单击 **[构建]** 按钮。需要执

行此操作才能重新构建环境。

  - 有关 CLM 的详细信息，请参见 **Administration Guide › Content Lifecycle**。

30.7. 对禁用 FQDNS grain 时出现的问题进行查错

FQDNS grain 会返回系统中所有完全限定 DNS 服务的列表。通常很快就能完成这些信息的收集，但如果 DNS 设置配置错误，花费的时间可能会长很多。在某些情况下，客户端会变成无响应状态或者会崩溃。

为了防止发生此问题，您可以使用 Salt 标志来禁用 FQDNS grain。如果禁用 grain，您便可以使用网络模块提供 FQDNS 服务，而不会面临客户端变成无响应状态的风险。

这仅适用于较旧的 Salt 客户端。如果您是最近注册 Salt 客户端的，FQDNS grain 默认会禁用。

在 Uyuni Server 上的命令提示符下，使用以下命令禁用 FQDNS grain：

```
salt '*' state.sls util.mgr_disable_fqdns_grain
```

此命令会重启每个客户端并生成服务器需要处理的 Salt 事件。如果您的客户端非常多，可以改用批处理模式执行该命令：

```
salt --batch-size 50 '*' state.sls util.mgr_disable_fqdns_grain
```

等待批处理命令执行完。请不要按 `Ctrl`+`C` 中断该过程。

30.8. 磁盘空间查错

磁盘空间不足可能会对 Uyuni 数据库和文件结构造成严重影响，在大多数情况下，这种影响不可恢复。Uyuni 会监控特定目录的可用空间，在有问题时会发出警报（可配置）。有关空间管理的详细信息，请参见 **Administration Guide › Space Management**。

Generally, container volumes share space with the host filesystem. When the **btrfs** filesystem becomes full, it can prevent you from deleting files. To resolve this, you can add a small amount of storage temporarily to the filesystem using the **btrfs device add** command. This will allow you to delete files to gain space for further filesystem maintenance. When done with the maintenance, you can remove the temporary storage with **btrfs device delete**. For more information about this topic, see https://www.suse.com/support/kb/doc/?id=000018779.

可以通过去除未使用的软件通道来恢复磁盘空间。

- 有关如何删除供应商通道的说明，请参见 **Administration Guide › Channel Management**。
- 有关如何删除自定义通道的说明，请参见 **Administration Guide › Custom Channels**。

您还可以检查自定义通道的同步频率。有关如何处理自定义通道同步的说明，请参见 **Administration Guide › Custom Channels › Custom Channel Synchronization**。

还可以通过清理未使用的激活密钥、内容生命周期项目和客户端注册来恢复磁盘空间。还可以去除多余的数据库项：

过程：去除多余的数据库项

1. Use the **spacewalk-data-fsck** command to list any redundant database entries.
2. Use the **spacewalk-data-fsck --remove** command to delete them.

30.8.1. Disk Check Warnings for Missing Directories

After migrating to containerized deployments from 5.0 or when using custom disk check configurations, you might see warnings for directories that no longer exist. This commonly occurs when the **DISKCHECKDIRS** environment variable references **/var/lib/pgsql** or its subdirectories.

In containerized deployments, the PostgreSQL database runs in its own container, and the **/var/lib/pgsql** directory no longer exists in the server container. The database files are stored in a separate database container volume.

If you see warnings like these in the logs or notifications:

```
DISKCHECK: Directory /var/lib/pgsql does not exist
DISKCHECK: Directory /var/lib/pgsql/data does not exist
```

This indicates that the disk check configuration includes directories that are no longer valid in containerized deployments.

30.8.1.1. Cause

Prior to containerization, customers could configure custom directories to monitor by setting the **spacecheck_dirs** parameter in **/etc/rhn/rhn.conf**. A common configuration included **/var/lib/pgsql** to monitor database disk space.

In containerized deployments:

- The **spacecheck_dirs** parameter in **/etc/rhn/rhn.conf** is still read, but is overridden by the **DISKCHECKDIRS** environment variable if set
- If **DISKCHECKDIRS** is not set, the system falls back to **spacecheck_dirs** from the configuration file
- The database runs in a separate container and has its own dedicated disk check

30.8.1.2. Fix

Remove the obsolete directory references from the **DISKCHECKDIRS** environment variable.

Procedure: Removing Obsolete Directory References

1. Check if you have a systemd override file with custom directory configuration:

   ```
   cat /etc/systemd/system/uyuni-server.service.d/diskcheck.conf
   ```

2. If the file contains **/var/lib/pgsql** or other non-existent directories, update it to use only valid directories:

```
cat > /etc/systemd/system/uyuni-server.service.d/diskcheck.conf << 'EOF'
[Service]
Environment="DISKCHECKDIRS=/var/spacewalk /var/cache /srv"
EOF
```

3. Apply the configuration by restarting Uyuni:

```
mgradm restart
```

30.8.1.3. 结果

The disk check warnings for missing directories will no longer appear. The server container will only monitor the directories that exist within the container.

The database container has its own separate disk check that monitors the PostgreSQL data directory (**/var/lib/pgsql/data**) within the database container. This check is automatically configured and does not require manual intervention.

For more information about disk space management in containerized deployments, see **Administration Guide › Space Management**.

30.9. 防火墙查错

If you are using a firewall that blocks outgoing traffic, it can either **REJECT** or **DROP** network requests. If it is set to **DROP** then you might find that synchronizing with the SUSE Customer Center times out.

之所以发生这种情况，是因为同步进程需要访问为非 SUSE 客户端（而不仅仅是 SUSE Customer Center）提供软件包的第三方储存库。当 Uyuni 服务器尝试访问这些储存库以检查它们是否有效时，防火墙会丢弃请求，并且同步进程会继续等待响应，直到超时。

如果发生这种情况，同步将持续很长时间，然后失败，并且您的非 SUSE 产品不会显示在产品列表中。

可以通过多种不同的方法解决此问题。

最简单的方法是将防火墙配置为允许访问非 SUSE 储存库所需的 URL。这样，同步进程就能访问这些 URL 并成功完成。

If allowing external traffic is not possible, configure your firewall to **REJECT** requests from Uyuni instead of **DROP**. This rejects requests to third-party URLs, so that the synchronization fails early rather than times out, and the products are not shown in the list.

如果您无权配置防火墙，可以考虑在 Uyuni Server 上单独设置一个防火墙。

30.10. 对通过 WAN 连接在 Uyuni Server 与 Proxy 之间同步时间过长的问题进行查错

Depending on what changes are executed in the WebUI or via an API call to distribution or system settings, **cobbler sync** command may be required to transfer files from Uyuni Server to Uyuni Proxy systems. To accomplish this, Cobbler uses a list of proxies specified in **/etc/cobbler/settings**.

Due to its design, **cobbler sync** is not able to sync only the changed or recently added files.

Instead, executing **cobbler sync** triggers a full sync of the **/srv/tftpboot** directory to all specified proxies configured in **/etc/cobbler/settings**. It is also influenced by the latency of the WAN connection between the involved systems.

The process of syncing may take a considerable amount of time to finish according to the logs in **/var/log/cobbler/**.

例如，该过程开始时间为：

```
Thu Jun  3 14:47:35 2021 - DEBUG | running python triggers from
/var/lib/cobbler/triggers/task/sync/pre/*
Thu Jun  3 14:47:35 2021 - DEBUG | running shell triggers from
/var/lib/cobbler/triggers/task/sync/pre/*
```

结束时间为：

```
Thu Jun  3 15:18:49 2021 - DEBUG | running shell triggers from
/var/lib/cobbler/triggers/task/sync/post/*
Thu Jun  3 15:18:49 2021 - DEBUG | shell triggers finished successfully
```

传输量大约为 1.8 GB。传输花费了将近 30 分钟。

By comparison, copying a single big file of the same size as **/srv/tftboot** completes within several minutes.

Switching to an **rsync**-based approach to copy files between Uyuni Server and Proxy may help to reduce the transfer and wait times.

可以从 https://suse.my.salesforce.com/sfc/p/1i000000gLOd/a/1i000000ll5B/B2AmvIJN2_JsAyjTQzCVP_x5ioVgd0bYN9X9NpMugS8 下载用于完成此任务的脚本。

The script does not accept command line options. Before running the script, you need to manually edit it and set correctly **MLMHOSTNAME**, **MLMIP** and **MLMPROXY1** variables for it to work correctly.

> 不支持调整该脚本的个别设置。该脚本及其包含的注释旨在提供传输过程以及要考虑的步骤的概述。如需进一步的帮助，请联系 SUSE 咨询部门。

对于以下情况，最好使用建议的脚本用法：

* Uyuni 代理系统是通过 WAN 连接的；
* **/srv/tftboot** contains a high number of files for distributions and client PXE boot files, in total several thousand files;
* Any proxy in **/etc/cobbler/settings** has been disabled, otherwise Uyuni will continue to sync content to the proxies.

```
#proxies:
# - "MLMproxy.MLMproxy.test"
# - "MLMproxy2.MLMproxy.test"
```

过程：分析新的同步速度

1. 为 Uyuni 与相关系统之间的 TCP 流量创建转储。
   * 在 Uyuni 服务器上：

     ```
     tcpdump -i ethX -s 200 host <SUSE Manager Proxy 的 IP 地址> and not ssh
     ```

   * 在 Uyuni 代理上：

     ```
     tcpdump -i ethX -s 200 host <SUSE Manager 的 IP 地址> and not ssh
     ```

   * 这样就只会捕获大小为 200 的软件包，但足以运行分析。
   * 将 ethX 调整为由 Uyuni 用来与代理通信的相应网络接口。
   * 最后，不会捕获 ssh 通信，从而进一步减少了软件包的数量。
2. Start a **cobbler sync**.
   * To force a sync, delete the Cobbler json cache file first and then issue **cobbler sync**:

     ```
     rm /var/lib/cobbler/pxe_cache.json
     cobbler sync
     ```

3. When {command]**cobbler sync** is finished, stop the TCPdumps.
4. Open the TCPdumps using Wireshark, go to **Statistics > Conversations** and wait for the dump to be analyzed.
5. 切换到“TCP”选项卡。此选项卡上显示的数字是在 Uyuni 服务器与 Uyuni 代理之间捕获的对话总数。
6. Look for the column **Duration**.
   * 首先按升序排序，找出传输文件所花费的最短时间。
   * 继续按降序排序，找出大文件传输时间的最大值，例如内核和 initrd 传输。

   > 忽略端口 4505 和 4506，因为它们用于 Salt 通信。

TCPdumps 分析结果表明，将大小约为 1800 字节的小文件从 Uyuni Server 传输到 Proxy 大约花费了 0.3 秒。

此处的大文件不多，而大量的小文件导致建立了大量连接，因为每传输一个文件就要创建新的 TCP 连接。

因此，在知道最短传输时间和所需连接数（在本示例中大约为 5000）的情况下，可以大致估算出总传输时间：5000 * 0.3 / 60 = 25 分钟。

30.11. 非活动客户端查错

某个 Taskomatic 作业会定期 ping 客户端以确保它们保持连接状态。有 24 小时或更长时间未响应 Taskomatic 签入的客户端被视为非活动客户端。要在 Web UI 中查看非活动客户端的列表，请导航到**系统 › 系统列表 › 非活动**。

客户端可能由于多种原因而进入非活动状态：

- 客户端无权使用任何 Uyuni 服务。
- 客户端位于不允许 HTTPS 连接的防火墙后面。
- 客户端位于配置不当的代理后面。
- 客户端正在与其他 Uyuni Server 通信，或者连接配置不当。
- 客户端不在可以与 Uyuni 服务器通信的网络中。
- 防火墙阻止了客户端与 Uyuni Server 之间的通信。
- Taskomatic 配置不当。

有关客户端与服务器的连接的详细信息，请参见 **Client Configuration Guide › Contact Methods Intro**。

有关配置端口的详细信息，请参见 **Installation and Upgrade Guide › Network Requirements › Ports**。

有关防火墙查错的详细信息，请参见 **Administration Guide › Troubleshooting › Tshoot Firewalls**。

30.12. 服务器间同步查错

服务器间同步使用缓存来管理 ISS 主服务器和从属服务器。这些缓存容易出现 bug，导致创建无效的项。在这种情况下，即使更新到解决了 bug 的版本，这些 bug 也还可能出现，因为缓存仍在使用无效的项。如果您升级到新版 ISS 后仍然遇到问题，请清除所有缓存，以确保不再有旧项造成问题。

缓存错误可能导致同步失败并出现各种错误，但错误消息通常会报告如下内容：

```
考虑去除 /var/cache/rhn/satsync/* 中的 satellite-sync 缓存并使用相同的选项重新运行
satellite-sync。
```

可以通过删除 ISS 主服务器和 ISS 从属服务器上的缓存来解决此问题，这样，同步即可成功完成。

> To access a shell inside the Server container run **mgrctl term** on the container host.

过程：解决 ISS 缓存错误

1. 在 ISS 主服务器上的命令提示符下，以 root 身份删除主服务器的缓存文件：

```
rm -rf /var/cache/rhn/xml-*
```

2. 重启该服务：

```
rcapache2 restart
```

3. 在 ISS 主服务器上的命令提示符下，以 root 身份删除从属服务器的缓存文件：

```
rm -rf /var/cache/rhn/satsync/*
```

4. 重启该服务：

```
rcapache2 restart
```

30.13. 本地颁发者证书查错

Some older bootstrap scripts create a link to the local certificate in the wrong place. This results in zypper returning an **Unrecognized error** about the local issuer certificate. You can ensure that the link to the local issuer certificate has been created correctly by checking the **/etc/ssl/certs/** directory. If you come across this problem, you should consider updating your bootstrap scripts to ensure that zypper operates as expected.

30.14. 登录超时查错

默认情况下，Uyuni Web UI 要求用户在 30 分钟后重新登录。根据您的环境，可能需要调整登录超时值。

To adjust the value, you need to make the change in both **rhn.conf** and **web.xml**. Ensure you set the value in seconds in **/etc/rhn/rhn.conf**, and in minutes in **web.xml**. The two values must equal the same amount of time.

For example, to change the timeout value to one hour, set the value in **rhn.conf** to 3600 seconds, and the value in **web.xml** to 60 minutes.

过程：调整 Web UI 登录超时值

1. 在容器主机上，在服务器容器内打开命令行：

```
mgrctl term
```

   a. Open **/etc/rhn/rhn.conf** and add or edit this line to include the new timeout value in seconds:

```
web.session_database_lifetime = <以秒为单位的超时值>
```

b. 保存并关闭该文件。

c. Open **/etc/tomcat/web.xml** and add or edit this line to include the new timeout value in minutes:

```
<session-timeout>Timeout_Value_in_Minutes</session-timeout>
```

d. 保存并关闭该文件。

2. 在容器主机上，重启服务器以强制应用新配置：

```
systemctl restart uyuni-server.service
```

30.15. 邮件配置查错

For secure mail communication, you can enable authentication, define username and password, and enable **SSL** or **STARTLS** in **/etc/rhn/rhn.conf**:

```
java.smtp_server = string (default: localhost)
java.smtp_port = integer (default: 25)
java.smtp_auth = true/false (default: false)
java.smtp_ssl = true/false (default: false)
java.smtp_starttls = true/false (default: false)
java.smtp_user = string (default: null)
java.smtp_pass = string (default: null)
```

To increase the connection timeout for SMTP server communication, the following parameters can be set in **/etc/rhn/rhn.conf**:

```
java.smtp_timeout = integer (default: 5000)
java.smtp_connection_timeout = integer (default: 5000)
java.smtp_write_timeout = integer (default: 5000)
```

30.16. 批量计算机 ID 重复问题

如果您使用 Uyuni 来管理虚拟机，您可能发现创建 VM 的克隆版本会很有用。克隆版本是使用主磁盘（与现有磁盘完全相同的副本）的 VM。

虽然克隆 VM 可以节省大量时间，但磁盘上复制的标识信息有时会导致问题。

如果您尝试注册两台计算机（其中一台是另一台的克隆机），通常希望 Uyuni 将它们识别为两个独立的客户端。但如果原始客户端与克隆机的计算机 ID 相同，Uyuni 会将两个客户端注册为同一个系统，且第二台计算机完成注册后，原有数据会被覆盖。

此问题可通过更改克隆机的计算机 ID 来解决，使 Uyuni 能够将它们识别为两个不同的客户端。有关更改计算机 ID 的详细信息及操作指南，请参见 **Administration Guide › Troubleshooting › Tshoot Registerclones**。

然而，在部分场景下，更改所有系统的计算机 ID 既不可行也不现实，因为这可能会影响依赖现有计算机 ID 运

行的其他应用程序。针对这类情况，Uyuni 提供了一种解决方案，可生成一个虚拟计算机 ID 供系统使用。

过程：在注册时生成 Uyuni 的计算机 ID

1. 创建引导脚本（如果已有该脚本，请跳过此步骤）。有关详细信息，请参见 **Client Configuration Guide › Registration Bootstrap**。
2. 打开该文件并更改相应参数：

```
GENERATE_OWN_MACHINEID=1
```

3. 启用此引导脚本后，计算机 ID 将不再发生冲突。

30.17. 对使用 noexec 挂载 /tmp 时出现的问题进行查错

Salt runs remote commands from **/tmp** on the client's file system. Therefore you must not mount **/tmp** with the **noexec** option. The other way to solve this issue is to override temporary directory path with the **TMPDIR** environment variable specified for the Salt service to make it pointing to the directory with no **noexec** option set. It is recommended to use systemd drop-in configuration file **/etc/systemd/system/venv-salt-minion.service.d/10-TMPDIR.conf** if Salt Bundle is used, or **/etc/systemd/system/salt-minion.service.d/10-TMPDIR.conf** if **salt-minion** is used on the client. The example of the drop-in configuration file content:

```
[Service]
Environment=TMPDIR=/var/tmp
```

30.18. 对使用 noexec 挂载 /var/tmp 时出现的问题进行查错

Salt SSH is using **/var/tmp** to deploy Salt Bundle to and execute Salt commands on the client with the bundled Python. Therefore you must not mount **/var/tmp** with the **noexec** option. It is not possible to bootstrap the clients, which have **/var/tmp** mounted with **noexec** option, with the Web UI because the bootstrap process is using Salt SSH to reach a client.

30.19. 对“磁盘空间不足”错误进行查错

Check the available disk space before you begin migration. We recommend locating **/var/spacewalk** and **/var/lib/pgsql** on separate XFS file systems.

When you are setting up a separate file system, edit **/etc/fstab** and remove the **/var/lib/pqsql** subvolume. Reboot the server to pick up the changes.

To get more information about an upgrade problem, check the migration log file. The log file is located at **/var/log/rhn/migration.log** on the system you are upgrading.

30.20. 通知查错

The default lifetime of notification messages is 30 days, after which messages are deleted from the database, regardless of read status. To change this value, add or edit this line in **/etc/rhn/rhn.conf**:

```
java.notifications_lifetime = 30
```

To enable or disable a notification type, add or edit a line as follows in **/etc/rhn/rhn.conf**:

```
java.notifications_type_disabled = OnboardingFailed,ChannelSyncFailed,\
  ChannelSyncFinished,CreateBootstrapRepoFailed,StateApplyFailed,\
  PaygAuthenticationUpdateFailed,EndOfLifePeriod,SubscriptionWarning
```

For default settings and configuration options, see the **usr/share/rhn/config-defaults/rhn_java.conf** template file.

30.21. 对软件包不一致问题进行查错

当客户端上的软件包被锁定时，Uyuni Server 可能无法正确确定适用的补丁集。如果发生这种情况，软件包更新会显示在 Web UI 中，但不会显示在客户端上，并且尝试更新客户端会失败。检查软件包锁定和排除列表，以确定是否在客户端上锁定或排除了软件包。

在客户端上，检查软件包锁定和排除列表，以确定是否已锁定或排除软件包：

- On SUSE Linux Enterprise and openSUSE, use the **zypper locks** command.

30.22. 对将 Grain 传递给启动事件时出现的问题进行查错

Every time a Salt client starts, it passes the **machine_id** grain to Uyuni. Uyuni uses this grain to determine if the client is registered. This process requires a synchronous Salt call. Synchronous Salt calls block other processes, so if you have a lot of clients start at the same time, the process could create significant delays.

为了解决此问题，Salt 中引入了一项新功能来避免进行单独的同步 Salt 调用。

要使用此功能，您可以在支持该功能的客户端上向客户端配置中添加一个配置参数。

To make this process easier, you can use the **mgr_start_event_grains.sls** helper Salt state.

> 这仅适用于已注册的客户端。如果您是最近注册 Salt 客户端的，系统默认会添加此配置参数。

On the Uyuni Server, at the command prompt, use this command to enable the **start_event_grains** configuration helper:

```
salt '*' state.sls util.mgr_start_event_grains
```

此命令会在客户端的配置文件中添加所需的配置，并在客户端重启时应用更改。如果您的客户端非常多，可以改用批处理模式执行该命令：

```
salt --batch-size 50 '*' state.sls mgr_start_event_grains
```

30.23. 代理连接和 FQDN 查错

有时，通过代理连接的客户端会显示在 Web UI 中，但不会显示它们是通过代理连接的。如果您连接时使用的不是完全限定的域名 (FQDN)，而代理对 Uyuni 而言是未知的，就可能发生此情况。

要更正此行为，请在代理上的客户端配置文件中指定其他 FQDN 作为 grain：

```
grains:
  susemanager:
    custom_fqdns:
      - name.one
      - name.two
```

30.24. 对注册克隆的客户端时出现的问题进行查错

如果您使用 Uyuni 来管理虚拟机，您可能发现创建 VM 的克隆版本会很有用。克隆版本是使用主磁盘（与现有磁盘完全相同的副本）的 VM。

虽然克隆 VM 可以节省大量时间，但磁盘上复制的标识信息有时会导致问题。

如果您有一个已注册的客户端，并创建了该客户端的克隆版本，在尝试注册该克隆版本时，您可能希望 Uyuni 将原始客户端和克隆版本注册为两个不同的客户端。但是，如果原始客户端和克隆版本中的计算机 ID 相同，则 Uyuni 会将这两个客户端注册为一个系统，并且现有客户端数据将由克隆版本的数据重写。

可以通过更改克隆版本的计算机 ID 来解决此问题，以便 Uyuni 将它们识别为两个不同的客户端。

> 此过程的每个步骤都在克隆的客户端上执行。此过程不会对原始客户端进行操作，原始客户端仍保持注册到 Uyuni。

过程：解决克隆的 Salt 客户端中的重复计算机 ID

1. **初始系统配置**
   a. 在克隆的计算机上，更改主机名和 IP 地址。 Make sure **/etc/hosts** contains the changes you made and the correct host entries.
2. **解决重复计算机 ID 问题**
   a. 对于支持 systemd 的发行套件：
      i. 如果您的计算机具有相同的计算机 ID，请以 root 身份删除每个重复客户端上的文件，然后重新创建这些文件：

```
rm /etc/machine-id
rm /var/lib/dbus/machine-id
rm /var/lib/zypp/AnonymousUniqueId
dbus-uuidgen --ensure
systemd-machine-id-setup
```

ii. If the cloned machine also has a folder in **/var/log/journal/** it needs to be renamed accordingly to the new machine ID. If names do not match, **journalctl** could not retrieve any log and **podman logs** would not show anything.

```
mv /var/log/journal/* /var/log/journal/$(cat /etc/machine-id)
```

b. 对于不支持 systemd 的发行套件：

i. 以 root 身份从 dbus 生成计算机 ID：

```
rm /var/lib/dbus/machine-id
rm /var/lib/zypp/AnonymousUniqueId
dbus-uuidgen --ensure
```

- 如果您要克隆一台后续需适配迁移至 SUSE Liberty Linux 系统的 Red Hat Enterprise Linux 8.10 服务器，则必须执行额外步骤修复内核配置文件。

Red Hat Enterprise Linux uses the machine ID to generate kernel entries in **/boot/loader/entries**. Not performing these steps will result in a mix of old and new kernel entries after the liberation, as SUSE Liberty Linux kernels will create new entries instead of replacing the old ones.

- 在更改计算机 ID 后、执行系统适配迁移前，请运行以下命令：

```
sudo rm -rf /boot/loader/entries/
sudo for ver in $(rpm -q kernel --qf '%{VERSION}-
%{RELEASE}.%{ARCH}\n'); do echo "Reinstalling kernel
$ver..."; sudo kernel-install add $ver /lib/modules/$ver;
done
sudo grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

- 有关释放 Red Hat Enterprise Linux 8.10 服务器的详细信息及示例，请参见 **Common Workflows Guide ›**

Workflow Liberate Rhel with Secureboot。

3. **重新配置 Salt 客户端**

   a. If your clients still have the same Salt client ID, delete the **minion_id** file on each client (FQDN is used when it is regenerated on client restart).

      i. 对于 Salt 受控端客户端：

      ```
      rm /etc/salt/minion_id
      rm -rf /etc/salt/pki
      ```

      ii. 对于 Salt 捆绑包客户端：

      ```
      rm /etc/venv-salt-minion/minion_id
      rm -rf /etc/venv-salt-minion/pki
      ```

   b. 从初始配置页面中删除接受的密钥，并从 Uyuni 中删除系统配置文件，然后使用以下命令重启客户端。

      i. 对于 Salt 受控端客户端：

      ```
      service salt-minion restart
      ```

      ii. 对于 Salt 捆绑包客户端：

      ```
      service venv-salt-minion restart
      ```

   c. Re-register the clients. Each client now has a different **/etc/machine-id** and should be correctly displayed on the **System Overview** page.

30.25. SL Micro 上的远程 root 登录

For enhanced security, new installations of SL Micro 6.2 and later do not allow password-based remote root login anymore, which affects server and proxy container hosts running on SL Micro and managed SL Micro clients. Also, SLE Micro 5.5 clients with password-based remote root login which will when be migrated to 6.1/6.2 will suddenly lose this access and must be newly configured. For more information, see SL Micro Release Notes 6.2 (https://documentation.suse.com/releasenotes/sle-micro/html/releasenotes_sle-micro_6.2/).

在部署 Uyuni 的组件（如 Uyuni 代理）时，默认情况下需要进行基于口令的远程 root 登录。您可通过以下步骤启用基于口令的远程 root 登录。

过程：在 SL Micro 上启用基于口令的 SSH root 登录

您可通过两种方式启用 SSH root 用户登录功能，两种方式均可生效，请根据实际设置环境选择合适的方案。

方案一：使用预配置的软件包

1. Install the package **openssh-server-config-rootlogin** from the UI/API on the client.
2. 在 UI 或终端中重引导容器主机以激活新配置

方案二：手动编辑 SSH 配置文件

1. Add a drop-in config **/etc/ssh/sshd_config.d/permit_root.conf** file, with the following content:

```
PermitRootLogin yes
```

2. 重新加载 SSH 服务器配置

```
systemctl reload sshd
```

3. 如果目前通过 SSH 连接服务器，请在断开连接前，新建一个 SSH 连接以验证服务器运行是否正常。

For more information about **transactional-update**, see https://documentation.suse.com/sle-micro/6.2/html/Micro-transactional-updates/index.html.

30.26. 对注册已删除的客户端时出现的问题进行查错

Use **mgrctl term** before running steps inside the server container.

有时可能无法重新注册已删除（已取消注册）的客户端。要解决此问题，应该先在 Uyuni 服务器（Salt 主控端）上删除一些 Salt 缓存文件，然后再尝试重新注册：

```
rm /var/cache/salt/master/thin/version
rm /var/cache/salt/master/thin/thin.tgz
```

30.27. 对在 Web UI 中注册失败且未显示任何错误的问题进行查错

在 Web UI 中进行初始注册时，所有 Salt 客户端使用的都是 Salt SSH。

由其性质决定，Salt SSH 客户端不会向服务器回报错误。

However, the Salt SSH clients store a log locally at **/var/log/salt-ssh.log** that can be inspected for errors.

30.28. 对 Red Hat CDN 通道和多个证书进行查错

有时，Red Hat 内容分发网络(CDN) 通道会提供多个证书，而 Uyuni Web UI 只能导入单个证书。如果 CDN 提供的证书与 Uyuni Web UI 已知的证书不同，即使该证书准确无误，验证也会失败，并且访问储存库的权限会被拒绝。收到的错误消息如下：

```
[错误]
储存库 '<repo_name>' 无效。
<repo.pem> 在指定的 URL 未找到有效元数据
历史记录:
 - [|] 尝试从 '<repo.pem>' 读取数据时出错
 - 访问 '<repo.pem>' 的权限被拒绝。
请检查为此储存库定义的 URL 是否指向有效储存库。
由于发生上述错误，正在跳过储存库 '<repo_name>'。
由于发生错误，无法刷新储存库。
HH:MM:SS RepoMDError: 无法访问储存库。可能未导入储存库 GPG 密钥
```

To resolve this issue, merge all valid certificates into a single **.pem** file, and rebuild the certificates for use by Uyuni:

过程：解析多个 Red Hat CDN 证书

1. On the Red Hat client, at the command prompt, as root, gather all current certificates from **/etc/pki/entitlement/** in a single **rh-cert.pem** file:

```
cat 866705146090697087.pem 3539668047766796506.pem redhat-entitlement- authority.pem
> rh-cert.pem
```

2. Gather all current keys from **/etc/pki/entitlement/** in a single **rh-key.pem** file:

```
cat 866705146090697087-key.pem 3539668047766796506-key.pem > rh-key.pem
```

现在，您可以按照**Client Configuration Guide › Clients Rh Cdn**中的说明将新证书导入 Uyuni Server。

30.29. RPC 超时查错

由于网络速度缓慢或网络链接断开，RPC 连接有时会超时。这会导致软件包下载或批处理作业挂起或运行时间超过预期。您可以通过编辑配置文件来调整 RPC 连接可以花费的最长时间。虽然这不能解决网络问题，但可以使进程失败而不是挂起。

过程：解决 RPC 连接超时

1. On the Uyuni Server, open the **/etc/rhn/rhn.conf** file and set a maximum timeout value (in seconds):

```
server.timeout =`数字`
```

On the Uyuni Proxy, open the **/etc/uyuni/proxy/config.yaml** file and set a maximum timeout value (in seconds). The proxy containers need to be restarted for the change to be effective:

```
timeout: `number`
```

2. On a SUSE Linux Enterprise Server client that uses zypper, open the **/etc/zypp/zypp.conf** file and set a maximum timeout value (in seconds):

```
## 有效值: [0,3600]
## 默认值: 180
download.transfer_timeout = 180
```

3. On a Red Hat Enterprise Linux client that uses yum, open the **/etc/yum.conf** file and set a maximum timeout value (in seconds):

```
timeout =`数字`
```

如果您将 RPC 超时限制为小于 **180** 秒，可能会中止完全正常的操作。

30.30. 对 Salt 客户端显示为关闭状态的问题和 DNS 设置进行查错

即使 Salt 客户端正在运行，软件包刷新或应用状态这样的操作也可能会标示为失败，并显示以下消息：

```
受控端已关闭或无法联系。
```

在此情况下，请尝试重新安排该操作。如果重新安排成功，发生问题的原因可能在于 DNS 配置有误。

To access a shell inside the Server container run **mgrctl term** on the container host.

When the Salt client is restarted, or in case the grains are refreshed, the client calculates its FQDN grains, and it is unresponsive until the grains are proceeded. When a scheduled action on Uyuni Server is going to be executed, Uyuni Server performs a **test.ping** to the client before the actual action to ensure the client is actually running and the action can be triggered.

By default, Uyuni Server waits for 5 seconds to get the response from **test.ping** command. If the response is not received within 5 seconds, then the action is set to fail with the message that the client is down or could not be contacted.

要解决此问题，请修复客户端上的 DNS 解析，使客户端在解析其 FQDN 时不会卡顿 5 秒时间。

If this is not possible, try to increase the value for **java.salt_presence_ping_timeout** in the **/etc/rhn/rhn.conf** file on the Uyuni Server to a value higher than 4.

例如：

```
mgrctl term
vim /etc/rhn/rhn.conf
java.salt_presence_ping_timeout = 6
```

然后运行以下命令：

```
mgradm restart
```

> 将此值增大会使 Uyuni 服务器花费更长时间检查受控端是否无法连接或无响应，导致 Uyuni 服务器总体速度更慢或响应能力更低。

30.31. Troubleshooting Salt SSH known hosts

If a machine or VM is reinstalled with a new SSH key, but the same IP address is used to onboard it, there is a chance that Salt will complain about a "man-in-the-middle attack". This happens because Salt's **.ssh/known_hosts** file still contains the old SSH fingerprint.

To remove a host from this list, run the following command:

```
mgrctl ssh knownhost remove <hostname>
```

If the host to remove is not using the default SSH port **22**, you can specify the port manually:

```
mgrctl ssh knownhost remove <hostname> <port>
```

30.32. 对纲要升级失败的问题进行查错

If the schema upgrade fails, the database version check and all the other spacewalk services do not start. Run **mgradm start** on the container host for more information and hints how to proceed.

> To access a shell inside the Server container run **mgrctl term** on the container host.

也可以直接在容器中运行版本检查：

```
systemctl status uyuni-check-database.service
```

或

```
journalctl -u uyuni-check-database.service
```

These commands print debug information if you do not want to run the more general **mgradm** command.

30.33. 同步查错

同步失败的原因有很多。要获取有关连接问题的详细信息，请运行以下命令：

```
export URLGRABBER_DEBUG=DEBUG
spacewalk-repo-sync -c <通道名称> <选项> > /var/log/spacewalk-repo-sync-$(date +%F-%R).log
2>&1
```

You can also check logs created by Zypper at **/var/log/zypper.log**

GPG 密钥不匹配

Uyuni does not automatically trust third party GPG keys. If package synchronization fails, it could be because of an untrusted GPG key. You can find out if this is the case by opening **/var/log/rhn/reposync** and looking for an error like this:

```
['/usr/bin/spacewalk-repo-sync', '--channel', 'sle-12-sp1-ga-desktop-
nvidia-driver-x86_64', '--type', 'yum', '--non-interactive']
RepoMDError: 无法访问储存库。可能未导入储存库 GPG 密钥
```

要解决该问题，需要将 GPG 密钥导入 Uyuni。有关导入 GPG 密钥的详细信息，请参见 **Administration Guide › Repo Metadata**。

GPG Key Removal from spacewalk-repo-sync

When a GPG key for repository has been manually imported using **spacewalk-repo-sync**, and this key is no longer needed (for example if the key was compromised, or was used for testing purposes only), it can be removed from the zypper RPM database used by **spacewalk-repo-sync** with the following command:

```
rpm --dbpath=/var/lib/spacewalk/reposync/root/var/lib/rpm/ -e gpg-pubkey-*
```

where **gpg-pubkey-*** is the name of the GPG key to be removed.

续订 GPG 密钥

如果您要续订某个 GPG 密钥，请先去除旧密钥，然后生成并导入新密钥。

校验和不匹配

If a checksum has failed, you might see an error like this in the **/var/log/rhn/reposync/*.log** log file:

```
Repo Sync Errors: (50, u'checksums did not match
326a904c2fbd7a0e20033c87fc84ebba6b24d937 vs
afd8c60d7908b2b0e2d95ad0b333920aea9892eb', 'Invalid information uploaded
to the server')
The package microcode_ctl-1.17-102.57.62.1.x86_64 which is referenced by
patch microcode_ctl-8413 was not found in the database. This patch has
been skipped.
```

You can resolve this error by running the synchronization from the command prompt with the **-Y** option:

```
spacewalk-repo-sync --channel <通道名称> -Y
```

此选项在同步之前校验储存库数据，而不是依赖于本地缓存的校验和。

连接超时

如果下载超时并出现以下错误：

```
28：操作速度太慢。过去 300 秒的传输速率小于 1000 字节/秒
```

You can resolve this error by specifying **reposync_timeout** and **reposync_minrate** configuration values in **/etc/rhn/rhn.conf**. By default, when less than 1000 bytes per second are transferred in 300 secs, the download is aborted. You can adjust the number of bytes per second with **reposync_minrate**, and the number of seconds to wait with **reposync_timeout**.

重新同步期间手动信任密钥

It is possible that in some cases when **reposync** is run, you may need to accept the GPG key manually. For example:

```
# spacewalk-repo-sync -c nvidia-compute-sle-15-x86_64-we-sp3
17:07:40 ======================================
17:07:40 | Channel: nvidia-compute-sle-15-x86_64-we-sp3
17:07:40 ======================================
17:07:40 Sync of channel started.
New repository or package signing key received:
  Repository:       nvidia-compute-sle-15-x86_64-we-sp3
  Key Fingerprint:  610C 7B14 E068 A878 070D A4E9 9CD0 A493 D42D 0685
  Key Name:         cudatools <cudatools@nvidia.com>
  Key Algorithm:    RSA 4096
  Key Created:      Thu Apr 14 16:04:01 2022
  Key Expires:      (does not expire)
  Rpm Name:         gpg-pubkey-d42d0685-62589a51
    Note: Signing data enables the recipient to verify that no modifications occurred
after the data
    were signed. Accepting data with no, wrong or unknown signature can lead to a
corrupted system
    and in extreme cases even to a system compromise.
    Note: A GPG pubkey is clearly identified by its fingerprint. Do not rely on the
key's name. If
    you are not sure whether the presented key is authentic, ask the repository
provider or check
    their web site. Many providers maintain a web page showing the fingerprints of the
GPG keys they
    are using.
Do you want to reject the key, trust temporarily, or trust always? [r/t/a/?] (r):
```

30.34. Taskomatic 查错

储存库元数据重新生成是一个资源消耗量相对较高的进程，因此 Taskomatic 可能需要几分钟才能完成。此外，如果 Taskomatic 崩溃，则储存库元数据重新生成可能会中断。

To access a shell inside the Server container run **mgrctl term** on the container host.

If Taskomatic is still running, or if the process has crashed, package updates can seem available in the Web UI, but do not appear on the client, and attempts to update the client fail. In this case, the **zypper ref** command shows an error like this:

```
在指定的 URL 上找不到有效元数据
```

要更正此问题，请确定 Taskomatic 是否仍在生成储存库元数据，或者是否发生了崩溃。等待完成元数据重新生成，或者在崩溃后重启 Taskomatic 以正常执行客户端更新。

过程：解决 Taskomatic 问题

1. On the Uyuni Server, check the **/var/log/rhn/rhn_taskomatic_daemon.log** file to determine if any metadata regeneration processes are still running, or if a crash occurred.
2. 重启 Taskomatic：

   ```
   service taskomatic restart
   ```

3. 在 Taskomatic 日志文件中，可以通过查看如下所示的开始行和结束行来识别与元数据重新生成相关的部分：

   ```
   <YYYY-DD-MM> <HH:MM:SS>,174 [Thread-584] INFO
   com.redhat.rhn.taskomatic.task.repomd.RepositoryWriter - Generating new repository
   metadata for channel 'cloned-2018-q1-sles12-sp3-updates-x86_64'(sha256) 550 packages,
   140 errata

   ...

   <YYYY-DD-MM> <HH:MM:SS>,704 [Thread-584] INFO
   com.redhat.rhn.taskomatic.task.repomd.RepositoryWriter - Repository metadata
   generation for 'cloned-2018-q1-sles12-sp3-updates-x86_64' finished in 4 seconds
   ```

30.35. 对 Web UI 无法加载的问题进行查错

有时，Web UI 在迁移后不会加载。如果新系统的主机名和 IP 地址与旧系统的相同，这种情况通常是由浏览器缓存所致。两个系统的主机名和 IP 地址相同可能会使一些浏览器产生混淆。

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